

TinyML Smart Plug for Complementary Protection Against Progressive Fire-Inducing Electrical Faults in Residential Circuits

by

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ABSTRACT

This study developed a TinyML Smart Plug as complementary socket-level protection for progressive fire-inducing electrical faults in 230V residential circuits. The prototype addressed sustained overload, degraded socket-contact heating, and load-limited series arc faults, which may remain hazardous before conventional breaker operation. It integrated voltage, current, and temperature sensing with an ESP32-S3 controller, relay-based load isolation, local audio-visual alerts, Firebase logging, and a Progressive Web App for monitoring and basic relay control. Local firmware handled calibration, thermal estimation, threshold protection, load-context classification, and Random Forest series-arc inference; the cloud interface supported visualization, logging, labeling, and operator interaction. Evaluation included sensor calibration and revalidation, controlled overload tests, socket-heating characterization and protection tests, induced series-arc data collection, waveform-feature extraction, manual labeling, and embedded model deployment. Revalidation produced mean absolute errors of 3.910V, 0.178A, and 1.02°C for the voltage, current, and temperature channels, respectively. The highest-stress overload case reached 17.30A mean current and 19.73A peak current before relay disconnection after 61.382s. Heating tests showed faster temperature rise in the degraded socket than in the normal socket under comparable loading. For trial-level series arc detection, the embedded model and firmware persistence logic achieved 97.50% accuracy, 98.97% precision, 96.00% recall, and 97.46% F1-score across tested loads. The results support controlled laboratory feasibility for calibrated sensing, deterministic threshold protection, and on-device TinyML inference in a smart plug that complements, but does not replace, mandatory residential electrical protection.

Keywords: TinyML, Smart Plug, Series Arc Fault Detection, Socket Heating, Overload Protection

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Chapter 1

INTRODUCTION

This chapter presents the foundation of the study by establishing the residential electrical-fire problem, the limitations of conventional protection in detecting progressive fault conditions, and the need for a complementary socket-level monitoring device. It also defines the problem addressed by the research, identifies the significance of the proposed system, states the objectives that guided the development and evaluation of the prototype, and clarifies the scope and limitations of the study.

1.1 Background of the Study

Electrical faults remained one of the leading causes of residential fires in the Philippines. Bureau of Fire Protection (BFP) records showed that electrical faults accounted for 27.09% of reported fire incidents, while related literature also identified undetected electrical abnormalities, faulty outlets, overloaded circuits, degraded wiring, and faulty appliances as recurring causes of residential fire events [1]-[3]. These incidents did not only result in property loss, but also caused injuries, displacement, and long-term economic hardship for affected households.

Residential electrical service in the Philippines, particularly in provincial and urban domestic settings such as those commonly found in Naga City, generally consisted of a 230V, single-phase, two-wire, 60Hz supply [4]. Although this configuration could technically accommodate more advanced protective devices, the actual residential protection landscape still depended primarily on miniature circuit breakers (MCBs), with wider adoption of specialized protective devices such as AFCIs and GFCIs remaining limited by cost, availability, and lack of mandatory implementation in residential premises [5]. As a result, many households still relied on protection systems that were effective mainly against high overcurrent and short-circuit events, but not against slowly developing pre-fire electrical conditions.

This created a critical safety gap. Conventional MCBs were designed primarily to

protect fixed building wiring from sustained overcurrent and short-circuit damage. They were not designed to warn occupants about gradual or localized electrical hazards before those hazards escalated into ignition. In particular, they did not directly detect progressive contact heating at a socket, persistent sub-trip overload behavior, or low-current series arc faults that occurred below the breaker's immediate trip region. In poor or degraded electrical connections, a small region of elevated resistance could generate intense localized heating under otherwise ordinary current flow. This condition could persist long enough to char surrounding material, deform outlet components, and eventually cause fire even without a classical breaker trip [6], [7].

At the same time, recent developments in non-intrusive electrical monitoring, embedded machine learning, and low-cost microcontrollers made it possible to design a more accessible protective device. Non-intrusive sensing enabled waveform-based monitoring without modifying existing building wiring [8]. Machine learning methods, especially Random Forest-based classifiers, had already demonstrated strong performance in electrical-fault classification tasks [9]. TinyML deployment on microcontrollers further showed that real-time inference could be achieved within tight memory, latency, and power constraints [10]. These developments suggested that a practical, socket-level protective device for residential environments was already technically feasible.

This study, therefore, developed an ESP32-based protective smart plug that combined voltage, current, and thermal sensing with hybrid fault detection. The system was designed as a complementary protection layer for Philippine single-phase residential circuits. It focused on three progressive fire-inducing fault conditions: overload-related hazardous overcurrent behavior, series arc faults from unstable electrical conduction, and contact heating at the plug-socket interface. Rather than replacing mandatory protection, the device was intended to provide earlier warning, localized monitoring, and automatic disconnection when hazardous conditions were detected before they escalated into more serious fire events.

1.2 Statement of the Problem

The preceding discussion shows that residential electrical fire prevention does not only depend on clearing severe short circuits or major overloads. It also requires attention to progressive electrical hazards that may remain energized while still producing thermal or waveform abnormalities. This study was therefore concerned with the protection gap between normal household operation and the point at which conventional circuit protection is forced to operate.

Residential electrical fires remained a persistent public safety concern because the electrical protection normally present in ordinary homes was not sufficient to address all dangerous fault mechanisms. The Philippine residential setup commonly relied on MCBs as the primary mandatory protective device [4] [5]. While effective against major overload and short-circuit events, these devices did not provide early warning for progressive hazards and did not directly detect the low-current or localized fault mechanisms that often preceded electrical fires.

A particular problem involved fault conditions that developed below the level of obvious breaker action. A household branch circuit could remain energized while a socket or terminal connection slowly degraded, while intermittent series arcing occurred during unstable conduction, or while a connected load remained in a hazardous but not immediately breaker-tripping overcurrent state. In such cases, occupants could remain unaware of the developing danger until visible damage, smoke, or ignition had already begun. This protection gap became more serious in settings where advanced commercial arc-fault protection was uncommon and where households continued to rely on low-cost, conventional protective hardware alone.

Although advanced protective technologies already existed, their direct adoption in ordinary Philippine homes remained limited. AFCIs and related commercial systems could improve safety, but they were not yet common in many residential settings because of cost, product availability, installation practice, and regulatory limitations [5]. Existing research prototypes also often relied on intrusive instrumentation, laboratory-grade measurement equipment, or computation-heavy processing that was not suitable for low-cost

embedded deployment. Moreover, arc-fault detection systems in both literature and practice still faced nuisance-alarm challenges when normal appliance behavior, switching events, or electrically noisy loads produced signatures similar to true faults.

Therefore, this study addressed the need for an affordable, socket-level, non-intrusive protective device that could operate alongside conventional circuit protection. Specifically, the device needed to detect and respond to three progressive fault classes relevant to residential fire safety: overload-related hazardous current behavior, induced series arc faults, and contact heating at the socket interface. The device also needed to operate on resource-constrained embedded hardware, provide local protection even without cloud access, and support logging as well as remote visibility when network connectivity is available.

1.3 Significance of the Study

This study is expected to provide substantial benefits to residential users, fire-safety stakeholders, engineering researchers, and policymakers by addressing a practical gap in household electrical protection. Since progressive electrical hazards may develop before conventional protective devices operate, the development of a socket-level smart plug offers a complementary approach for earlier detection, localized monitoring, and automatic response. The study also supports the principles of Sustainable Development Goal 11 by contributing to safer and more resilient communities through accessible residential safety technology [11]. The key beneficiaries and their potential gains are outlined below.

1.3.1 Residential Households

Residential households are the primary beneficiaries of this study. The proposed smart plug provides an added layer of protection against electrical hazards that are not easily visible to ordinary occupants, such as sustained overload, socket-contact heating, and series arc faults. By monitoring voltage, current, and temperature at the outlet level, the device can alert users and disconnect the connected load when hazardous conditions are detected. This gives households earlier awareness of possible electrical danger before the condition

develops into visible damage, smoke, or fire.

The study is especially relevant to residential users who rely mainly on conventional circuit breakers for electrical protection. While breakers remain necessary, they may not directly detect localized socket heating or low-current arc behavior. The prototype therefore demonstrates how a plug-through device can supplement existing household protection without requiring permanent rewiring of the home.

1.3.2 Local Government Units

Local government units, disaster risk reduction offices, and fire-safety stakeholders may benefit from the study because it presents a possible technology-based approach for improving household-level fire prevention. Since electrical faults remain a recurring cause of residential fire incidents, devices capable of detecting early electrical abnormalities can support broader fire-risk reduction efforts.

The study may also help fire-safety personnel and community safety programs explain the importance of monitoring progressive electrical hazards, not only severe short circuits. By showing that overload, contact heating, and series arc faults can be monitored at the socket level, the research provides a technical basis for promoting more proactive residential electrical-safety awareness.

1.3.3 Electrical Practitioners and Safety Inspectors

Electrical practitioners, technicians, and safety inspectors may benefit from the study because it highlights fault conditions that are not always addressed by ordinary visual inspection or conventional overcurrent protection. The prototype demonstrates how localized sensing can identify abnormal behavior at the plug-socket interface, where degraded contact, poor fit, corrosion, or thermal buildup may occur.

The study may also support future diagnostic practices by showing the value of combining current behavior, temperature behavior, and fault-event logging. Although the device is not a substitute for professional electrical inspection, it provides a possible tool for observing early warning patterns that may guide maintenance, troubleshooting, or further assessment of residential electrical installations.

1.3.4 Engineering and Academic Community

The engineering and academic community may benefit from the study through its documented integration of embedded sensing, TinyML inference, threshold-based protection, relay actuation, and cloud-assisted monitoring in a single residential prototype. The study provides a development workflow that includes sensor calibration, waveform-feature extraction, series arc data collection, model training, embedded deployment, and controlled fault evaluation.

This contribution is relevant to future research in embedded electrical safety, smart protection devices, TinyML, non-intrusive monitoring, and Internet-of-Things-based residential systems. It may also serve as a reference for researchers who aim to improve arc-fault classification, socket-heating estimation, overload response, or real-time protection on resource-constrained microcontrollers.

1.3.5 Policymakers and Standards Stakeholders

Policymakers, standards stakeholders, and residential-safety advocates may benefit from the study because it demonstrates the feasibility of treating intelligent fault detection as a complement to existing electrical protection. The prototype does not replace code-mandated devices such as circuit breakers, but it shows how additional outlet-level protection may help address hazards that are difficult to detect using conventional protection alone.

The findings may contribute to future discussions on affordable residential safety devices, smart-home electrical protection, and the gradual modernization of household fire-prevention strategies. By presenting a locally developed prototype for Philippine residential conditions, the study may help inform future research, policy planning, or pilot implementation of complementary electrical-safety technologies.

1.4 Objectives

This study aimed to develop an intelligent ESP32-S3-based protective smart plug that provides complementary, socket-level protection for Philippine single-phase residential

circuits by detecting progressive fire-inducing electrical faults through hybrid on-device sensing, analysis, and fault response.

Specifically, the study aimed to:

1. Design and develop an ESP32-S3-based smart plug prototype integrating voltage, current, and temperature sensing, embedded processing, relay-based load disconnection, and local audio-visual fault indication for residential single-phase applications;
2. Calibrate and validate the voltage, current, and temperature sensing subsystems against appropriate reference instruments to establish measurement accuracy suitable for threshold-based protection and machine learning-based analysis;
3. Implement a hybrid fault-detection system that uses on-device TinyML classification for series arc-fault detection and deterministic threshold-based logic for overload and socket-heating protection;
4. Develop a complementary protection, alert, and monitoring mechanism that operates alongside existing safety devices through local audio-visual alerts, relay disconnection, fault-event logging, and a Firebase-integrated Progressive Web App for real-time visualization and basic remote-control functions; and
5. Evaluate the developed system under controlled laboratory conditions involving overload, induced series arc fault, and socket/contact-heating scenarios in terms of detection behavior, system response, measurement reliability, and operational feasibility.

1.5 Scope and Delimitations

The scope and delimitations define the boundary within which the prototype was designed, tested, and interpreted. Since the study involved electrical safety, embedded sensing, machine learning, and controlled fault simulation, the boundaries are separated into the actual coverage of the work and the methodological, technical, and practical limitations that affected the interpretation of the findings.

1.5.1 Scope of the Study

This study focused on the design, development, calibration, and controlled evaluation of an intelligent protective smart plug for Philippine residential electrical systems. The prototype was intended for 230V, single-phase, two-wire, 60Hz circuits typical of ordinary household branch connections. It was developed as a plug-through, socket-level device that could be inserted into an existing outlet without requiring permanent modification of the building wiring.

The study covered the complete prototype process, including hardware integration, firmware development, signal acquisition, sensor calibration, feature extraction, TinyML model deployment, threshold-based protection logic, relay actuation, local notification, cloud logging, and remote data visualization. The final system architecture used a ZMPT101B-based voltage sensor, an ACS37035 current sensor digitized through an MCP3204 external SAR ADC, and a TTC05103JSY 10k Ω thermistor for temperature measurements. Current analysis was executed on the ESP32-S3 Xiao using embedded sampling, windowed feature computation, and on-device inference for real-time protection decisions.

The developed prototype monitored three primary fault conditions relevant to residential fire risk: overload current behavior, induced series arc faults, and contact heating at the plug-socket interface. With this scope, the device was implemented as a complementary safety function rather than a replacement for breaker protection. Overload protection used current-based warning and trip logic, socket-heating protection used estimated socket-temperature behavior and a conservative 40°C control threshold, and series arc-fault detection used on-device Random Forest inference based on extracted current features and lab-generated labeled data.

The study evaluated the prototype under controlled laboratory scenarios involving resistive overload simulation, controlled series arc generation, and simulated contact heating using a degraded socket sample. The emphasis of the evaluation was on technical feasibility, detection behavior, protection response, measurement reliability, and prototype operability in a socket-level residential use case. It did not seek to replace utility protection,

distribution-panel protection, or full-house fault-monitoring systems.

1.5.2 Methodological Limitations

The study was conducted under controlled laboratory conditions rather than in occupied residential homes. Although this was necessary for safety, repeatability, and controlled comparison, laboratory conditions did not fully reproduce the variability of actual household environments. Factors such as wiring age, long-term humidity exposure, appliance diversity, plug fit, environmental contamination, user behavior, and branch-circuit variability in actual homes were not comprehensively represented. The findings therefore demonstrated controlled technical feasibility rather than complete real-house field performance.

The series arc-fault dataset used for training, validation, and detection was based on controlled fault inducement rather than naturally occurring household arc events. The arc generator used a repeatable electrode separation and recontact mechanism to create series arcs under known conditions, and the collected labels were derived from induced test events. This approach was appropriate for safe data collection and reproducibility, but naturally occurring residential arc faults may exhibit wider variability in duration, intermittence, load interaction, and noise than the controlled setup reproduced. Consequently, model generalization to all real-world arc behaviors remained limited.

The training and validation scope was also limited by the load families included in the experiments. The device was tested using selected load categories that were safe and practical for induced-fault experimentation. It did not comprehensively include all possible household load types, such as sensitive SMPS loads, phase-angle-controlled loads, or mixed electronic loads, because some of these devices could be damaged during induced fault testing. The resulting model performance should therefore be interpreted within the conditions and load classes represented in the collected dataset.

1.5.3 Technical Limitations

The contact-heating function was constrained by the non-intrusive thermal architecture of the prototype. The thermistor channel did not directly measure the true internal

contact-spot temperature inside the socket interface. Instead, it measured a delayed external thermal response that the firmware interpreted as an estimated socket-temperature proxy. For that reason, the implemented 40°C threshold should be understood as a conservative, device-specific control threshold for an indirect external sensor that could under-read the true hidden hotspot. This limitation was influenced by thermistor placement, thermal lag, enclosure conduction, and the presence of cooling hardware inside the prototype.

The prototype also remained susceptible to nuisance detections under certain operating conditions. As with AFCI research and other arc-detection systems, normal events such as switching transitions, relay actuation, brushed-motor behavior, intermittent appliance conduction, or electrically noisy loads could generate arc-like signatures. Although the firmware and model pipeline were designed to reduce false positives, nuisance alarms and classification ambiguity remained possible, particularly for unseen load types or fault-adjacent events not fully represented in the training data.

The device provided only local, socket-level sensing. It monitored the branch segment and connected load at the outlet where it was installed; it was not designed to provide whole-house monitoring, multi-branch coordination, or exact upstream fault localization. It therefore could not be treated as a substitute for panel-level protective infrastructure, wiring inspection, or full-building electrical diagnostics. In addition, very low-power loads were outside the reliable operating range of the arc-detection pipeline because low current levels may reduce signal-to-noise ratio and weaken waveform-feature reliability.

Remote visualization, cloud logging, and web-based control also depended on Wi-Fi and internet availability. Core protection functions were implemented locally on the device, but convenience features such as Firebase logging, Progressive Web App visualization, and remote monitoring were not fully available when network connectivity was unstable or unavailable.

1.5.4 Practical Limitations

The prototype was constrained by practical hardware and packaging limitations. Because it used a relay, heatsink, fan, sensing modules, and protective enclosure, the device

was larger and heavier than an ordinary household plug accessory and was not yet optimized for consumer-ready ergonomics. The available GPIO, memory, enclosure space, and power arrangement also limited future expansion of hardware features.

The study was also limited to prototype performance and technical feasibility. It did not comprehensively address long-term field aging, certification, mass-manufacturing readiness, regulatory approval, market adoption, user acceptance, maintenance practice, or full economic comparison with commercial AFCI/GFCI solutions. Latched fault handling also required user awareness and intervention for confirmation, reset, or continued operation after certain protective events.

For these reasons, the prototype was not presented as a complete solution to residential electrical fire prevention. It was instead framed as a complementary protective layer that could operate alongside existing technologies and provide earlier local awareness of progressive fire-inducing electrical faults under the tested conditions.

Chapter 2

REVIEW OF RELATED LITERATURE

This chapter reviews the literature that supports the development of the proposed TinyML Smart Plug as a socket-level protection device for progressive electrical-fire precursors. Rather than treating electrical faults as a single undifferentiated class of events, the review is organized around the specific hazards that motivated the prototype: overload-related thermal stress, localized contact heating at the plug-socket interface, series arc faults, and abnormal supply conditions that may intensify equipment stress or affect protection behavior. It also reviews related literature on non-intrusive load monitoring, TinyML deployment on microcontrollers, current-signal feature extraction, embedded fault detection, and validation practices for protection-oriented devices. The chapter is divided into three major parts: theoretical background, related studies, and synthesis.

2.1 Theoretical Background

This section presents the theoretical concepts needed to understand the fault conditions addressed by the study. It begins with the residential electrical-fire problem and then narrows the discussion to electrical fault mechanisms, conventional protective devices, contact heating, external temperature measurement, waveform disturbance, conduction intermittence, and load-family behavior. These concepts establish why the proposed system was designed as a complementary socket-level protection device rather than as a replacement for existing circuit protection.

2.1.1 Residential Fire-Risk and Protection Gap

Electrical faults remain a persistent residential safety problem because not all hazardous conditions develop as instant catastrophic failures. In the Philippines, [1] identified electrical issues as the leading cause of recorded fire incidents, emphasizing that domestic electrical safety remains a practical and not merely theoretical concern. This is reinforced by NFPA fire statistics showing that electrical failure or malfunction continues to account for a substantial share of home structure fires, injuries, and losses [2].

The core implication for this study is that many dangerous electrical events evolve gradually. Some are cleared quickly by conventional protection, but others develop in a sub-breaker regime in which heat, contact degradation, and intermittent conduction are already becoming hazardous while the branch circuit still appears electrically acceptable. This gap between ordinary operation and fully developed failure is the operating space addressed by the proposed smart plug.

Since residential fire risk can arise from different electrical mechanisms, the next section distinguishes among the fault classes that may occur in household circuits and clarifies which of these are most relevant to the proposed device.

2.1.2 Electrical Fault Mechanisms

Overload is a sustained current demand above the intended operating capacity of a conductor, outlet, or protection device. It is fundamentally a thermal process because Joule heating rises with current and time. Conventional low-voltage breakers are intentionally inverse-time devices: lower overcurrents are tolerated longer, while larger overcurrents trip more quickly [12]. This design is necessary to avoid nuisance trips, but it also means that a localized plug, relay, receptacle, or appliance cord can remain under significant thermal stress before upstream protection is forced to operate. For common PVC-insulated low-voltage cables, the 70°C conductor temperature remains a meaningful thermal reference, and repeated thermal stress accelerates insulation aging and deterioration [13], [14].

Short circuits and ground faults are more abrupt fault classes. A short circuit produces a low-impedance path and very high current, while a ground fault diverts current through an unintended path to earth or grounded metal. For personnel protection, the danger of ground-fault current is severe because fibrillation can occur at relatively low current levels, especially when body resistance is reduced by moisture [15]. In practice, these fault classes are already well served by breakers and residual-current or ground-fault protective devices, so they are reviewed here mainly to clarify why they were not the primary target of the present smart plug.

Arc faults occupy a more difficult middle ground. A parallel arc can resemble a

short circuit and may trip conventional overcurrent protection, but a series arc is limited by the load current and may therefore remain below the normal breaker trip region. AFCI literature consistently identifies this as the main reason arc-fault protection was developed: the waveform can be thermally dangerous without becoming a classical overcurrent event [16]. For a smart plug intended to protect the immediate load connection, this distinction is crucial because many plug-side and cord-side failures are series-arc-like before they ever become hard faults.

High-resistance connection faults are especially relevant to this study because they explain why socket-level temperature monitoring is needed. A loose, corroded, worn, oxidized, or mechanically weakened contact increases local resistance, so the same operating current generates disproportionate local heating according to $P=I^2R$. Classic NIST work on glowing electrical connections and later studies on high-resistance mechanisms show that such faults can persist for long periods, ignite nearby polymers, and remain effectively invisible to ordinary breakers or GFCIs because the total current may stay within an otherwise normal appliance range [17], [18].

Overvoltage and undervoltage are not the main fire mechanisms targeted by the research prototype, yet they remain important because they influence equipment stress and protection logic. Transient overvoltages can damage connected electronics, which is why surge protective devices are used to limit overvoltage at the load side [19], [20]. Sustained undervoltage can also be harmful because some loads, especially motors, may overheat or operate inefficiently under depressed supply conditions. In the context of this study, voltage supervision therefore serves two purposes: direct protection against abnormal supply conditions and better interpretation of current-based fault signatures.

After identifying the major electrical fault mechanisms, it is necessary to examine how existing protective devices respond to these conditions. This establishes why some faults are already adequately handled by conventional protection, while others require complementary monitoring.

2.1.3 Conventional Protective Devices and Their Limits

Conventional protection is effective because it is selective, but that same selectivity creates blind spots. Miniature circuit breakers primarily protect branch wiring against overload and short circuit. Their thermal-magnetic behavior, including fast tripping at high multiples of rated current and delayed tripping at lower overcurrent levels, is intentionally coordinated around conductor protection rather than local receptacle or plug-blade temperature [3], [21], [22].

Ground-fault circuit interrupters compare outgoing and returning current and trip on small imbalances, typically around the milliampere range associated with leakage or shock hazards [23]. AFCIs were introduced to detect wiring arcs that overcurrent protection can miss, especially series arcs in branch circuits [16]. Surge protective devices, meanwhile, are intended to clamp transient overvoltages and prevent damage to downstream equipment [19], [20].

Taken together, these devices provide important layers of protection, but none directly measure the thermal condition of the plug-socket interface. They are therefore not designed to identify a local hotspot created by poor contact pressure, oxidation, or partial separation while the total current remains modest. This is the central theoretical justification for the proposed prototype: not replacement protection, but complementary protection at the socket level, where early-stage heating and unstable conduction can begin.

Because conventional protection does not directly observe the plug-socket interface, the next section focuses on the thermal behavior of poor contacts and degraded sockets, which form one of the main fault targets of the study.

2.1.4 Thermal Behavior of Poor Contacts and Degraded Sockets

The literature on glowing contacts, poor plug-receptacle interfaces, and overheated outlets converges on a single physical principle: electrical-fire precursors are often thermally local before becoming electrically global. The *Development and Analysis of Electrical Receptacle Fires*, authored by [24], documented that loose terminal conditions resulted in substantial overheating, visible damage, and failure in receptacle systems. Later fire stud-

ies showed that glowing connections can persist, consume surrounding non-metal parts, and ignite plastics even though conventional protection may not operate [7], [17].

The importance of a socket-level monitor is therefore not merely that it measures temperature, but that it measures temperature near the region where hidden degradation manifests first. The thermal process may be modeled as a first-order system in which the local node temperature rises progressively toward a steady-state value set by injected Joule heating, thermal resistance to ambient, and thermal capacitance. In this view, dangerous faults do not need to create instant flame to be serious; they only need enough sustained local power to initiate oxidation, polymer softening, charring, and further growth of contact resistance.

Standards and fire studies also show that external body temperature is not equivalent to true contact temperature. IEC 60884-1 classifies household plugs and socket-outlets for ordinary ambient temperatures not normally exceeding 40°C and uses temperature-rise criteria at internal conductive points rather than prescribing one universal external body limit [25]. IEC 60227 similarly preserves 70°C as a meaningful conductor-temperature reference for common PVC-insulated cable families, since this region is associated with insulation stress and visible wire degradation [14]. UL 498 also provides temperature-rise limits for attachment plugs and receptacles under test conditions [26]. When these standards are interpreted together with thermal lag and hidden contact-spot behavior, an early external intervention threshold becomes more defensible for a non-intrusive prototype.

2.1.5 External Temperature Measurement and Hotspot Underestimation

For embedded socket monitoring, the NTC thermistor is useful because it is low-cost, sensitive in the temperature range of interest, easy to interface with, and suitable for continuous protection duty [27], [28]. However, the same literature also warns that NTC measurements must be interpreted with care. [27] defines thermal time constant as the time required for the sensor to traverse 63.2% of a sudden temperature change, while [28] notes that the thermistor is useful for over-temperature protection but does not directly measure the hottest internal point of the system. [29] further explains that poor sensor placement

increases lag because the controller only reacts to the temperature it “sees” through the measurement point.

This matters directly for receptacle monitoring because the hottest point is usually inside the contact interface, and not at the external surface touched by the thermistor. [30] argued that surface temperature is a poor severity indicator when overheated contact spots are thermally separated from external surfaces. The study by [31] on indirect thermography indicates that the internal temperature of electromechanical relay contacts can significantly exceed the package temperature. In other words, an external sensor reading near 40°C can correspond to a hidden contact zone already much hotter than the measured point.

For that reason, the 40°C threshold used in this study is best interpreted as a conservative early intervention threshold for a non-intrusive, thermally lagged sensor used to estimate a hidden hotspot. It sits below the internal temperature region associated with clearly abnormal accessible heating and well below severe material-stress levels, while remaining sufficiently above normal ambient temperatures. This provides a safety margin for sensor lag, thermal gradients, and placement uncertainty.

With the thermal aspects of the fault problem established, we now turn our attention to series arc faults. Unlike contact heating, which is mainly thermal and localized, series arcs need to be detected based on their waveform behavior, as they can occur at current levels that are limited by the load.

2.1.6 Waveform Disturbance and Conduction Intermittence

Arc fault detection is possible only because arcing modifies the current waveform in ways that healthy operation does not. The literature repeatedly describes arc currents as random, discontinuous, pulse-rich, and rich in abnormal harmonic or high-frequency energy, but it also warns that the exact appearance depends strongly on the connected load [32]-[34]. This has two important implications for the study.

First, it justifies a waveform-disturbance family of features. These include descriptors related to harmonic distortion, spectral change, residual impulsiveness, and band-energy redistribution. In the literature, THD, high-frequency noise, residual or singular

components, and spectral fluctuation are regularly used because arcs inject non-stationary energy that does not behave like ordinary steady-state appliance current [34]-[39].

Second, it justifies a conduction-intermittence family of features. Series arcs and unstable contacts often create partial conduction, zero-current gaps, restrikes, half-cycle asymmetry, and irregular zero-cross behavior. Accordingly, the literature emphasizes zero-current features, pulse counting, current-fluctuation descriptors, and zero-cross interval instability because they directly represent the intermittent-recontact nature of arcing and poor electrical connection [34], [36], [37].

Viewed together, waveform disturbance and conduction intermittence form a multi-domain profile of an arc fault. However, these indicators cannot be interpreted in isolation because many normal household loads are also electrically nonlinear. The next section therefore discusses the importance of non-intrusive load monitoring and load-family context in reducing false detections.

2.1.7 Non-Intrusive Load Monitoring and Load Families

A further theoretical issue is that many healthy appliances are already nonlinear. Motors produce inrush and commutation effects; rectifier and SMPS loads draw peaky current; and phase-angle-controlled loads intentionally chop the waveform. NILM research demonstrates that these differences are not just nuisance artifacts but informative load signatures that can be extracted from current and voltage waveforms for appliance identification and load disaggregation [8], [40]-[42].

This is precisely why load family matters in fault detection. A waveform that looks abnormal for a resistive heater may be normal for a universal motor or rectifier load. Consequently, the detector must distinguish fault-induced non-stationarity from load-intrinsic nonlinearity. Context-aware inference, whether framed as NILM, load-family recognition, or appliance-family conditioning, is therefore not optional but a false-positive control strategy.

TinyML makes this context-aware approach feasible on embedded hardware. Reviews of machine learning for microcontroller-class hardware and TensorFlow Lite Micro

show that lightweight models can run within tight power, memory, and latency budgets when the feature space and runtime are carefully designed [32], [43]. This makes a hybrid embedded architecture realistic: deterministic thresholds for hard safety limits, with lightweight inference for ambiguous waveform cases such as load-dependent series arcs.

To summarize the theoretical boundary of the study, *Table 2.1* compares the main electrical fault classes reviewed in this section. The table identifies the primary hazard associated with each fault, whether conventional protection generally handles it, and how each fault class relates to the proposed smart plug.

Table 2.1. Summary of electrical faults

Fault class	Primary hazard	Conventional protection	Relevance
Overload	Thermal stress, insulation degradation, fire risk	Partial	Central target because branch breakers protect wiring, not necessarily the plug-socket interface.
Short circuit	Immediate high-current damage and fire	Yes	Reviewed for scope delimitation; not the main target of the smart plug.
Ground fault	Shock and leakage hazard	Yes	Reviewed to distinguish personnel-protection devices from the proposed socket-level fire-prevention focus.
Series arc fault	Ignition-capable arcing below ordinary breaker trip levels	Not always	Central target because its current is load-limited and often waveform-dependent.
High-resistance connection	Localized contact heating, glowing, melting, ignition	No	Central target because the contact can overheat without causing a classical overcurrent event.
Overvoltage	Equipment damage and stress	Partial	Included as voltage supervision for supply-state interpretation.
Undervoltage	Equipment stress, motor overheating, unreliable operation	Partial	Included as a supporting measurement-quality and supply-state condition.

Table 2.1 shows that the proposed smart plug is not intended to replace breakers,

GFCIs, AFCIs, or surge protective devices. Instead, it is positioned as a complementary protection layer for progressive fire-inducing faults that may remain dangerous before conventional protection is forced to operate. Standard circuit breakers are already effective at mitigating immediate, high-current events; therefore, this research excludes short circuits and high-current parallel arc faults as primary targets. The scope instead concentrates on more insidious fault conditions such as overload behavior that creates slow cumulative heat damage, series arc faults that operate below ordinary breaker thresholds, and contact heating caused by degraded plug-socket interfaces.

It also clarifies why the study uses the term contact heating rather than treating high-resistance connection as the final hazard itself. A high-resistance connection is the electrical condition, but the fire-relevant consequence is localized heating. Contact heating is therefore defined in this study as the hazardous thermal state that occurs when a poor or loose localized connection at the plug-socket interface generates intense heat from ordinary operating current. This is the specific thermal pathway the prototype aims to detect because it can develop without immediate conventional protection response.

2.2 Related Studies

With the theoretical basis of residential faults and the limitations of standard protective devices established, this section evaluates existing research to determine the most appropriate technical direction for the study. The review begins with non-intrusive load monitoring because load context affects how waveform disturbances should be interpreted. It then moves to TinyML feasibility, arc-fault detection features, threshold-based overload protection, receptacle heating studies, TinyML training strategy, and validation practices.

2.2.1 Non-Intrusive Load Monitoring and Load-Family Identification Studies

Recent non-intrusive load monitoring studies support the idea that useful appliance information can be extracted from centralized electrical measurements without installing dedicated sensors on every device. [8] shows that NILM can be used not only for energy disaggregation but also for automation-oriented services in residential buildings, while the

PLAID dataset in [40] demonstrates that plug-load identification from sampled voltage and current waveforms is feasible across many household appliances and operating instances. Together, these studies justify the use of one electrical measurement stream for both protection and context recognition [8], [40].

Feature-based NILM studies are especially relevant because they show that load identification does not always require large end-to-end deep-learning models. [41] uses steady-state current waveform features in a rule-based framework, while [42] demonstrates that percentage THD can serve as a meaningful descriptor of load patterns in smart-home environments. These findings support classifying appliances into load families, such as resistive, inductive, rectifier/SMPS, phase-angle-controlled, and brush-motor loads, before interpreting abnormal waveform behavior [41], [42].

The importance of load-family recognition in this study is therefore protective, not merely descriptive. Healthy household loads can naturally produce commutation spikes, conduction gaps, or harmonic distortion that may resemble fault behavior if they are evaluated without context. The NILM literature supports the claim that family recognition provides a necessary interpretive layer before a protection-oriented detector decides whether a waveform disturbance is intrinsic to the load or evidence of a developing electrical fault [8], [40]-[42].

Since load-family recognition requires embedded interpretation of waveform features, the next section examines whether machine-learning inference can realistically be performed on a microcontroller-class smart plug.

2.2.2 TinyML Feasibility on Resource-Constrained Microcontrollers

The feasibility of running machine-learning models on microcontroller-class hardware is well established in the TinyML literature. The review in [32] explains that successful deployment depends on a hardware-aware workflow in which data preparation, feature design, model selection, and resource limits are considered together rather than separately. This is directly relevant to the study, since the proposed smart plug must perform sensing, signal processing, inference, actuation, and communication inside a constrained embedded

system [32].

This feasibility is reinforced by [43], which presents TensorFlow Lite Micro as an inference framework designed for systems with severe memory and runtime constraints, and by [10], which shows that TinyML-style anomaly detection can be built around low-cost runtime operations when the processing pipeline is designed carefully. These studies support a compact, feature-based embedded pipeline over a large end-to-end model that would be harder to justify on a small microcontroller [10], [43].

The practical implication is that the strongest embedded design is not necessarily the most complex one, but the one that preserves physical interpretability and remains deployable under real runtime limits. This supports the study's hybrid philosophy: deterministic threshold logic for conditions with clear physical limits, and lightweight inference for ambiguous waveform cases in which feature interaction and load context materially improve detection quality [10], [32], [43].

Having established that TinyML deployment is feasible, the review now focuses on the most difficult fault class for the proposed system: series arc detection. Unlike overload and socket heating, arc faults often require multi-feature interpretation because their current magnitude alone may not be sufficient.

2.2.3 Arc-Fault Detection Studies and the Relevance of Multi-Domain Features

Recent arc-fault literature consistently shows that low-voltage series arcs are difficult to detect because they are load-dependent, stochastic, and often masked by healthy nonlinear appliances. The review in [33] identifies recurring challenges in AI-based arc-fault detection, including limited data, false alarms, missed detections, and poor generalization across loads. This is important to the study because it shows that arc detection should be framed as a constrained, multi-feature classification problem rather than as a single-threshold decision on one waveform quantity [33].

Feature-oriented studies point in the same direction. [34] combines current-fluctuation features with zero-current features to capture the concealed and random character of series arcs. [35] examines THD as a discriminant for series-arc behavior

in single-phase networks, while [36] combines high-frequency noise sensing with zero-crossing period analysis in a multi-load setting. More recent works such as [37], [38], and [39] also rely on multi-feature fusion, even though their final models differ. The common message across these studies is that useful arc evidence comes from combining complementary indicators of distortion, randomness, interruption, and context rather than from relying on current magnitude alone [34]-[39].

Another consistent lesson from this literature is that useful arc information is not confined to a single frequency band or a single analysis domain. Some studies exploit high-frequency content, some rely on zero-current or cycle-randomness descriptors, and others combine several feature families before classification. For the study, this supports organizing the current-signal review into two coherent groups: waveform disturbance and conduction intermittence. These groups are not independent mechanisms, but complementary views of the same arc process [34]-[39], [49].

2.2.3.1 Waveform Disturbance

Waveform-disturbance features describe how arcing injects broadband distortion, cycle asymmetry, and unstable spectral content into the current waveform. A useful starting point is the family of frequency-band energy features. [49] performs FFT-based integration of current content above 100Hz and shows that arc states increase integrated spectral content, while [50] uses time-frequency decomposition and sub-band energy to characterize serial arc behavior. These studies support the logic of the midband residual ratio, which isolates spectral energy in a selected midband after suppressing the dominant fundamental structure, then normalizes that energy against a baseline current reference so that the resulting ratio is more comparable across loads [49], [50].

A second waveform-disturbance descriptor is spectral change over time. [49] treats cycle-to-cycle difference and current-cycle similarity as ways to capture arc randomness, while [51] includes spectral flux in a broader waveform-monitoring feature space. For the study, spectral flux is therefore best understood as a frame-to-frame measure of spectral instability rather than as a direct measure of arc magnitude. This makes it useful for identi-

fying the non-stationary spectral injections associated with unstable conduction [49], [51].

Waveform disturbance also includes half-cycle shape distortion. [49] explicitly notes that, under arc conditions, the current can exhibit missing waveforms, positive-negative irregularity, and semi-axis asymmetry. That observation supports the use of half-cycle asymmetry in the study, where the positive and negative portions of the waveform are compared as a compact way to quantify unequal conduction between half cycles [49].

For harmonic distortion, the literature supports a contextual rather than absolute interpretation. [35] investigated THD for series-arc detection, while [42] shows that THD is also informative for distinguishing healthy load patterns, and [52] explicitly uses THD together with the amplitudes of the third and fifth harmonics, RMS, RMS change rate, and pulse-derived features. This means THD should not be treated as a standalone proof of arcing; instead, it is better used as one disturbance feature inside a broader decision process that already accounts for the natural nonlinearity of certain appliance families [35], [42], [52].

Finally, high-frequency energy elevation is one of the most common spectral signatures of series arcing. [53] designed a band-pass filter specifically to isolate arc-related content and then developed an algorithm using arc-signal energy and pulse counts, while [36] also relies on high-frequency noise as part of its detection logic. These studies support the study's use of a high-frequency energy-change feature that tracks whether the upper-band energy share increases relative to a baseline or broader band, thereby capturing the broadband elevation that often accompanies arcing [36], [53].

Waveform-disturbance features are useful because they capture the spectral and harmonic effects of arcing. However, arcing is also a conduction problem: the current may extinguish, restrike, and become discontinuous.

2.2.3.2 Conduction Intermittence

Conduction-intermittence features describe how series arcs repeatedly extinguish and restrike, producing irregular current continuity even when the overall load current re-

mains modest. [34] is the clearest support for this line of reasoning because it explicitly computes a zero-current time-proportional coefficient and combines it with a current-fluctuation measure. This makes [34] the proper anchor for the study’s low-current runtime and low-current ratio features: both are intended to quantify how much of the observation window is spent in weak or interrupted conduction.

The same physical logic supports zero-dwell behavior around the zero-current region. In a healthy waveform, the near-zero portion of the cycle is relatively regular, whereas in an arc state, extinction and restrike can increase or destabilize the time spent near zero current. [36] relies directly on zero-crossing period analysis, and [34] likewise centers its method on zero-current behavior.

Changes in RMS from one analysis interval to the next also belong to the conduction-intermittence family when they are interpreted as evidence of unstable conduction rather than simple load switching. [52] explicitly includes RMS and the change rate of RMS between adjacent cycles in its lightweight arc-fault method. This provides direct support for the study’s use of an absolute delta-of-RMS feature to capture cycle-to-cycle instability that is too persistent and irregular to be explained by healthy steady-state conduction [52].

[54] introduces a moving Z-score method calculated from dual FFT bands. Building on this concept, the current study employs an absolute z-deviation from a rolling RMS baseline, adapting the normalization principle outlined in [54]. This feature generates a load-relative deviation score that helps maintain meaningful thresholds across appliances with significantly different baseline currents [54].

The arc-fault literature supports feature-based classification because series arcs are ambiguous and load-dependent. In contrast, overload is more directly tied to current magnitude and heating duration.

2.2.4 Studies on Overload Monitoring and Threshold-Based Protection

Threshold-based domestic protection remains relevant because overload is a physical stress problem before it is a machine-learning problem. [44] presents a smart overload

trip system that measures current and disconnects supply when the current exceeds a preset value, while [45] presents an ESP32-based smart multiplug with current and temperature monitoring, remote control, and protective thresholding. These studies support the study's use of direct and interpretable current-limit logic for overload protection [44], [45].

Their importance to the study is architectural. They show that the smart-plug form factor can already accommodate sensing, communication, and protective switching. The present study therefore extends an already credible embedded architecture, but directs it toward a broader protection role by combining threshold protection with context-aware fault interpretation rather than stopping at current monitoring alone [44], [45].

Since overload protection addresses current-related thermal stress, the next related fault concern is localized socket heating. Unlike overload, contact heating may occur even when total load current remains within a range that does not immediately force upstream breaker operation.

2.2.5 Receptacle Heating Studies and Early Socket-Temperature Trip

The most important heating studies for this project are the ones showing that poor contact is a local and progressive fire mechanism. [24] demonstrated that loose terminal conditions can produce severe overheating and receptacle damage across many test configurations. [17] showed that receptacles and attachment plugs subjected to glowing conditions could suffer severe material damage, while the current path itself could continue functioning. These findings strongly support the idea that a connection can remain electrically functional even when thermally dangerous.

[46] simulated poor electric contact between plug and receptacle and showed that contact pressure strongly influences temperature rise and fire risk. [47] experimentally studied the ignition process caused by poor electrical contact and described a staged mechanism progressing from resistive heating and pyrolysis to arc-accelerated ignition. [7] also investigated glowing contact in electrical fires and highlighted that glowing phenomena may persist for hours while transferring hazardous heat to surrounding materials. Taken together, these studies show that socket heating is not a secondary symptom but a primary ignition

pathway.

The temperature-threshold question is best interpreted through studies of hidden versus measured temperature. [30] argued that there can be a large difference between overheated contact-spot temperature and external surface temperature. [31] again demonstrated that external package temperature can under-read internal contact temperature. This is the strongest scientific support for an early trip threshold such as 40°C at an external thermistor sensor. The argument is not that all outlets melt at 40°C, but that a 40°C external reading can already correspond to a hidden contact region operating far above that value. Considering the literature on standards, degradation, thermal lag, and indirect sensing, along with the limitations of the NTC sensor, the 40°C threshold is treated in this study as a conservative estimated tripping temperature.

The related studies on overload and socket heating justify deterministic threshold protection, while the arc-fault studies justify feature-based inference. The next section connects these two approaches by discussing how TinyML training should be structured for embedded fault detection.

2.2.6 TinyML Training

The reviewed literature also supports the overall training strategy used by the study. First, feature nomination should begin from fault physics rather than from blind feature accumulation. NILM and arc studies consistently favor features that describe harmonic content, waveform stability, conduction gaps, restrikes, and cross-cycle irregularity because these are closer to the physical differences between healthy nonlinear loads and unstable electrical faults [34]-[37], [41], [42]. This supports the study's decision to treat waveform disturbance and conduction intermittence as two coherent feature families.

Second, the training data should represent the operating regimes the detector will actually see. That means not only arc windows, but also normal steady-state windows, turn-on transients, and load-family diversity. The literature on load dependence and false positives makes this especially important: a detector trained only on clean resistive examples will not generalize well to motors, rectifiers, or phase-angle loads [8], [33], [40]. Thus,

the training pipeline used in the project, which captures start, steady, and arc segments, follows a structure already justified by the literature.

Third, model selection in TinyML is not solely about raw test accuracy. The reviews by [32] and [43] both imply that model choice must account for runtime footprint, deterministic behavior, and deployment simplicity. For this reason, a practical workflow is to screen plausible feature subsets using a relatively fast ensemble method, then finalize a compact classifier such as a Random Forest or other lightweight model for deployment. This kind of staged search is consistent with the broader TinyML literature because it balances exploratory breadth with embedded realism.

Finally, the load-family context should be trained separately or jointly in a way that informs the final arc decision. The literature strongly suggests that context should influence thresholds, persistence rules, or posterior interpretation rather than being treated as an isolated auxiliary label [8], [33], [41]. In practical terms, this means a smart plug should recognize whether it is observing a resistive heater, a motor, a rectifier, or a phase-angle-controlled load before deciding how aggressively to interpret impulsive waveform disturbances.

After establishing how the model should be trained, the next concern is how the system should be validated. Because the proposed device addresses different fault classes, the validation procedure must also be divided according to the physical behavior of each fault.

2.2.7 Testing and Validation

The validation literature supports using controlled, repeatable fault-induction methods rather than waiting for naturally occurring household faults. [48] constructs a dedicated AC arc-fault platform with a fixed electrode, a moving electrode driven by a stepper motor, and multiple household loads, which is the kind of repeatable setup needed for arc-data collection and algorithm evaluation. This supports the use of controlled arc-generation procedures in the study [48].

Validation should also extend beyond one appliance type. The load-diversity ar-

gument in NILM and arc-fault work implies that good performance on one heater or one motor is insufficient. A protection-oriented smart plug should therefore be validated across several load families, nuisance events, and non-fault transients so that its specificity is tested alongside its sensitivity [33], [40], [41], [48].

For overload and contact-heating validation, the literature supports a layered approach. Threshold-based smart-plug studies validate whether trip logic responds consistently when safety limits are exceeded, while poor-contact studies justify separate thermal validation for localized heating risk. The resulting implication for the study is that overload, arc detection, and contact heating should not be merged into one generic accuracy test; they require distinct procedures, reference measurements, and nuisance checks [44]-[48].

Table 2.2 summarizes how the reviewed literature maps to the feature and decision families used in the study. It groups the quantities into waveform disturbance, conduction intermittence, and deterministic stress variables to clarify why the prototype combines feature-based TinyML inference with threshold-based protection logic.

Table 2.2. Feature families relevant in the study

Feature family	Representative quantities	Literature connection
Waveform disturbance	Midband residual ratio; mid/high-frequency spectral flux; half-cycle asymmetry; current THD; high-frequency energy delta	Arc studies report random harmonic and high-frequency disturbance, while NILM studies show that healthy loads also have family-specific spectral signatures [34]-[39], [41], [42], [49]-[53].
Conduction intermittence	Low-current run time; low-current ratio; zero-dwell ratio; absolute RMS-current delta; baseline z-deviation	These quantities represent restrikes, current interruption, partial conduction, and unstable near-zero conduction timing, which are physically consistent with series arcs and weak-contact behavior [34], [36], [37], [52], [54].
Deterministic stress variables	RMS current; estimated socket temperature; voltage sag or abnormal supply condition	These variables encode thermal or electrical stress and remain essential even when machine learning is added. They improve interpretability and provide hard safety overrides for overload and heating events [3], [27]-[29], [44], [45].

Table 2.2 shows that the proposed feature set is not arbitrary. The waveform-disturbance features represent spectral and harmonic effects of arcing, while the conduction-intermittence features represent the extinguish-and-restrike behavior of unstable current flow. The deterministic stress variables remain necessary because some faults, particularly overload and socket heating, are better handled through direct physical thresholds than through classification alone. This table therefore supports the hybrid architecture of the study: machine learning is used where waveform interpretation is ambiguous, while deterministic logic is retained where physical safety limits are clear.

2.3 Synthesis

The reviewed literature supports the proposed smart plug most strongly when the device is framed as a complementary socket-level protection layer. Existing smart-plug and overload-monitoring studies already show that compact embedded devices can measure current and temperature, communicate with users, and actuate protective switching [44], [45]. At the same time, poor-contact fire studies show that the plug-receptacle interface is itself a meaningful origin of heating and ignition risk [46], [47]. Taken together, these findings justify a protection device that monitors hazards locally at the socket rather than relying only on upstream protection behavior [44]-[47].

The literature also shows that overload, contact heating, and series arcing are related but not equivalent fault classes. Overload is fundamentally a current-stress problem and is therefore well-suited to deterministic threshold logic. Contact heating is a localized thermal problem caused by degraded or poor electrical contact and therefore requires direct local temperature supervision. Series arcing, by contrast, is load-dependent and stochastic; it often appears as a mixture of spectral disturbance, zero-current behavior, and cycle irregularity rather than as a dramatic increase in current magnitude. This is why the most defensible design is a hybrid one: hard thresholds for direct electrical or thermal stress, and feature-based inference for ambiguous waveform cases [34]-[39], [44]-[47].

A second synthesis point concerns context. NILM and load-identification studies establish that healthy appliances have distinct waveform signatures, while arc-fault studies

show that many healthy nonlinear behaviors can resemble fault signatures if load context is ignored. Therefore, load-family recognition is not merely an added analytics feature; it is part of the protection logic itself because it helps separate load-intrinsic nonlinearity from fault-induced non-stationarity [8], [33]-[42].

A third synthesis point concerns implementation feasibility. The TinyML literature shows that on-device inference is realistic on microcontroller-class hardware when the pipeline is built around compact features, controlled memory use, and deployment-aware model selection. That makes the proposed architecture technically plausible: a smart plug can combine deterministic rules with a lightweight model, provided the feature computation, inference path, and safety logic are kept small, interpretable, and robust [10], [32], [43].

The gap addressed by the study is therefore clear. Prior work has separately shown the feasibility of smart-plug monitoring, the need for local supervision of poor-contact heating, the value of load-aware waveform interpretation, and the practicality of TinyML on constrained hardware. What remains comparatively underdeveloped is a single embedded device that unifies these ideas into one protection-oriented platform. The proposed TinyML Smart Plug addresses that gap by combining socket-level thermal supervision, threshold-based overload protection, load-family recognition, and multi-feature series-arc interpretation in one embedded system [8], [10], [32]-[47].

Chapter 3

METHODOLOGY

This chapter describes the design, implementation, computation, training, deployment, and evaluation procedures used for the TinyML Smart Plug. The methodology begins with the system model and firmware flow, then proceeds to the physical prototype, progressive web application (PWA), TinyML workflow, sensor calibration, waveform-feature computation, protection logic, and test procedures. This order follows the flow of the prototype itself: the hardware measures voltage, current, and temperature; the firmware converts these measurements into calibrated electrical and thermal variables; the TinyML pipeline classifies load context and series-arc behavior; and the protection manager decides whether to warn, log, or disconnect the load.

3.1 System Model

This section defines the prototype as a research system before the firmware, hardware, and test procedures are discussed in detail. It connects the measured inputs, embedded computations, cloud interface, and protection outputs so that the later methodology can be traced back to the same system boundaries.

3.1.1 Conceptual Framework

The prototype was modeled as an input–process–output system. Its inputs were the mains-voltage sensor, current sensor, thermistor, and wireless connection; its processing stage was the ESP32-S3 firmware with calibration, thermal estimation, waveform-feature extraction, TinyML inference, relay control, and cloud logging; and its outputs were warnings, PWA monitoring, stored event records, and relay disconnection when a protection condition was latched.

Figure 3.1 summarizes this research model.

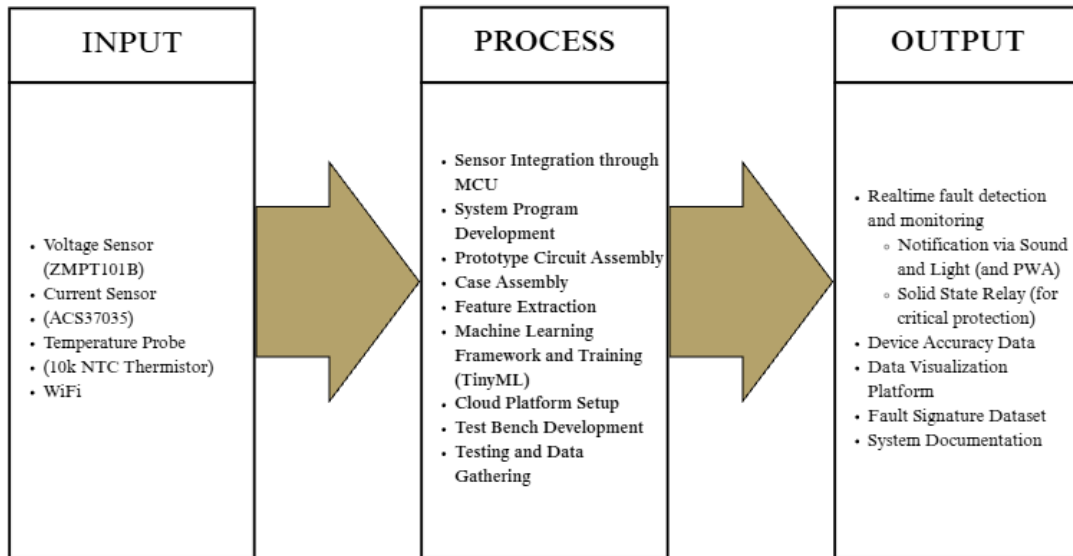


Figure 3.1. Conceptual framework of the TinyML Smart Plug.

Figure 3.1 presents the conceptual framework of the TinyML Smart Plug using an input-process-output structure. The input stage consists of the sensing and communication components used by the prototype, including the voltage sensor, current sensor, temperature probe, and Wi-Fi module. These inputs allow the system to acquire electrical and thermal conditions from the socket and load during operation. In the process stage, the acquired signals are integrated into the system program, converted into protection variables and waveform features, and used for machine-learning-based fault evaluation. This stage also includes prototype assembly, case assembly, cloud platform setup, firmware and model training, and testing and data gathering, which together support the development and validation of the smart plug. The output stage represents the expected functions and deliverables of the system, including real-time fault detection and monitoring, sound and light notification, relay-based circuit isolation, device accuracy data, data visualization, platform development, fault-signature dataset generation, and system documentation. The framework shows that the prototype was designed not only as a monitoring device, but as a complementary protection system capable of sensing abnormal socket conditions, processing them locally, notifying the user, logging relevant data, and opening the relay when a confirmed critical fault is detected.

The design variables used by the system are summarized in *Table 3.1*. The table

separates direct measurements from computed variables so that later equations can be traced to the firmware implementation.

Table 3.1. Main firmware variables used by the prototype

Variable group	Source	Application
RMS voltage	ZMPT101B channel and voltage calibration	Voltage monitoring, mains-present logic, undervoltage and overvoltage protection
RMS current	ACS37035 channel, MCP3204 ADC, anti-aliasing filter, and current calibration	Load monitoring, overload protection, thermal load term, and arc-feature extraction
Device temperature measurement	10k Ω thermistor and divider computation	Baseline thermal measurement for socket-temperature estimation
Waveform features	Current samples processed by the arc-detection module	Context classification and Random Forest arc inference
Protection state	Protection manager and relay driver	Warning, latching, relay disconnection, and event logging

Table 3.1 shows that current is used by several subsystems: it supports ordinary overload detection, thermal compensation, and the TinyML feature vector. Therefore, the current channel was treated as a shared measurement path whose calibration and sampling behavior must remain consistent across protection functions.

3.1.2 System Architecture

The same behavior can be described as a layered embedded system. The sensing layer captures voltage, current, and socket temperature; the firmware layer computes calibrated variables and protection states; the TinyML layer provides context-aware series-arc inference; and the PWA layer displays live and historical data using Firebase services [55], [56].

Figure 3.2 presents the system architecture used to connect these layers.



Figure 3.2. System architecture of the TinyML Smart Plug.

Figure 3.2 places the ESP32-S3 at the center of the design. The microcontroller receives conditioned analog signals, runs the protection and TinyML routines locally, drives the relay and indicators, and uploads selected telemetry. Firebase and the PWA also carry basic remote control fields, such as fault clearing and relay-control commands, but these commands do not replace the local protection logic. Keeping the protection decision on the microcontroller prevents the relay action from depending on internet availability, while the cloud and PWA remain useful for visualization, logging, labeling, review, and operator interaction.

3.2 Program and Firmware Flow

This section introduces the firmware flow that connects the measured hardware signals to the protection and reporting behavior. The discussion follows the same order used by the embedded program: initialization, repeated sampling and feature computation, state evaluation, relay actuation, and cloud/PWA updates.

3.2.1 Initialization and Sampling

The firmware follows a repeated acquisition, computation, inference, and protection cycle. On startup, the ESP32-S3 initializes the sensors, relay driver, OLED display, buzzer, Wi-Fi, Firebase interface, and local protection state. In the main loop, it samples the voltage, thermistor, and current waveform, applies calibration and filtering, computes thermal and waveform features, updates the context classifier, evaluates the arc classifier, and passes the resulting variables to the protection manager.

Figure 3.3 shows the operating sequence implemented by the firmware.

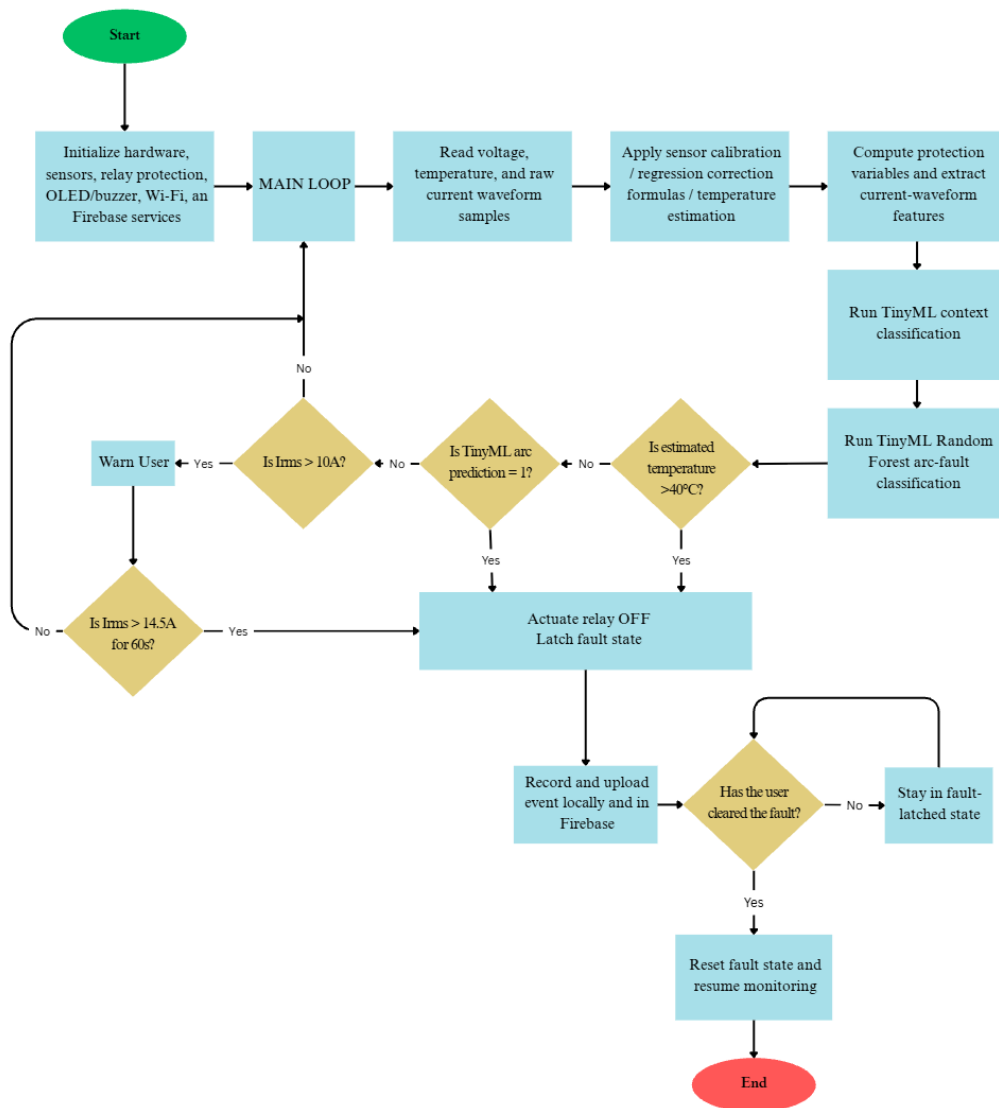


Figure 3.3. Program flowchart of the TinyML Smart Plug firmware.

The flowchart separates slow supervision from waveform-based inference. RMS voltage, RMS current, and thermistor temperature are updated for ordinary monitoring and deterministic protection. In parallel, the current waveform is sampled at the higher ADC rate, conditioned, framed, and converted into candidate waveform features. Usable feature frames update the load-context classifier first. Then, the accepted load-family code and confidence are appended to the arc-model input vector before Random Forest inference. The result is not used as a single-frame trip command. It is passed to the protection manager, where persistence counters, blanking intervals, and fault latches determine whether the relay should remain energized, warn the user, or disconnect the load. Another is that since the

system targets fire-inducing faults, thermistor temperature is treated with the highest priority because it is the most direct indicator of socket heating. Arc-fault inference is evaluated next because waveform discontinuities and intermittent conduction can produce localized heating even before a sustained current burden is established. Overload protection is handled after these conditions because excessive current is treated as a time-dependent stress that requires persistence before relay action. Thus, the protection manager resolves the fault sequence by prioritizing temperature, followed by the arc fault, and finally, the overload.

3.2.2 Protection-State Evaluation

Figure 3.3 shows that a relay trip can be caused by a sustained overload, an estimated socket over-temperature, or an arc-fault classification. The overload path issues a warning above 10.0A, but relay disconnection is based on persistence above 14.5A. This persistence is implemented as a score that increases every 10s while the condition remains active and trips when the score reaches 6, equivalent to approximately 60s of sustained overload. This makes the overload behavior closer to inverse-time protection logic than to an instantaneous cutoff [22].

The socket-heating path compares the estimated socket temperature with thermal warning and trip thresholds. The voltage path provides abnormal-supply supervision and relay response where implemented, but it is not presented as a replacement for a surge protective device, AVR, GFCI, AFCI, branch-circuit breaker, or certified wiring protection. The firmware also latches fault states. After relay disconnection, the event is recorded locally and uploaded when the network is available. The plug remains in the fault state until the user clears the fault through the device or PWA. This mandatory acknowledgment ensures that the user is aware of the issue, prompting them to actively identify and resolve the root cause of the fault before resuming operation.

3.2.3 Timing Constants

The sampling, protection, and reporting constants used by the firmware are listed in *Table 3.2*. These values are from the firmware configuration file and the current-feature implementation.

Table 3.2. Selected firmware timing and sampling constants

Constant	Value	Role
Target current sampling rate	27,000Hz	Current waveform acquisition for feature extraction
Arc runtime frame length	512 samples	FFT and time-domain feature window, approximately 18.96ms at the target sampling rate
Arc runtime hop length	256 samples	Overlap between consecutive feature frames, approximately 9.48ms per hop
Feature emission cadence	Every 3 hops	Approximate 35.16Hz feature output rate for logging and inference
MCP3204 SPI clock	1.2MHz	12-bit current ADC transfer clock
Voltage sampling interval	250 μ s	Firmware voltage-window acquisition interval
Load-context acquisition window	5s	Time allowed for stable load-family estimation before latching context
Overload warning threshold	10.0A	Local warning level before sustained overload trip scoring
Sustained overload trip threshold	14.5A	Current level used by the relay-trip persistence score
Sustained overload score timing	6 steps at 10s	Approximately 60s of sustained overload before relay trip
Socket-temperature threshold	trip 40°C	Estimated socket-temperature threshold used by thermal protection
Arc persistence counter	+2, -6, trip at 8	Leaky counter that requires repeated arc-positive frames before relay action
Arc hold interval	800ms	Short hold after arc indication to stabilize the protection state
Cloud live update interval	10s normal, 5s fault	Firebase/PWA telemetry cadence, slower than local protection evaluation

Table 3.2 explains why the current path is handled separately from slow telemetry. Arc-fault detection requires millisecond timing, while Firebase and PWA displays operate slower for human monitoring. Consequently, the local firmware triggers relay protection instantly, rather than waiting for Firebase updates. It independently evaluates safety states, leaving the PWA solely to record and visualize conditions.

3.2.4 Data Logging and PWA Communication

The monitoring system was implemented as a PWA connected to Firebase. The PWA was not placed in the protection path; the ESP32-S3 still made relay decisions locally. The PWA instead provided live measurements, protection status, event history, CSV download, labeling support, and review of waveform-derived features. This separation allowed the prototype to remain protective during network loss while still supporting data collection and analysis.

Figure 3.4 shows the PWA and Firebase data path.

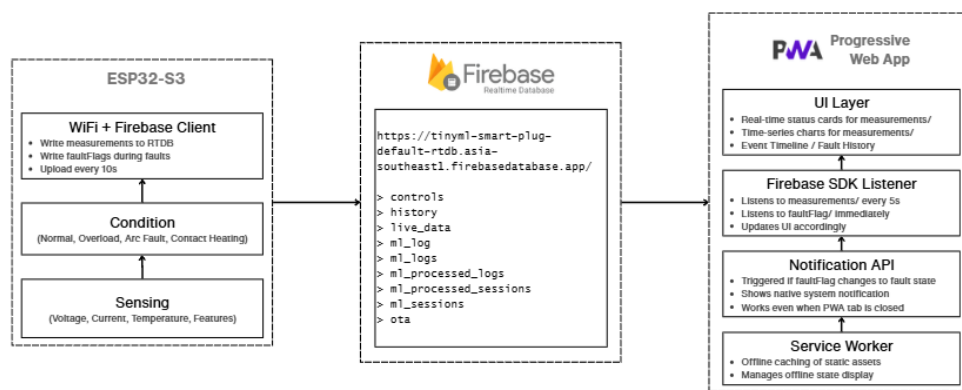


Figure 3.4. PWA block diagram and Firebase data path.

Figure 3.4 shows the communication path among the ESP32-S3 firmware, Firebase Realtime Database, and the browser-based PWA. The firmware serves as the main decision-making unit by acquiring sensor data, evaluating protection conditions, and publishing selected voltage, current, temperature, fault-state, and logging data to Firebase.

The PWA subscribes to these Firebase paths to display live measurements, time-series charts, fault history, and logging status. It may also write operator commands, such as relay control or fault clearing, but these commands are still validated and executed by the ESP32-S3. This setup improves remote monitoring and data recording while keeping time-critical protection actions, such as overload, arc-fault, and socket-heating response, local to the device and less dependent on browser availability or network delay [55], [56].

The live dashboard used during testing is shown in *Figure 3.5*.

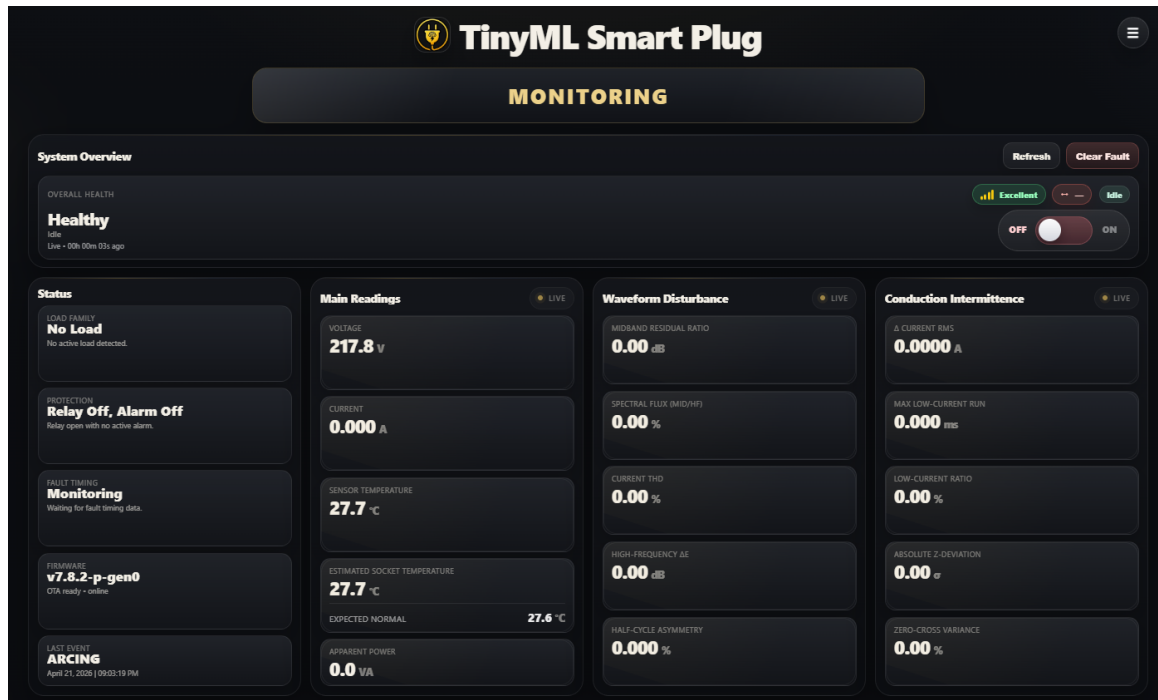


Figure 3.5. Actual PWA live dashboard.

Figure 3.5 displays the status and protection state, the main measurements, displays waveform-disturbance features, and displays conduction-intermittence features. The dashboard intentionally distinguishes the device temperature measurement from estimated socket temperature because the relay-facing thermal decision uses the compensated estimate, not the raw thermistor reading alone.

The event-history page is shown in Figure 3.6.

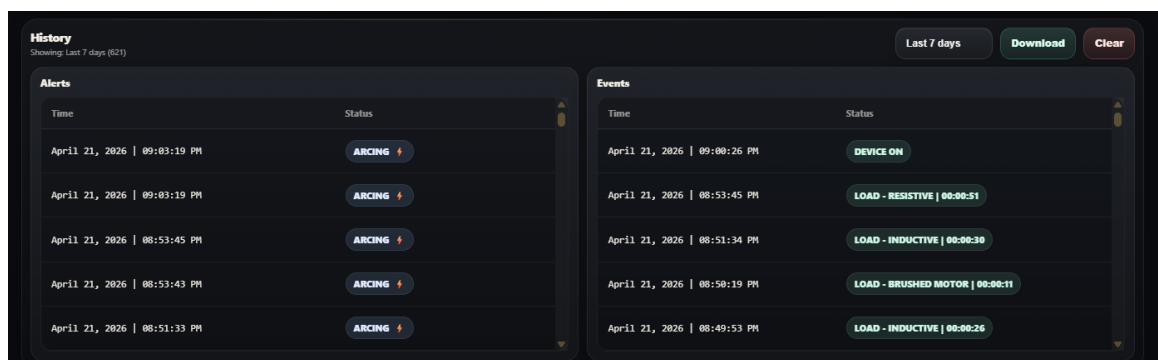


Figure 3.6. Actual PWA history-log view.

Figure 3.6 documents the alert and event list. The history function was useful for checking whether the firmware sequence matched the expected warning, persistence, and latching behavior during overload, heating, and arc-fault trials.

3.2.4.1 Telemetry Fields

The PWA displayed only a selected subset of firmware variables. The important telemetry fields are listed in *Table 3.3*.

Table 3.3. Selected telemetry fields shown in the PWA

Field	Firmware source	Interpretation
Voltage and current	Calibrated sensor routines	Electrical state of the connected load
Device temperature measurement	Thermistor routine	External temperature at the sensor position
Estimated socket temperature	Thermal estimator	Protection-facing estimate of socket temperature
Arc probability and class	Random Forest inference	TinyML arc-fault decision support
Context family and confidence	Context prototype model	Load-family estimate used by arc-threshold selection
Relay and fault state	Protection manager	Whether the load is energized or latched off

Table 3.3 shows that the PWA was both an operator display and a research instrument. During data collection, these fields allowed the researchers to compare raw measurements, computed estimates, and protection outputs in the same session record.

3.3 Physical Setup and Hardware Implementation

The physical setup translates the system model into the actual prototype used for testing. This section introduces the hardware blocks, full schematic, signal-conditioning path, enclosure arrangement, and component roles before the calibration and protection computations are presented.

3.3.1 Sensing Subsystems and Hardware Blocks

The physical implementation combined a line-side plug, protected outlet, solid-state relay module, ESP32-S3 controller, voltage sensor, current sensor, NTC thermistor, anti-aliasing circuit, backup supply, OLED display, buzzer, and enclosure. The hardware was organized so that mains conductors, sensing circuits, low-voltage logic, and user-

interface wiring could be traced separately during testing.

Figure 3.7 gives the hardware block-level view before the detailed schematic is introduced.

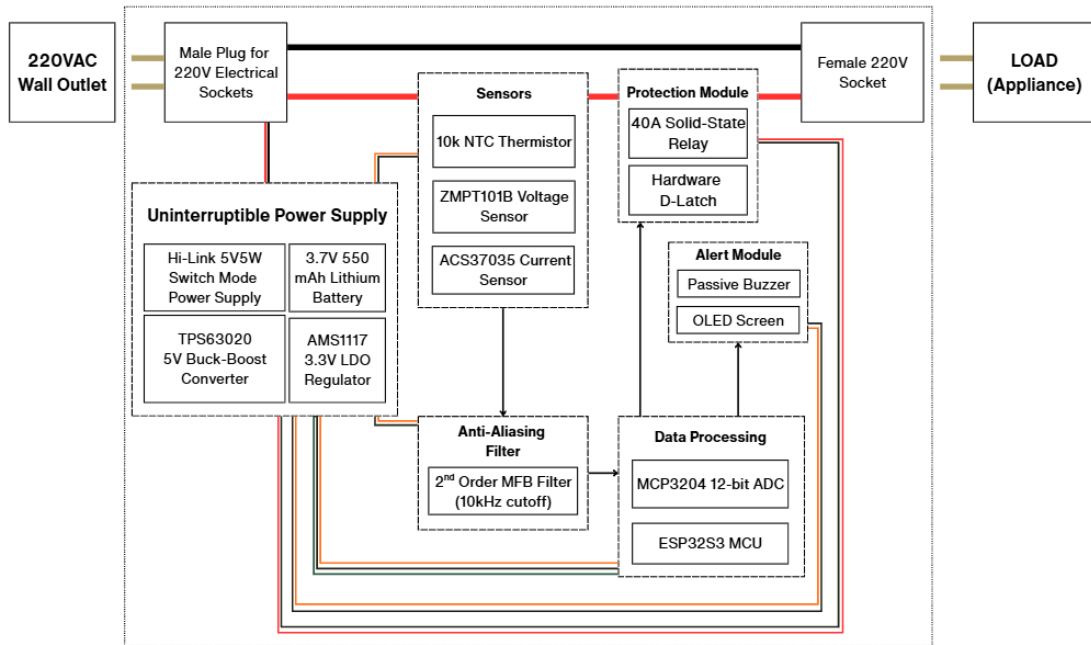


Figure 3.7. Block diagram of the prototype hardware.

Figure 3.7 shows the main hardware path of the prototype before the detailed schematic is introduced. The $220V_{AC}$ input enters through the male plug and is routed to the protected female socket through the 40A solid-state relay, which serves as the final protective actuator before the load. The voltage, current, and temperature sensors monitor the socket condition, while the anti-aliasing filter and MCP3204 SAR ADC prepare the current waveform for processing by the ESP32-S3. The isolated power supply provides the required 5V, 3.3V, and backup battery rails for the sensing, processing, relay, and alert modules. This arrangement allows the firmware and hardware latch to remove power from the protected outlet and activate the buzzer and OLED notification once a protection condition is confirmed.

3.3.2 Full Schematic Discussion

Figure 3.8 shows the full circuit schematic.

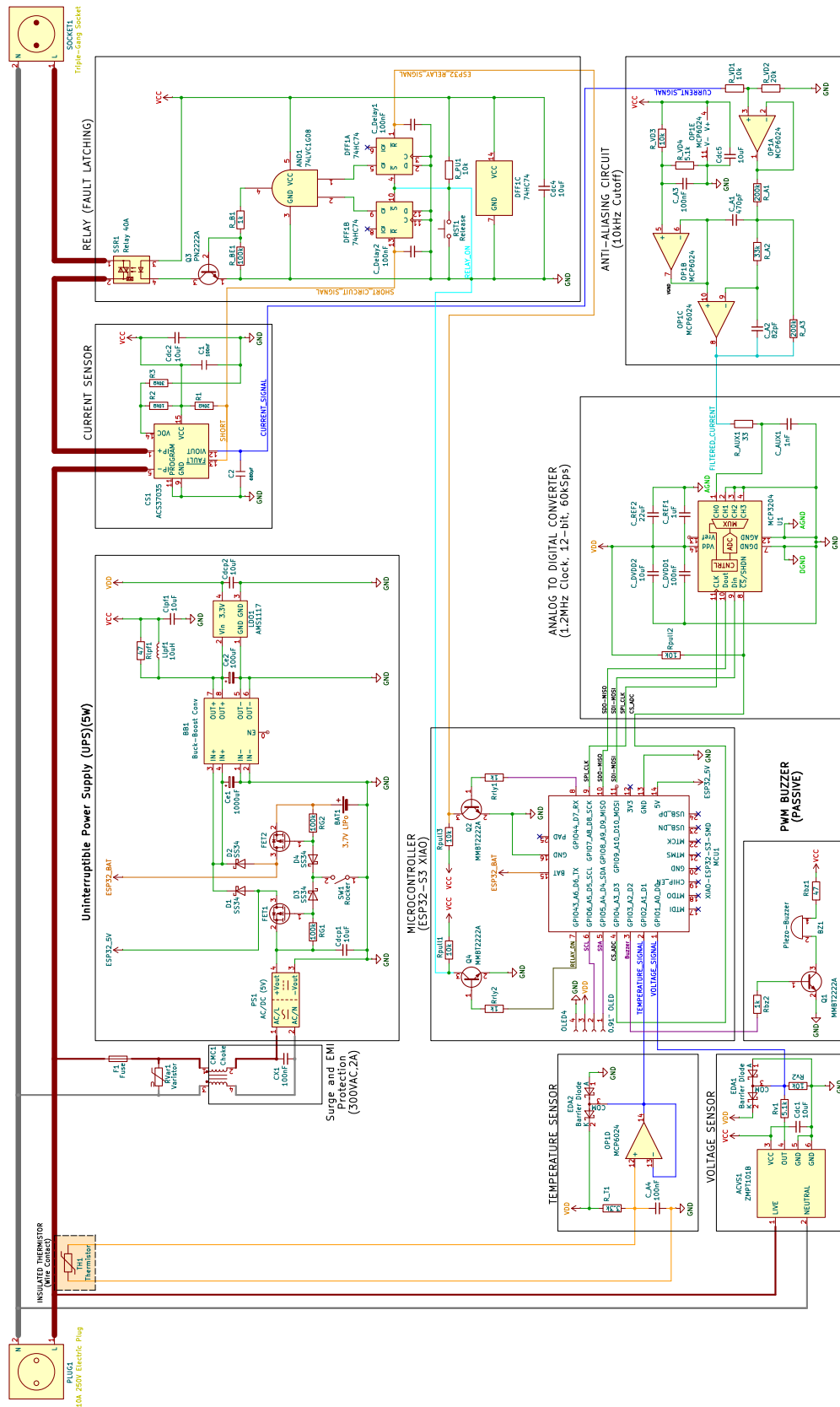


Figure 3.8. Full schematic of the TinyML Smart Plug prototype.

Figure 3.8 presents the complete hardware implementation of the TinyML smart plug prototype. The schematic is discussed according to the circuit flow: AC input protection, low-voltage power conversion, voltage, current, and temperature sensing, signal conditioning, analog-to-digital conversion, ESP32-S3 processing, relay fault latching, and user notification. *Appendix B* provides the detailed board-layout views, while this section explains how each block contributes to sensing, protection, and prototype operation.

The AC input enters through the line-side plug and is routed first to the input-protection stage. A fuse interrupts abnormal overcurrent conditions, while a varistor suppresses transient overvoltage events on the line. After this stage, the live and neutral conductors continue toward the relay-controlled outlet and the internal power-supply circuit. This places the first hardware protection layer before the load and before the low-voltage electronics.

The power-supply section converts the mains input into the regulated rails used by the controller and sensing circuits. These rails supply the ESP32-S3, ADC, operational amplifiers, sensors, OLED display, buzzer, and relay-control logic. Diode-isolated routing and local regulation help provide stable low-voltage power. The rocker-switch path also uses MOSFET switching to reduce the exposure of the regulated control rail to mechanical switch sparks, contact bounce, and switching transients during manual switching.

The voltage-sensing path uses the ZMPT101B voltage sensor to provide a scaled representation of the AC line. This signal is used by the firmware for RMS-voltage monitoring and mains-present detection. Through this path, the ESP32-S3 can determine whether the outlet is energized, whether the mains has vanished, and whether voltage has returned after an interruption, supporting relay-state handling, fault recovery, and data interpretation.

The current-sensing path uses the ACS37035 current sensor placed in series with the load. Its analog waveform is the main input for RMS-current computation, overload monitoring, waveform-feature extraction, and arc-related analysis. The same sensor also provides a hardware short-circuit output. In this prototype, that output was configured at $28.28A_{pk}$, equivalent to $20A_{RMS}$ for a sinusoidal waveform. This pin was included as a backup relay-trip layer independent of the firmware decision, although it was not formally

tested during validation.

The measured current waveform is routed to the anti-aliasing and signal-conditioning stage before digitization. This stage limits unwanted high-frequency components and prepares the waveform level for the ADC input. The conditioned signals are digitized by the MCP3204 analog-to-digital converter and transferred to the ESP32-S3 through the SPI interface lines SDI-MOSI, SDO-MISO, SPI-CLK, and CS-ADC. This bridges the analog sensing circuits and the digital protection firmware.

The ESP32-S3 XIAO is the central processing section of the prototype. It receives digitized waveform data, evaluates voltage and current conditions, reads the temperature input, executes the local protection logic, controls the relay command, drives the OLED display, activates the buzzer, and handles wireless monitoring and logging. The thermistor is installed near the plug-side wire-contact region to observe localized socket heating. During installation, the sensing point was insulated with Kapton tape, coupled with the thermistor using thermal paste, secured with another Kapton tape layer, and covered with electrical tape to reduce ambient-air influence on the reading.

The relay circuit acts as the output actuator that connects or disconnects the protected socket from the input line. It is not driven directly as a simple ESP32-S3 output. Instead, the relay command passes through a hardware fault-latching circuit so that a detected fault remains stored after the relay opens. This is necessary because opening the relay can remove the measured fault from the sensor inputs. Without a latch, the device could repeatedly close the relay, detect the same fault, and reopen the relay.

The relay-latching circuit uses two sections of the 74HC74 D flip-flop and an AND-gate stage. One latch stores the ESP32-side relay or fault command, while the other stores the short-circuit command from the current-sensor side. The relay is allowed to close only when both paths indicate that no fault is present. Thus, the AND-gate stage implements the De Morgan equivalent of a NOR-style fault decision: the relay remains enabled only when there is no firmware-side fault and no current-sensor short-circuit fault.

The notification section uses a transistor-driven buzzer controlled by the ESP32-S3. The buzzer was driven using PWM so that different duty cycles and pulse patterns

could indicate different alert states instead of relying on a single fixed warning tone. Several delicate low-voltage nodes also include BAT54S clamp-diode protection to limit abnormal voltage excursions before they reach the microcontroller, ADC, or sensing interfaces.

The schematic in *Figure 3.8* and the board layout in *Appendix B* should be interpreted as prototype-development documents for bench validation. They do not by themselves certify creepage distance, clearance distance, enclosure safety, thermal safety, regulatory compliance, or suitability as a replacement for listed protective devices in the reviewed literature.

3.3.3 Signal Conditioning and Enclosure Implementation

The anti-aliasing circuit and its frequency response are shown in *Figures 3.9* and *3.10*.

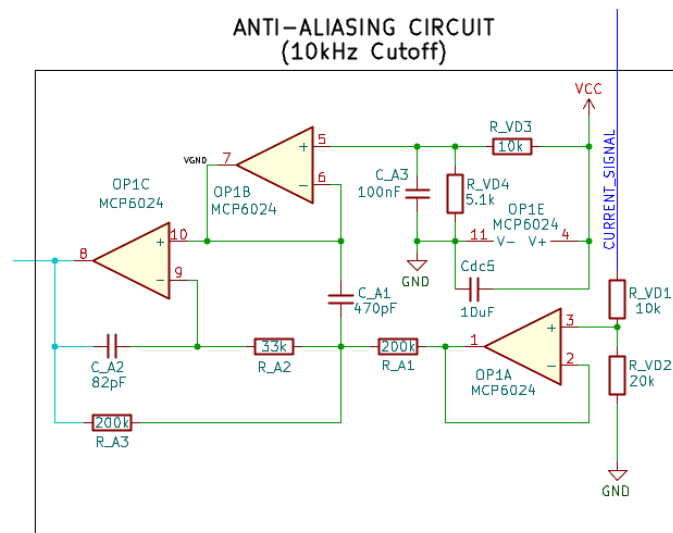


Figure 3.9. Second-order multiple-feedback low-pass anti-aliasing circuit.

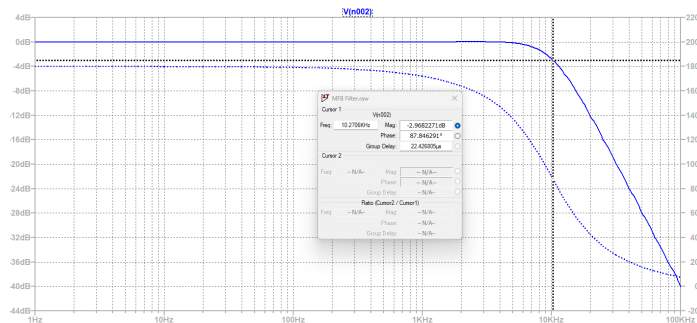


Figure 3.10. Bode response of the anti-aliasing filter.

Figure 3.9 shows the analog low-pass filter placed between the current sensor and the ADC. *Figure 3.10* verifies that the filter attenuates higher-frequency content before sampling. The implemented firmware also applies a digital soft anti-aliasing filter in the current-feature path, so the analog and digital filters work together rather than as isolated design elements [57].

The frequency-setting values shown in *Figure 3.9* were checked using the standard second-order low-pass relationship in (3.1):

$$f_0 = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}. \quad (3.1)$$

Substituting the schematic values $R_1 = 200\text{k}\Omega$, $R_2 = 33\text{k}\Omega$, $C_1 = 470\text{pF}$, and $C_2 = 82\text{pF}$ into (3.1) gives

$$f_0 = \frac{1}{2\pi\sqrt{(200\text{k}\Omega)(33\text{k}\Omega)(470\text{pF})(82\text{pF})}} \approx 9.98\text{kHz}. \quad (3.2)$$

The computed result in (3.2) agrees with the plotted cutoff region in *Figure 3.10*, where the simulated response is approximately -3dB near 10kHz. Since a second-order Butterworth response has $Q \approx 0.707$ and reaches the -3dB point at its natural frequency, the plotted response was treated as a near-Butterworth anti-aliasing stage rather than a high-Q peaking filter. The firmware's additional digital filter uses a similar low-peaking setting ($Q = 0.73$) before feature extraction.

The op amp for this stage must support 3.3V single-supply operation, input and output swing around the mid-supply bias, unity-gain-stable active-filter operation, and enough bandwidth and slew-rate margin for the 10kHz cutoff [57]. For a conservative 3.3V peak-to-peak signal, the maximum amplitude about the mid-supply bias is 1.65V. Since the steepest part of a sinusoid occurs at the zero crossing, the required minimum slew rate is

$$SR_{\min} = 2\pi f V_p = 2\pi(10\text{kHz})(1.65\text{V}) \approx 0.104\text{V}/\mu\text{s}. \quad (3.3)$$

The MCP6024-class device indicated in the schematic satisfies this requirement because its

datasheet specifies a typical slew rate of $7.0\text{V}/\mu\text{s}$, which is about 67 times higher than the computed minimum [76]. This margin allows the op amp to reproduce a 10kHz waveform without slew-rate limiting, avoiding slope-limited or triangular distortion. The device is also a rail-to-rail input/output, unity-gain-stable, 10MHz single-supply op amp family, making it suitable for the 3.3V active-filter stage used in the prototype [76].

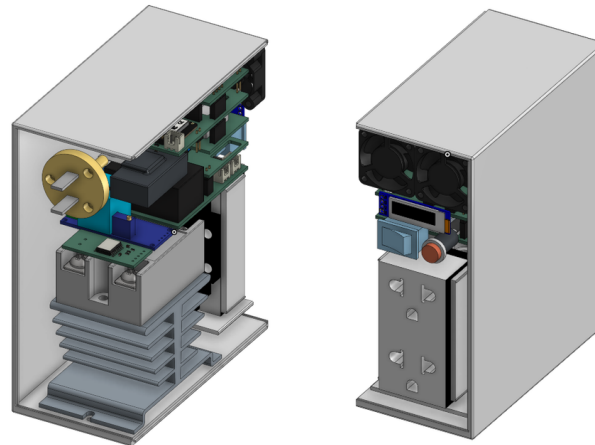


Figure 3.11. Three-dimensional model of the prototype without the case.

Figure 3.11 presents the relative placement of the PCB, outlet, relay, fan, and front-panel controls before final enclosure assembly.

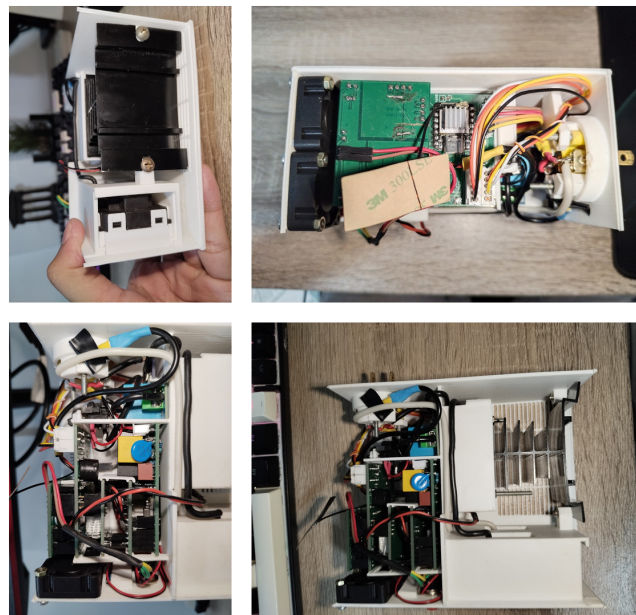


Figure 3.12. Actual prototype assembly without the case.

Figure 3.12 confirms that the physical wiring followed the planned internal organization. The exposed view was used during bench testing to probe the sensing, supply, relay, and communication subsystems.

The enclosure model is shown in *Figure 3.13*. It shows the intended user-facing arrangement: outlet access, front controls, ventilation, and rear plug connection. The enclosure therefore served both a mechanical and an operator-interface purpose.



Figure 3.13. Three-dimensional model of the prototype with the case.



(a) Plug side



(b) Outlet and control side

Figure 3.14. Actual prototype with the case installed.

Figure 3.14 shows the completed enclosed prototype. *Figure 3.14(a)* shows the plug side of the completed device, while *Figure 3.14(b)* shows the outlet, fan vents, and front-panel controls. This documents the final form of the prototype used for testing.

Table 3.4. Major hardware elements of the prototype

Subsystem	Components	Application
Microcontroller	ESP32-S3	Local computation, Wi-Fi communication, TinyML inference, relay control
Voltage sensing	ZMPT101B module	RMS voltage measurement and voltage protection input
Current sensing	ACS37035 with MCP3204 ADC	RMS current measurement and waveform-feature source
Temperature sensing	10k Ω thermistor	Device temperature measurement for thermal estimation
Protection actuator	Solid-state relay module	Load disconnection after latched protection condition
User interface	OLED, buzzer, LEDs, PWA	Local and remote warning and monitoring

Table 3.4 summarizes the main hardware elements used in the prototype and their corresponding roles in the system. The ESP32-S3 serves as the local controller, while the voltage, current, and temperature sensing circuits provide the measured inputs needed for protection and feature extraction. The relay module acts as the protection actuator, and the OLED, buzzer, LEDs, and PWA provide local and remote user feedback. These elements show that the prototype combines sensing and switching hardware with local embedded inference. However, the final protection behavior is defined by the firmware processes and computations discussed in the following sections.

3.4 TinyML Development and Deployment Workflow

The TinyML workflow used logged current-waveform features to train two embedded models: a load-context model and a series arc-fault model. The context model estimated the appliance family during the early stable portion of a load run. Its load-family output was then used as an input to the arc model, where it was encoded as context fields. The arc model combined selected arc-sensitive features with that context output to classify whether a frame was consistent with a series arc fault. Random Forest classification was selected because it can model nonlinear feature interactions while remaining exportable as a compact tree ensemble for microcontroller execution [68], [69].

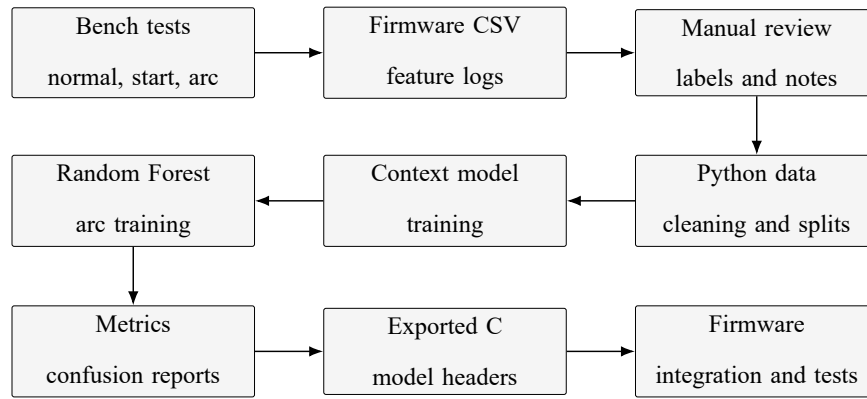


Figure 3.15. Data logging, labeling, and TinyML deployment workflow.

Figure 3.15 shows that the training workflow was not performed directly on the microcontroller. The firmware generated CSV records from bench tests, the records were reviewed and labeled, and Python scripts then cleaned the rows, formed training and test splits, trained the context and arc models, computed confusion reports, and exported C headers. The final embedded test was therefore a deployment check of the exported headers inside the firmware, not a new training step on the ESP32-S3. This follows the usual TinyML pattern of offline training followed by resource-constrained on-device execution [43], [73].

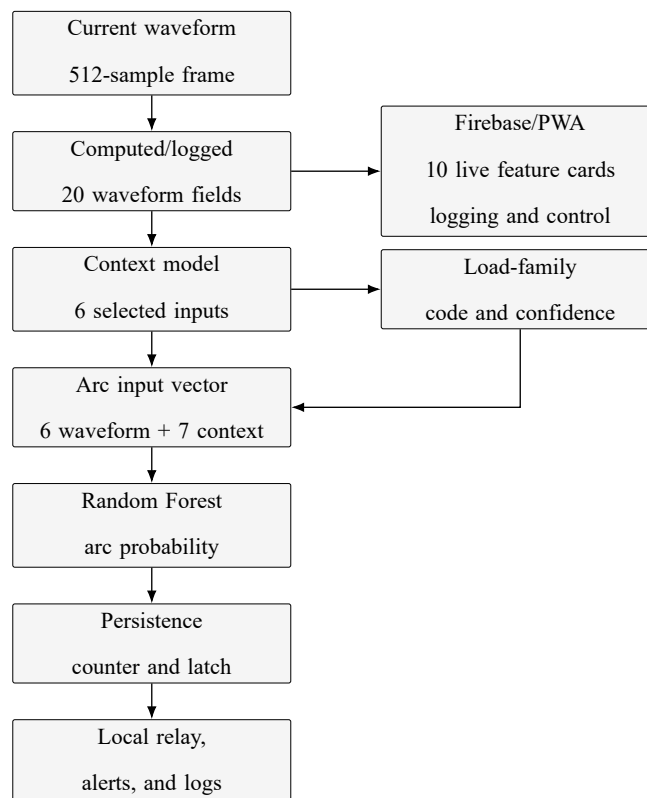


Figure 3.16. TinyML processing sequence from feature logging to embedded inference.

Figure 3.16 shows that the firmware computes feature fields first, then updates context, then evaluates the arc classifier. This order is important because the deployed arc header includes both waveform features and context fields. The load-family result is therefore not only a display label; when confidence is sufficient, it becomes part of the input context supplied to the arc model. The flowchart also separates three feature sets: the candidate fields computed and logged by the firmware, the six base waveform features selected for the deployed arc model, and the 10 live feature cards shown by the PWA. A broader 14-feature order remains in the PWA metadata and session viewer for historical review and CSV inspection.

Arc labels were assigned from logged waveform sessions. The labeling interface is shown in Figure 3.17.

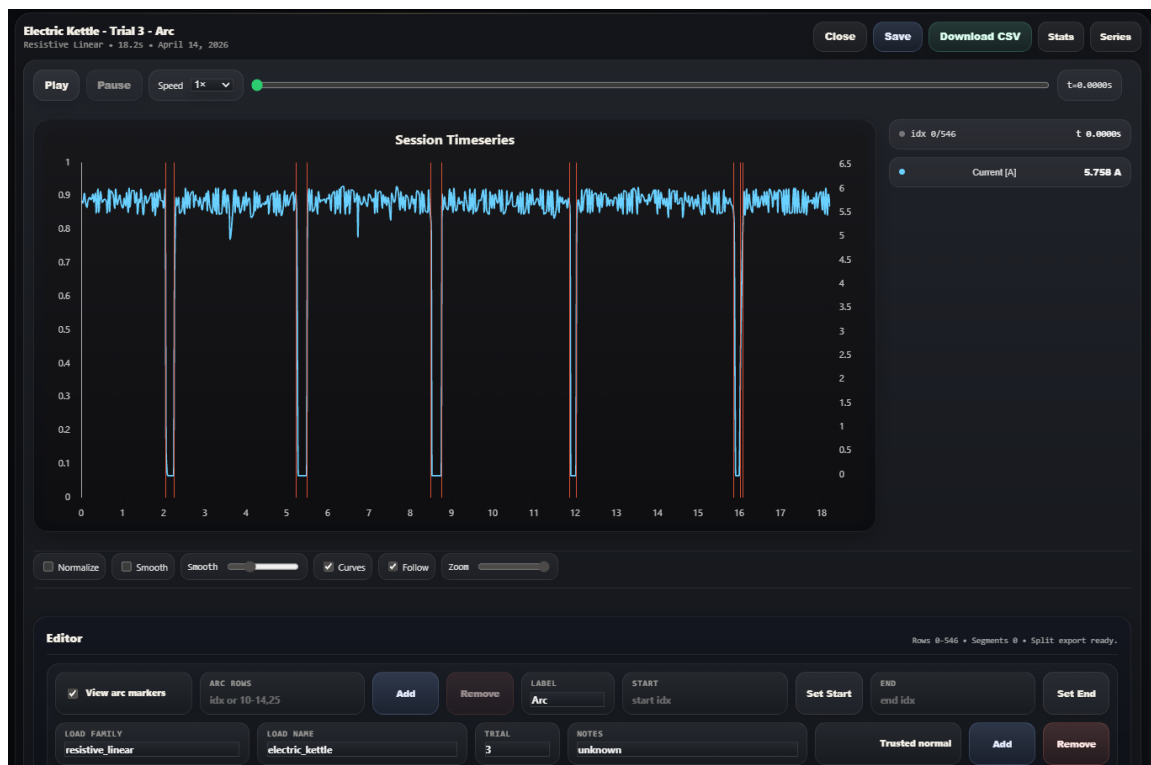


Figure 3.17. Arc labeling using logged waveform data.

Figure 3.17 shows the interface used to mark arc events from the logged waveform sessions. The labels were based on the observed test action and on the expected behavior of series arcs during contact separation and restriking. Because the marking was done manually after the test, the selected points are not exact arc-onset locations, but time-aligned

approximations near the RMS waveform changes associated with the disconnection and recontact events.

To account for this labeling uncertainty, the training set did not treat only the marked row as the arc event. Neighboring rows around each arc marker were also given soft arc weights so that brief timing delays in the manual marking would not force nearby arc-related frames to be learned as normal behavior. This is presented in *Appendix F* and is also shown in *Figure 3.18*.

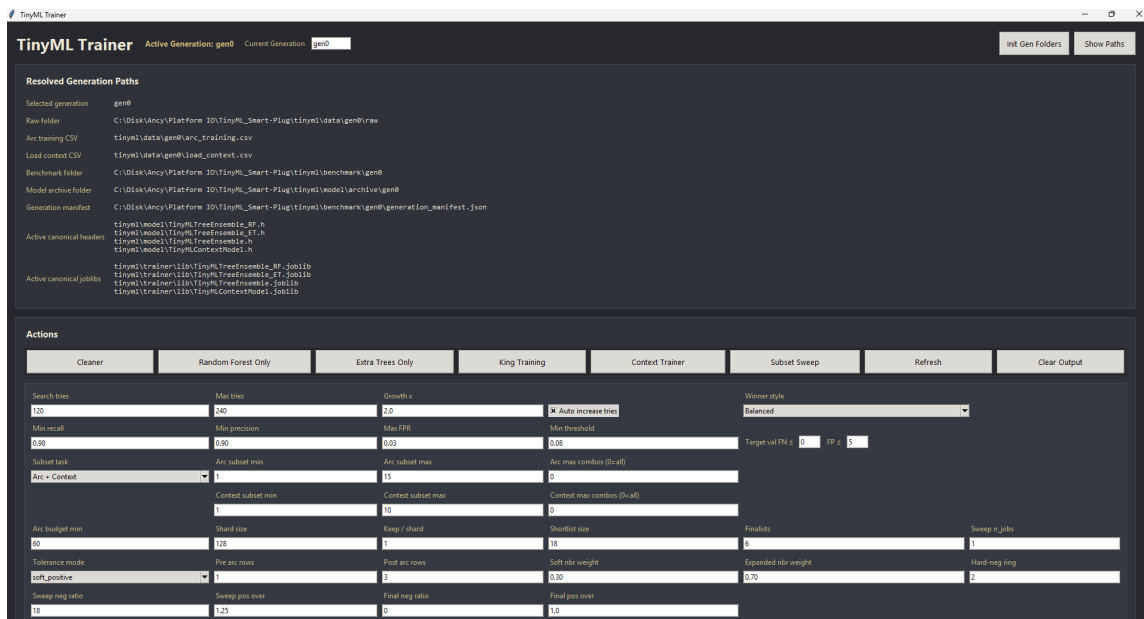


Figure 3.18. Local Python TinyML trainer.

Figure 3.18 documents the offline training environment used to prepare the exported embedded model files. The workflow used separate data preparation, context-model training, Random Forest training, shared TinyML utility, and generation-path routines as the source of truth for feature selection, data splits, hyperparameter search, metric computation, and embedded export. The dataset-preparation, training-report, and exported embedded-model summaries are provided in *Appendix F.2*.

3.4.1 Context Model

The load context model is a centroid classifier, not a Random Forest. The context model defines six input features: ΔI_{rms} , half-cycle asymmetry, midband residual ratio, current THD, mid/high-frequency spectral flux, and high-frequency energy delta. These

features were used because they describe the waveform behavior of the connected load rather than a single fault event. For example, different load families may exhibit different harmonic content, asymmetry, and current-change patterns even during normal operation.

In this study, the context model was used to provide load-family information to the arc classifier, but its experimental coverage was still limited by the available tested loads. As stated in the limitations of the study, the actual collected and tested load signatures were limited to resistive loads, inductive loads, and brushed motor loads.

During runtime, the firmware averages usable feature frames over the acquisition interval, standardizes each averaged feature, computes the squared Euclidean distance to each family centroid, and applies a softmax to the negative distances:

$$p_c = \frac{\exp(-d_c^2)}{\sum_{j=1}^C \exp(-d_j^2)}, \quad d_c^2 = \sum_{i=1}^m \left(\frac{x_i - \mu_i}{\sigma_i} - q_{c,i} \right)^2. \quad (3.4)$$

In (3.4), p_c is the context confidence for family c , x_i is the averaged runtime feature, μ_i and σ_i are the training-set mean and standard deviation exported in the header, $q_{c,i}$ is the standardized centroid coordinate for family c , $m = 6$ is the context input dimension, and $C = 6$ is the number of load families. A family is accepted only when the best confidence is at least 0.45; otherwise, the context remains unknown. This threshold prevents weak or ambiguous feature patterns from being forced into a load family.

The context acquisition constants in the firmware require a load to remain active before the context is latched. A provisional context may be used after the minimum acquisition period, and the final context is latched after the configured 5s acquisition window when confidence is sufficient. This prevents brief startup artifacts from immediately forcing a load family into the arc classifier. Once the context is latched, it remains stable for the continuous load run, so the arc model receives a consistent family condition during normal monitoring. The context is returned to unknown only after the load has been inactive long enough to indicate that the previous operating condition has ended.

3.4.2 Arc Model

The deployed series-arc classifier is the Random Forest used by the embedded arc model. The deployed Random Forest arc model defines 13 inputs: six base waveform features, six one-hot context-family fields, and one context confidence field. The base features are maximum low-current run, zero-dwell ratio, low-current ratio, absolute RMS-current z-score relative to baseline, midband residual ratio, and mid/high-frequency spectral flux. These were selected because series arcs can produce current interruption, restriking spikes, cycle-to-cycle waveform disturbance, and broadband spectral changes [34], [35], [50], [53]. Since the reviewed literature also shows that series arcs are load-dependent and difficult to detect using current magnitude alone, the arc decision was implemented as a multi-feature classifier rather than as a single-threshold current decision [33], [34], [36].

The first three base features describe current dropouts and low-current intervals that may occur when a contact separates. The RMS-current z-score compares the present current behavior with the recent baseline, allowing the model to detect abrupt deviations from the expected operating level. The midband residual ratio and mid/high-frequency spectral flux describe waveform disturbance and spectral changes that may occur during restriking. Together, these features allow the model to observe both the time-domain interruption and the frequency-domain disturbance caused by a possible arc event. This follows the reviewed arc-fault studies, where zero-current behavior, current-fluctuation features, high-frequency noise, and spectral changes were used as indicators of series arcing [34]-[39], [49]-[53].

The context-family inputs allow the arc classifier to interpret the same waveform features differently depending on the connected load type. This is important because some normal loads, such as motors or phase-controlled devices, can naturally produce distorted or non-sinusoidal current waveforms. The context confidence is included so that the model can also account for how reliable the assigned load family is. When the context is unknown or weak, the arc decision is treated more conservatively to reduce false operation during unstable startup or ambiguous load behavior. This follows the reviewed NILM and load-family literature, where load signatures were shown to affect how waveform features should be interpreted [8], [40]-[42].

Random Forest was used because the reviewed TinyML literature supports compact feature-based models that can be deployed on constrained embedded hardware when the feature space and runtime are carefully designed [32], [43]. In this study, the Random Forest therefore serves as the embedded decision layer that combines the selected arc-related features and the estimated load context.

Table 3.5. Deployed Random Forest arc-model settings

Item	Deployed value	Source or role
Model type	Random Forest	Exported tree ensemble from the local training workflow
Input dimension	13	Six waveform features plus seven context fields
Base feature count	6	Arc-sensitive features used before context fields
Default arc threshold	0.4800	Probability threshold for supported context families
Context confidence minimum	0.4500	Minimum accepted context confidence
Unknown-context threshold	0.7800	More conservative threshold when context is unknown
Unknown minimum feature votes	3	Additional runtime safeguard for unknown-context decisions

Table 3.5 shows that the deployed inference was context-aware. The model used lower thresholds for selected known families and a higher threshold for unknown contexts, reducing nuisance trips when the load family could not be established confidently. The firmware then passed the arc classification to the protection manager, where a leaky counter required persistence before relay disconnection.

3.4.3 Feature Selection for the Deployed Arc Model

The exported firmware header is the source of truth for the deployed feature selection. It defines 13 arc-model inputs: six waveform features and seven context fields. Therefore, this paper treats the context one-hot fields and context confidence as model inputs, but not as additional waveform features. The selected waveform features are maximum low-current run, zero-dwell ratio, low-current ratio, absolute RMS-current z-score

relative to baseline, midband residual ratio, and mid/high-frequency spectral flux. The full computed/logged feature contract is given in *Section 3.7*.

3.4.4 Training Split and Model Evaluation

The Random Forest report, as documented in *Appendix F*, shows that the final arc dataset was split into 35,512 training rows, 9,434 validation rows, and 10,268 test rows. The final training report used 5 cross-validation splits during search and reported the held-out test metrics after model selection. The best exported hyperparameters were 160 trees, maximum tree depth of 16, minimum leaf size of 4, minimum split size of 12, maximum feature fraction of 0.35, bootstrap sampling enabled, and entropy criterion. Train/test splitting, cross-validation, and confusion-matrix metrics are standard model-evaluation practices for supervised classification [69], [74].

The held-out test-set confusion matrix in the exported report contained 10,173 true negatives, 5 false positives, 25 false negatives, and 65 true positives. The corresponding frame-level accuracy, precision, recall, and F1-score were 99.7078%, 92.8571%, 72.2222%, and 81.25%, respectively. Event-level evaluation was also reported because arc faults occur in grouped episodes rather than as isolated independent frames; the event-level recall was 84.0%, with one false-alarm event in the test sessions.

3.5 Sensor Calibration and Computations

Sensor calibration converted raw sensor-derived values into engineering units used by the protection routines. Polynomial calibration was used for the voltage and current channels because the measured sensor outputs were not perfectly proportional to the reference instruments over the tested range. Ordinary least-squares regression estimates coefficients by minimizing the sum of squared residuals between the measured and fitted values [71]. The calibration coefficients below were verified from the firmware configuration. The raw calibration records are retained in *Appendix A*, specifically *Tables A.1*, *A.2*, and *A.3*.

3.5.1 Voltage Calibration

The voltage calibration setup is shown in *Figure 3.19*.

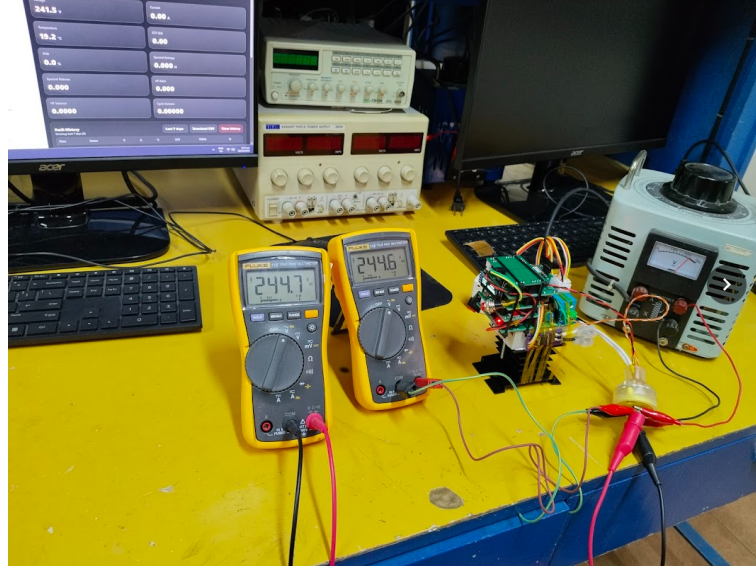


Figure 3.19. Data-collection setup for voltage calibration.

Figure 3.19 shows the prototype voltage channel being compared with reference multimeters while the applied voltage was varied. The calibration target was the reference RMS voltage, not the uncorrected sensor output.

The fitted voltage regression is shown in Figure 3.20.

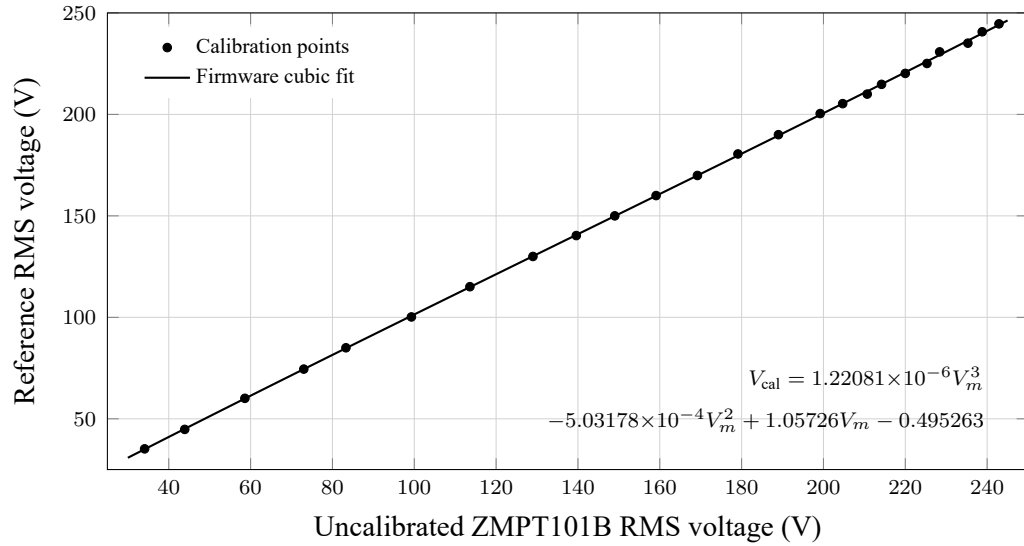


Figure 3.20. Cubic regression for voltage calibration.

Figure 3.20 supports the cubic correction used by the firmware. The implemented voltage calibration is

$$V_{\text{cal}} = 1.22081 \times 10^{-6} V_m^3 - 5.03178 \times 10^{-4} V_m^2 + 1.05726 V_m - 0.495263. \quad (3.5)$$

In (3.5), V_m is the measured RMS voltage before the cubic correction and V_{cal} is the calibrated RMS voltage used for display and voltage protection. The polynomial correction is needed because threshold decisions such as undervoltage, overvoltage, and mains-present detection depend on the calibrated value.

3.5.2 Current Calibration

The current calibration setup and load combinations are shown in *Figures 3.21 and 3.22*.

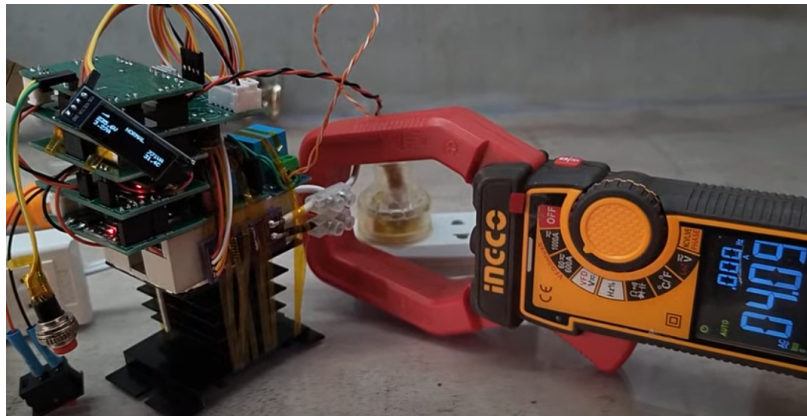


Figure 3.21. Data-collection setup for current calibration.



Figure 3.22. Load setups used for current calibration.

Figure 3.21 shows the comparison between the prototype current channel and a reference instrument. *Figure 3.22* shows the different resistive and appliance loads used to span the current range needed for overload and arc-feature testing.

The fitted current regression is shown in *Figure 3.23*.

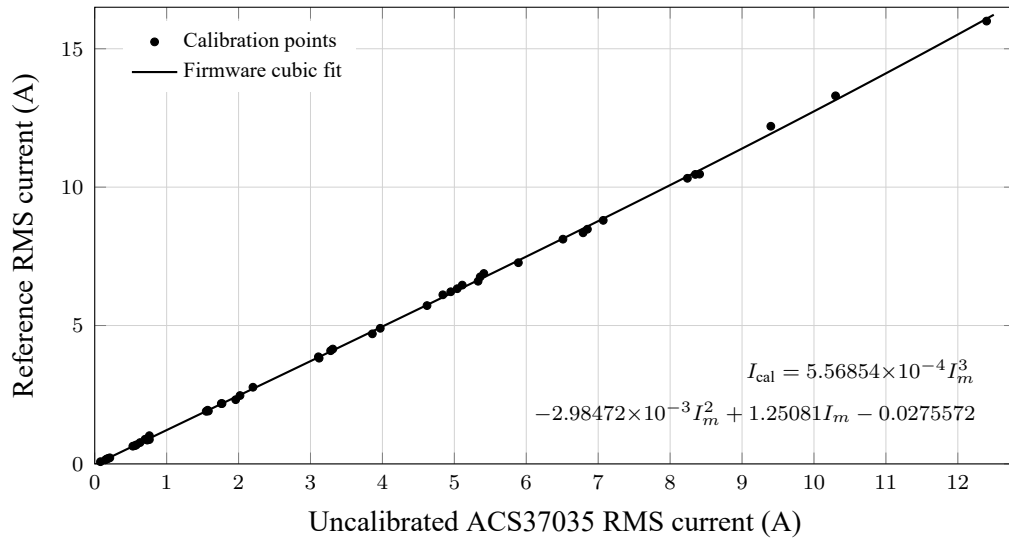


Figure 3.23. Cubic regression for current calibration.

The implemented current calibration is

$$I_{\text{cal}} = 5.56854 \times 10^{-4} I_m^3 - 2.98472 \times 10^{-3} I_m^2 + 1.25081 I_m - 0.0275572. \quad (3.6)$$

In (3.6), I_m is the measured RMS current before the cubic correction and I_{cal} is the calibrated RMS current used by the overload logic, socket-temperature model, and waveform-feature extractor. Negative fitted values are clamped to zero in firmware.

3.5.3 Device Temperature Measurement Calibration



Figure 3.24. Data-collection setup for device temperature measurement calibration.

Figure 3.24 shows the external thermistor measurement being compared with a reference temperature measurement. Thermistor placement is important because an external sensor can lag or under-read the hottest contact point inside a degraded socket [27], [29], [30]. The acquired temperature-resistance calibration points are listed in Table A.3.

The firmware computes the NTC resistance from the voltage-divider output:

$$R_{\text{NTC}} = \frac{V_{\text{NTC}} R_D}{V_{\text{CC}} - V_{\text{NTC}}}. \quad (3.7)$$

In (3.7), V_{NTC} is the measured divider voltage, $R_D = 3.3\text{k}\Omega$ is the fixed divider resistor, and $V_{\text{CC}} = 3.3\text{V}$ is the divider supply. The resistance is then converted to temperature with the Beta equation:

$$T_{\text{NTC}} = \left[\frac{1}{T_0} + \frac{1}{\beta} \ln \left(\frac{R_{\text{NTC}}}{R_0} \right) \right]^{-1} - 273.15. \quad (3.8)$$

In (3.8), T_{NTC} is in degrees Celsius, $T_0 = 298.15\text{K}$, $R_0 = 10\text{k}\Omega$, and $\beta = 4050$. The computed temperature is clamped from -20°C to 125°C and filtered with a 3s time constant in the temperature-sensing routine.

3.6 Socket-Heating Temperature Models

The socket-heating computation uses two related models. The first is the expected normal socket temperature, which estimates the socket temperature expected under healthy contact behavior. The second is the estimated socket temperature, which is the protection-facing compensated value intended to respond to degraded-contact heating and external device temperature under-reading. This separation is necessary because a raw external thermistor does not directly measure the internal contact hotspot [30], [46], [47].

The expected normal model serves as the reference condition, while the estimated socket temperature serves as the value used for protection. This follows the reviewed literature, which shows that poor plug-socket contact can produce localized heating and ignition risk even when the current path still continues to operate [17], [24], [46], [47]. The model also accounts for the reviewed limitation that an external NTC thermistor does not directly measure the hottest internal contact point and may respond with thermal lag [27]-[31]. For

this reason, the firmware does not rely on the raw NTC reading alone. Instead, it combines current-dependent heating behavior, ambient tracking, and direct thermistor compensation before the protection manager evaluates the thermal state.

3.6.1 Expected Normal Temperature Model

The firmware first decides whether the load is active:

$$L = \begin{cases} 1, & \text{if mains is present and } I_{\text{rms}} \geq 0.35\text{A}, \\ 0, & \text{otherwise.} \end{cases} \quad (3.9)$$

In (3.9), L is the load-active flag and I_{rms} is the calibrated RMS current. When $L = 0$, the current dependent thermal rise is forced toward zero.

The current-dependent shape term is

$$s_I = \left[\text{clamp} \left(\frac{I_{\text{rms}}}{12.0}, 0, 1.25 \right) \right]^2. \quad (3.10)$$

In (3.10), s_I is the normalized squared current term. The 12.0A denominator and 1.25 clamp are firmware constants. The square term reflects that heat generation associated with resistive loss rises with current squared.

The ambient or baseline term is tracked from the device temperature measurement only when the load is off or the current is below 0.18A:

$$T_a[k] = T_a[k-1] + \alpha_a (T_{\text{NTC}}[k] - T_a[k-1]), \quad \alpha_a = \frac{\Delta t}{240\text{s}}. \quad (3.11)$$

In (3.11), T_a is the inferred ambient or baseline temperature, Δt is the elapsed firmware update time, and T_{NTC} is the measured thermistor temperature. The 240s time constant prevents the baseline from following short-term load heating.

The target normal rise is

$$\Delta T_{\text{normal,target}} = \begin{cases} 0.20 + 12.0s_I, & L = 1, \\ 0, & L = 0. \end{cases} \quad (3.12)$$

In (3.12), 0.20°C is the normal base rise and 12.0°C is the normal current-rise gain. These constants define the expected socket-temperature rise for a healthy contact at the measured load current. This keeps the normal reference dependent on load current while still separating expected heating from abnormal degraded-contact heating.

The firmware filters this target with separate heating and cooling time constants:

$$\Delta T_{\text{normal}}[k] = \Delta T_{\text{normal}}[k-1] + \alpha_n (\Delta T_{\text{normal,target}}[k] - \Delta T_{\text{normal}}[k-1]). \quad (3.13)$$

In (3.13), $\alpha_n = \Delta t/900\text{s}$ when the target rise is increasing and $\alpha_n = \Delta t/1800\text{s}$ when the target rise is decreasing. These time constants represent the slower thermal response of the socket compared with the electrical measurements.

The expected normal socket temperature is then

$$T_{\text{normal,exp}} = \text{clamp}(T_a + \Delta T_{\text{normal}}, -20, 125). \quad (3.14)$$

In (3.14), $T_{\text{normal,exp}}$ is the baseline or normal thermal estimate. It represents what the socket temperature should be under expected contact behavior for the measured current history.

3.6.2 Estimated Socket Temperature Model

The measured device temperature value is compensated directly:

$$T_{\text{comp}} = \text{clamp}(T_{\text{NTC}} + \Delta T_{\text{meas}}, -20, 125), \quad \Delta T_{\text{meas}} = \begin{cases} 0.30 + 1.20s_I, & L = 1, \\ 0, & L = 0. \end{cases} \quad (3.15)$$

In (3.15), T_{comp} is the compensated thermistor-based estimate, 0.30°C is the measured base delta, and 1.20°C is the measured current gain.

Finally, the protection-facing socket estimate is

$$T_{\text{socket,est}} = \text{clamp}(\max(T_{\text{comp}}, T_{\text{normal,exp}}), -20, 125). \quad (3.16)$$

In (3.16), $T_{\text{socket,est}}$ is the estimated socket temperature used by the protection manager. The

maximum operation makes the estimator conservative: it does not allow the compensated device temperature estimate to fall below the expected normal socket estimate when load current indicates that some heating should be present.

The verified thermal constants are summarized in *Table 3.6*.

Table 3.6. Socket-heating constants verified from firmware

Constant	Value	Function
Load-on current threshold	0.35A	Enables current-dependent heating model
Ambient tracking current limit	0.18A	Allows baseline tracking only at very low load
Ambient time constant	240s	Slow baseline update
Normal reference current	12.0A	Denominator for current-shape term
Normal base rise	0.20°C	Minimum expected rise when load is active
Normal current gain	12.0°C	Expected normal heating gain
Current-ratio clamp	1.25	Limits current-shape term
Heating time constant	900s	Rise filter for normal estimate
Cooling time constant	1800s	Fall filter for normal estimate
Measured base delta	0.30°C	Direct thermistor compensation offset
Measured current gain	1.20°C	Direct thermistor compensation gain
Estimated-temperature trip threshold	40.0°C	Thermal relay trip threshold
Thermal warning and clear thresholds	39.0°C / 38.6°C	Warning hysteresis

Table 3.6 distinguishes the thermal estimator from the warning and trip thresholds. The estimator produces $T_{\text{socket,est}}$, while the protection manager decides warning and relay action from that estimate.

3.6.3 Least-Squares Gain and Regression Derivation

The calibration path for the socket-heating constants compares the device temperature measurement and reference terminal readings against ambient, then fits the compen-

sation terms from the characterization records.

This was used because the external thermistor follows the general heating of the socket body, but it does not directly represent the hottest internal contact point. The reference terminal reading therefore serves as the target temperature rise, while the device temperature measurement represents the embedded measurement available to the firmware.

For a gain-only rise model, the least-squares gain is

$$k = \frac{\sum_i x_i y_i}{\sum_i x_i^2}, \quad (3.17)$$

where x_i is the predictor rise above ambient, such as device temperature measurement rise or current-shaped rise, and y_i is the reference terminal rise above ambient.

This gain-only form is appropriate when the rise is assumed to approach zero at ambient condition. In this case, the model only adjusts the scale of the measured or predicted temperature rise.

When an offset is required, the corresponding linear model is

$$\Delta T_{\text{ref}} = a \Delta T_{\text{device}} + b. \quad (3.18)$$

In (3.18), ΔT_{ref} is the infrared thermometer or terminal-contact rise and ΔT_{device} is the external thermistor rise. The coefficient a scales the external thermistor rise, while b accounts for a fixed difference between the measured device temperature and the reference terminal temperature. This is useful when the external thermistor consistently under-reads the terminal region due to placement and thermal separation.

When time lag is observed, the time series is aligned before fitting so that the compared points represent the same stage of the heating process. This prevents sensor delay from being treated as a permanent temperature error. The resulting fitted constants were then used as the basis for the thermal compensation values summarized in *Table 3.6*.

3.6.4 Firmware Implementation of Constants

As an example, consider a load-active frame at $I_{\text{rms}} = 9.04\text{A}$. The current-shape term in (3.10) is

$$s_I = \left(\frac{9.04}{12.0} \right)^2 = 0.568.$$

The direct thermistor-compensation rise in (3.15) is therefore $0.30 + 1.20(0.568) = 0.982^\circ\text{C}$. If the device temperature measurement for that frame is 38.2°C , the direct compensated estimate is $T_{\text{comp}} = 39.18^\circ\text{C}$ before the final maximum operation. For the same current, the normal target rise in (3.12) is $0.20 + 12.0(0.568) = 7.02^\circ\text{C}$. This value does not mean that 7.02°C is added directly to the measured device temperature of 38.2°C . Instead, it represents the expected steady rise above the inferred ambient or baseline temperature for a healthy socket contact carrying the same current. The firmware approaches this value gradually through the normal-rise filter in (3.13), so it is reached only after the load has remained active long enough for the socket thermal state to settle. For example, if the inferred ambient temperature were 30.0°C , then the expected normal socket temperature would approach $30.0 + 7.02 = 37.02^\circ\text{C}$ after sufficient heating time, not immediately at the first frame. The final protection-facing estimate therefore depends on both the direct compensated thermistor value and the filtered normal thermal history, which prevents the method from relying only on a single instantaneous current and device temperature reading.

3.7 Waveform Feature Computation and Selection

The arc-detection firmware computes waveform features from current samples rather than from RMS current alone. This is necessary because a series arc can remain below a conventional breaker trip level while still producing low-current gaps, restrike pulses, irregular zero-crossing behavior, and mid/high-frequency energy [34], [35], [53], [54]. The reviewed literature also shows that series arcs are load-dependent and stochastic, so the detector must use multiple waveform indicators rather than a single current-magnitude threshold [33]-[39].

The firmware uses a 512-sample frame at a 27kHz target sampling rate, with

a 256-sample hop. A Hann window is applied before the FFT to reduce spectral leakage, which is standard practice in discrete Fourier analysis [70], [75]. The resulting features are interpreted using the two feature families established in the reviewed literature: waveform-disturbance indicators and conduction-intermittence indicators [34]-[39], [49]-[54]. Waveform-disturbance features include `midband_residual_ratio`, `spectral_flux_midhf`, `thd_i`, `hf_energy_delta`, `halfcycle_asymmetry`, and other quantities that describe spectral, harmonic, residual, or cycle-shape changes. Conduction-intermittence features include `low_current_ratio`, `max_low_current_run_ms`, `zero_dwell_ratio`, `delta_irms_abs`, and `abs_irms_zscore_vs_baseline`, which describe weak conduction, interruption, RMS-current change, and deviation from the recent current baseline.

The equation subsections below group the computations according to their main physical meaning. Current-magnitude and conduction-gap features mainly support the conduction-intermittence family, while spectral, residual, waveform-shape, and harmonic features mainly support the waveform-disturbance family. The feature `abs_irms_zscore_vs_baseline` can also support both interpretations because it measures how far the present RMS current deviates from the adaptive baseline, whether the deviation is caused by a normal load transition or by arc-related interruption.

3.7.1 Feature Set Used Across Firmware, Trainer, and PWA

The firmware, trainer, exported model headers, and PWA use related but not identical feature lists because each part of the system has a different role. The firmware computes and logs the complete waveform-feature set used for inference, session review, CSV inspection, and training analysis. The deployed Random Forest arc model uses only the selected waveform inputs required by the exported model, then appends the context-family fields and context confidence to form the final input vector. The PWA displays the main monitoring features, while the session viewer retains additional feature columns for review and labeling.

Table 3.7 summarizes the feature set used in this study. A *Yes* in the arc-model

column means that the feature is one of the waveform inputs used by the deployed Random Forest arc model. Context features are listed separately because they support the load-context model and the context fields appended to the arc-model vector, but they are not direct arc-model base inputs unless explicitly marked.

Table 3.7. Collected waveform features and deployed model feature selection

Feature name	Logged	Used by arc model	PWA card	Physical interpretation
abs_irms_zscore_vs_baseline	Yes	Yes	Live/CSV	Current magnitude departure from the adaptive healthy baseline.
delta_irms_abs	Yes	Context input	Live/CSV	Frame-to-frame absolute RMS-current step.
halfcycle_asymmetry	Yes	Context input	Live/CSV	Difference between positive and negative half-cycle RMS values.
suspicious_run_energy	Yes	No	CSV only	Leaky internal score combining arc-like indications.
delta_hf_energy	Yes	No	CSV only	Frame-to-frame change in high-frequency energy.
delta_flux	Yes	No	CSV only	Frame-to-frame change in spectral flux.
midband_residual_ratio	Yes	Yes; also context input	Live/CSV	Non-harmonic midband disturbance relative to baseline current.
zcv	Yes	No	Session/CSV	Zero-crossing interval variation used for zero-region timing review.
spectral_flux_midhf	Yes	Yes; also context input	Live/CSV	Abrupt mid/high-frequency spectral change between frames.
peak_fluct_cv	Yes	No	Session/CSV	Cycle-peak coefficient of variation.
residual_crest_factor	Yes	No	Session/CSV	Peak-to-RMS ratio of the residual disturbance.
thd_i	Yes	Context input	Live/CSV	Current total harmonic distortion from the fundamental and harmonics.
hf_energy_delta	Yes	Context input	Live/CSV	High-frequency energy share relative to the learned baseline.
edge_spike_ratio	Yes	No	Session/CSV	Edge or restrike spike strength relative to baseline current.
v_sag_pct	Yes	No	CSV only	Voltage sag relative to the healthy RMS-voltage baseline.
cycle_nmse	Yes	No	CSV only	Cycle-to-cycle normalized waveform mismatch.
pulse_count_per_cycle	Yes	No	Session/CSV	Residual pulse count normalized per mains cycle.
zero_dwell_ratio	Yes	Yes	Live/CSV	Portion of near-zero samples around zero crossings.
low_current_ratio	Yes	Yes	Live/CSV	Portion of the frame below the low-current threshold.
max_low_current_run_ms	Yes	Yes	Live/CSV	Longest low-current run expressed in milliseconds.

Table 3.7 shows that the prototype does not compute only the 10 features shown on the PWA dashboard. It exports 20 candidate waveform-related feature fields, deploys six as base waveform inputs to the arc model, uses six context-model features to produce the context fields, displays 10 monitoring feature cards, and keeps additional feature columns for session review. This separation allows the deployed model to remain compact while still preserving the feature values needed for interpretation, label review, and training analysis. The next sections describe the physical meaning and firmware computation of these features.

3.7.2 Collected Candidate Features

The current RMS value used in the feature frame is computed as

$$I_{\text{rms}} = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} i[n]^2}. \quad (3.19)$$

In (3.19), $i[n]$ is the calibrated current sample after the firmware's current conversion path and N is the number of samples in the frame. RMS current is not counted as one of the 20 exported feature IDs, but it is the physical reference used by overload logic and by several arc features.

3.7.2.1 Current-Magnitude and Baseline Features

The RMS-current baseline z-score is

$$Z_I = \frac{|I_{\text{rms}} - \mu_I|}{\sigma_I + 10^{-6}}. \quad (3.20)$$

In (3.20), μ_I and σ_I are the adaptive baseline mean and standard deviation maintained by the firmware. The feature `abs_irms_zscore_vs_baseline` is dimensionless, clipped by firmware, used by the deployed arc model, and shown in the PWA. It helps detect current magnitude departures that can accompany interruption and restrike.

The absolute RMS-current step is

$$\Delta I_{\text{rms}} = |I_{\text{rms}}[k] - I_{\text{rms}}[k-1]|. \quad (3.21)$$

The feature `delta_irms_abs` is expressed in amperes, supports the context model, and is shown in the PWA. It helps distinguish load turn-on, load-family changes, and abrupt waveform events from steady operation.

The voltage-sag feature is computed from the healthy voltage baseline:

$$V_{\text{sag}} = 100 \max \left(0, \frac{B_V - V_{\text{rms}}}{B_V} \right). \quad (3.22)$$

In (3.22), B_V is the voltage baseline and V_{rms} is the calibrated RMS voltage. The firmware logs `v_sag_pct` for event interpretation, but the deployed arc model does not use it as a base feature.

3.7.2.2 Conduction-Gap and Pulse Features

The low-current ratio measures the portion of the frame close to zero current:

$$L_{\text{ratio}} = \frac{100}{N} \sum_{n=0}^{N-1} \mathbf{1}(|i_c[n]| \leq \max(0.035, 0.06 \max(I_{\text{rms}}, 0.10))). \quad (3.23)$$

In (3.23), $i_c[n]$ is the cleaned current sample and $\mathbf{1}(\cdot)$ is an indicator function. The result is a percentage. The feature `low_current_ratio` is used by the deployed arc model and shown in the PWA because a high value can indicate current interruption during a series arc.

The maximum low-current run is computed as

$$t_{\text{low,max}} = \frac{1000 N_{\text{run,max}}}{f_s}. \quad (3.24)$$

In (3.24), $N_{\text{run,max}}$ is the longest consecutive low-current run and f_s is the sampling rate. The output is in milliseconds. The feature `max_low_current_run_ms` is used by the deployed arc model and shown in the PWA because series arcs can create observable extinction gaps.

The zero-dwell ratio measures near-zero dwelling around zero crossings. The firmware uses zero-crossing windows when crossings are stable and falls back to the low-current ratio when a robust dwell estimate is unavailable:

$$D_0 = 100 \frac{N_{\text{near zero}}}{N_{\text{zc window}}}. \quad (3.25)$$

In this paper, `zero_dwell_ratio` refers to the near-zero conduction behavior used by the arc model, while `zcv` is treated separately as zero-crossing interval variation.

The residual pulse count per cycle is

$$P_{\text{cycle}} = \frac{N_{\text{pulses}}}{N/f_s \cdot f_0}. \quad (3.26)$$

In (3.26), N_{pulses} is the detected residual-pulse count, f_s is the sampling rate, and f_0 is the mains frequency. The feature `pulse_count_per_cycle` is retained in the PWA session metadata and logged because short residual pulses are physically consistent with arc restrike, although the deployed arc model does not use this feature directly.

3.7.2.3 Spectral and Residual Features

The mid/high-frequency spectral flux feature compares the normalized magnitude spectrum of the present frame with the previous frame:

$$F_{\text{midhf}} = 50 \sum_{k=k_0}^{k_1} \left| \frac{M_k}{\sum_{r=k_0}^{k_1} M_r} - \frac{M_{k,\text{prev}}}{\sum_{r=k_0}^{k_1} M_{r,\text{prev}}} \right|. \quad (3.27)$$

In (3.27), M_k is the FFT magnitude in the 900Hz to 9000Hz flux band after skipping bins around mains harmonics. The feature `spectral_flux_midhf` is dimensionless, used by both the context model and deployed arc model, and shown in the PWA.

The high-frequency energy delta compares the high-frequency energy share with its baseline:

$$\Delta E_{\text{HF}} = 10 \log_{10} \left(\frac{E_{\text{HF}} / (E_{\text{HF}} + E_{\text{UM}}) + \epsilon}{B_{\text{HF}} + \epsilon} \right). \quad (3.28)$$

In (3.28), E_{HF} is the 4500-9000Hz band energy, E_{UM} is the 900-4500Hz upper-midband energy, B_{HF} is the baseline high-frequency share, and ϵ is a small numerical offset. The result is in decibels, supports the context model, and is shown in the PWA as `hf_energy_delta`. The separate `delta_hf_energy` is the frame-to-frame absolute change of this high-frequency measure and is logged for analysis.

The midband residual ratio is computed from FFT energy in the 220-3200Hz midband after excluding harmonic bins:

$$R_{\text{mid}} = 20 \log_{10} \left(\frac{\sqrt{E_{\text{mid,res}}/N}}{\max(B_I, 0.10)} \right). \quad (3.29)$$

In (3.29), $E_{\text{mid,res}}$ is the residual midband energy, N is the frame length, and B_I is the baseline RMS current. The output is in decibels and is used by both the context model and the deployed arc model.

The residual crest factor is

$$C_{\text{res}} = 20 \log_{10} \left(\frac{\max_n |r[n]|}{\sqrt{\frac{1}{N} \sum_{n=0}^{N-1} r[n]^2}} \right), \quad (3.30)$$

where $r[n]$ is the residual waveform after low-frequency content is removed. The feature `residual_crest_factor` is retained in the PWA session metadata and logged to describe spike-like residual behavior.

The edge-spike ratio is

$$S_{\text{edge}} = 20 \log_{10} \left(\frac{A_{\text{edge}}}{\max(B_I, 0.10)} \right), \quad (3.31)$$

where A_{edge} is the firmware's edge-burst amplitude estimate. The feature `edge_spike_ratio` is retained in the PWA session metadata and helps identify restrike-like edges, but it is not a deployed arc-model base input.

The firmware also logs `delta_flux`, the frame-to-frame change in spectral flux, and `suspicious_run_energy`, a leaky internal score that increases when several arc-like cues occur together. These features support logging, interpretation, and training analysis; they are not used directly by the deployed Random Forest base vector.

3.7.2.4 Waveform Shape and Harmonic Features

The firmware computes current THD from the fundamental and harmonic tone powers:

$$\text{THD}_I = 100 \sqrt{\frac{\sum_{h=2}^{10} P_h}{P_1}}. \quad (3.32)$$

In (3.32), P_1 is the tracked fundamental tone power and P_h is the tone power of harmonic h . The feature `thd_i` is a percentage, supports the context model, and is shown in the PWA.

The half-cycle asymmetry is

$$A_{\text{half}} = 100 \frac{|I_{\text{pos}} - I_{\text{neg}}|}{0.5(I_{\text{pos}} + I_{\text{neg}})}. \quad (3.33)$$

In (3.33), I_{pos} and I_{neg} are the positive and negative half-cycle RMS values. The feature

halfcycle_asymmetry is dimensionless in percent, supports the context model, and is shown in the PWA.

The peak-fluctuation coefficient of variation is

$$C_{V,\text{peak}} = 100 \frac{\sigma(A_{\text{pk}})}{\mu(A_{\text{pk}})}, \quad (3.34)$$

where A_{pk} is the set of cycle peak magnitudes in the frame. The feature peak_fluct_cv is retained in the PWA session metadata and describes irregular cycle peaks.

The cycle normalized mean-squared error is represented as

$$E_{\text{cycle}} = 100 \text{ median} \left(\frac{\sum_m (c_j[m] - c_{j-1}[m])^2}{\sum_m c_{j-1}[m]^2 + \epsilon} \right), \quad (3.35)$$

where $c_j[m]$ and $c_{j-1}[m]$ are adjacent cycle segments aligned by the firmware. The feature cycle_nmse is logged for waveform comparison but is not a deployed arc-model base input.

3.7.3 Selected Features Used by the Deployed Arc Model

The deployed arc model uses max_low_current_run_ms, zero_dwell_ratio, low_current_ratio, abs_irms_zscore_vs_baseline, midband_residual_ratio, and spectral_flux_midhf. These six waveform features cover conduction gaps, current-magnitude departure, and mid/high-frequency disturbance. The model then appends six context-family one-hot inputs and one context-confidence input. The context inputs improve interpretation across appliance families, but they are not additional waveform features.

3.7.4 Features Shown in the PWA

The PWA dashboard contains 10 waveform feature cards: abs_irms_zscore_vs_baseline, delta_irms_abs, halfcycle_asymmetry, max_low_current_run_ms, low_current_ratio, zero_dwell_ratio, midband_residual_ratio, spectral_flux_midhf, thd_i, and hf_energy_delta. The PWA metadata and session viewer still list 14 review features for CSV inspection,

so this paper distinguishes the live dashboard display from the full logging and review interface. This display set is broader than the deployed arc-model base vector so that the researchers can inspect useful diagnostic signals without implying that every displayed feature contributes directly to the Random Forest decision.

3.8 Fault Detection and Protection Logic

The protection manager receives calibrated voltage, calibrated RMS current, estimated socket temperature, and TinyML arc output. It then assigns warning and trip states according to thresholds and persistence rules in the firmware. The relay is opened only after a trip condition is accepted by the protection logic; warnings alone do not disconnect the load.

3.8.1 Voltage Supervision

Voltage supervision uses calibrated RMS voltage and mains-present state. The firmware defines the normal operating band as 200-250V. Instantaneous undervoltage protection is set at 170V, delayed undervoltage is set below 200V for 7s, delayed overvoltage is set above 250V for 7s, and instantaneous overvoltage is set at 260V. Recovery requires the voltage to return to 212-250V and remain stable for 5min.

These thresholds are treated as design choices for supply-state supervision rather than validated voltage-protection performance, since the experimental validation focused on fire-inducing fault types. The implemented values are summarized in *Table 3.8*.

Table 3.8. Voltage-protection thresholds verified from firmware

Condition	Firmware value	Interpretation
Normal lower limit	200V	Lower edge of normal operation
Normal upper limit	250V	Upper edge of normal operation
Instant undervoltage	170V	Immediate severe undervoltage trip
Delayed undervoltage	200V for 7s	Persistent low-voltage trip
Delayed overvoltage	250V for 7s	Persistent high-voltage trip
Instant overvoltage	260V	Immediate severe overvoltage trip
Recovery band	212-250V for 5min	Stable voltage required before recovery

Table 3.8 shows that the voltage logic includes both instantaneous and delayed decisions. This avoids treating every brief voltage excursion as a latched trip while still allowing severe voltage conditions to disconnect promptly.

3.8.2 Overload Protection

The overload logic separates warning from relay trip. A warning begins when current exceeds 10.0A after the warning delay and clears below 9.6A. A sustained-overload trip requires the current average to remain above 14.5A long enough for the overload score to reach the trip score.

The persistence score is represented as

$$S_{OL}[k] = \min(S_{\max}, S_{OL}[k-1] + 1), \quad I_{\text{avg}} \geq 14.5\text{A}, \quad (3.36)$$

and

$$S_{OL}[k] = \max(0, S_{OL}[k-1] - 1), \quad I_{\text{avg}} \leq 14.1\text{A}. \quad (3.37)$$

In (3.36) and (3.37), S_{OL} is the sustained-overload score, I_{avg} is the firmware's 1s averaged current, and $S_{\max} = 6$. The score is updated every 10s, and relay trip occurs at $S_{OL} = 6$. Therefore, a persistent overload near or above 14.5A trips after about 60s, while shorter excursions can produce warnings without disconnection. Persistence-based behavior is consistent with the principle that overcurrent protection should distinguish brief transients from sustained thermal stress [22].

3.8.3 Socket-Heating Protection

Socket-heating protection uses the estimated socket temperature $T_{\text{socket,est}}$ from (3.16). The firmware uses a warning threshold of 39.0°C, a warning-clear threshold of 38.6°C, and a trip threshold of 40.0°C. The warning hysteresis reduces rapid warning toggling near the threshold, while the trip threshold creates a conservative response to possible contact heating.

The trip rule is

$$H = \begin{cases} 1, & T_{\text{socket,est}} \geq 40.0^\circ\text{C}, \\ 0, & T_{\text{socket,est}} < 40.0^\circ\text{C}. \end{cases} \quad (3.38)$$

In (3.38), H is the heating-trip state. This trip criterion is intentionally based on the compensated socket estimate rather than the direct device temperature measurement because the external thermistor can under-read the contact hotspot during degraded-contact heating [30], [46], [47].

3.8.4 Series Arc-Fault Protection

The Random Forest arc model produces a frame-level arc class. The protection manager does not trip from a single isolated model output. Instead, the firmware uses a leaky counter:

$$C_{\text{arc}}[k] = \min(16, C_{\text{arc}}[k-1] + 2), \quad \hat{y}_{\text{arc}} = 1, \quad (3.39)$$

and

$$C_{\text{arc}}[k] = \max(0, C_{\text{arc}}[k-1] - 6), \quad \hat{y}_{\text{arc}} = 0. \quad (3.40)$$

In (3.39) and (3.40), C_{arc} is the arc persistence counter and $\hat{y}_{\text{arc}}$ is the TinyML model output. The relay trips when $C_{\text{arc}} \geq 8$. The counter therefore requires repeated arc-positive frames and quickly decays when the classifier output returns to normal.

The firmware also includes a current-return verification path. If current falls below 0.03A before the arc counter reaches the trip threshold, the system waits up to 200ms to verify whether current returns before treating the event as an arc trip. This guards against disconnecting on a simple load turn off while still allowing fast response to arc-like interruption and restrike.

3.8.5 Protection Priority and Latching

When multiple protection conditions are present, the firmware resolves the reported fault according to protection priority. Socket heating is given the highest priority because temperature is the most direct indicator of thermal fire risk, followed by arc fault,

sustained overload, and voltage-related protection states. The relay-latching circuit is pulsed by the firmware after a trip decision, and the fault state remains latched until the user clears it. This prioritization does not change the underlying detection equations; it only determines which fault reason is reported when conditions overlap.

The principal thresholds are summarized in *Table 3.9*.

Table 3.9. Principal protection thresholds verified from firmware

Protection function	Warning or model threshold	Relay-trip condition
Voltage supervision	Outside 200–250V band	Instant 170/260V or delayed 7s low/high voltage
Overload	Warning above 10.0A; clear below 9.6A	1s current average at or above 14.5A for about 60s
Socket heating	Warning at 39.0°C; clear at 38.6°C	Estimated socket temperature at or above 40.0°C
Series arc fault	RF probability threshold depends on context	Arc counter reaches 8 from repeated arc-positive frames

Table 3.9 shows that the prototype uses a hybrid method: deterministic thresholds for voltage, overload, and temperature, and TinyML-supported classification for series arcs. The hybrid design is intended to complement existing protection devices by observing socket-level thermal and waveform behavior that may not be sufficient to trip a branch breaker.

3.9 Test and Evaluation Procedures

The evaluation followed the same order as the implementation: sensor validation, overload testing, socket-heating characterization, socket-heating protection validation, series arc-fault testing, and TinyML model evaluation. Full calibration records are placed in *Appendix A*; socket-heating characterization and protection data are placed in *Appendices C and D*; arc-fault testing and data collection logs are placed in *Appendices E and F*. The letters, safety proposal, project schedule, and cost documentation are placed in *Appendices G, H, and I*.

3.9.1 Sensor-Validation Metrics

Sensor validation compared prototype readings against reference instruments. The mean absolute error was computed as

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (3.41)$$

The mean squared error was computed as

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (3.42)$$

The root mean squared error was computed as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (3.43)$$

The mean absolute percentage error was computed as

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|. \quad (3.44)$$

In (3.41)-(3.44), y_i is the reference measurement, $\hat{y}_i$ is the prototype measurement, and n is the number of test points. MAE gives the average magnitude of error, MSE and RMSE penalize larger errors more strongly, and MAPE expresses the error relative to the reference value [72].

3.9.2 Overload Test Procedure

Overload testing used high-power load combinations to determine whether the prototype warned above the warning threshold and tripped only when the sustained-overload condition persisted. The block diagram of the overload test is shown in *Figure 3.25*.

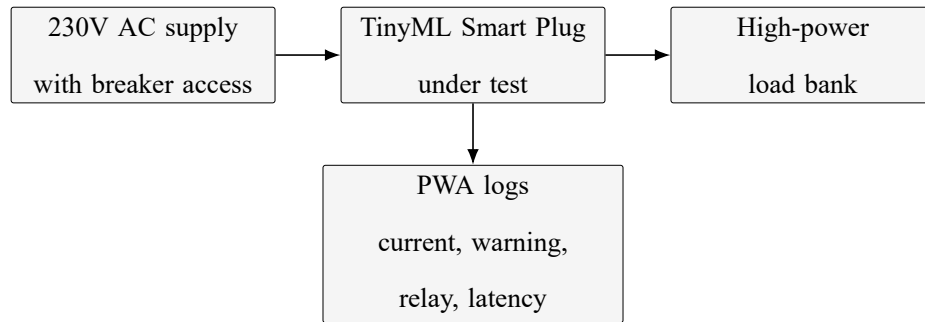


Figure 3.25. Testing setup block diagram for overload evaluation.

Figure 3.25 shows the prototype placed between the supply and load bank so that it could measure current and disconnect the load through its relay. This matches the current path used in normal operation.

The actual overload test setup is shown in *Figure 3.26*.

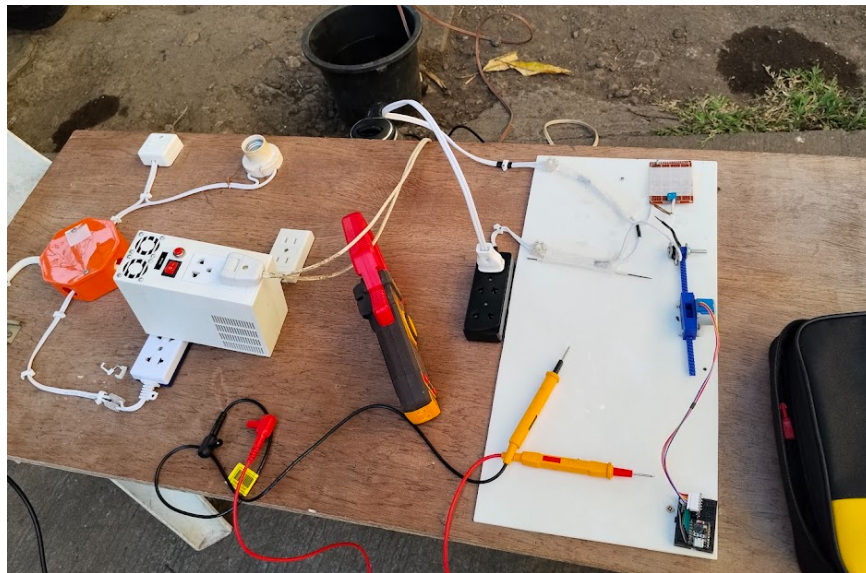


Figure 3.26. Actual test setup for overload evaluation.

Figure 3.26 documents the physical load arrangement used to create warning and sustained-overload conditions. During each trial, mean voltage, mean current, peak current, warning state, relay state, and trip latency were recorded.

3.9.3 Socket-Heating Test Procedure

Socket-heating tests compared normal and intentionally degraded socket contacts under similar load levels. The objective was to determine whether degraded contacts produced faster and higher temperature rise, and whether the external device temperature mea-

surement and compensated estimator could provide useful warning before dangerous contact heating progressed.

Figure 3.27 shows the socket-heating test block diagram.

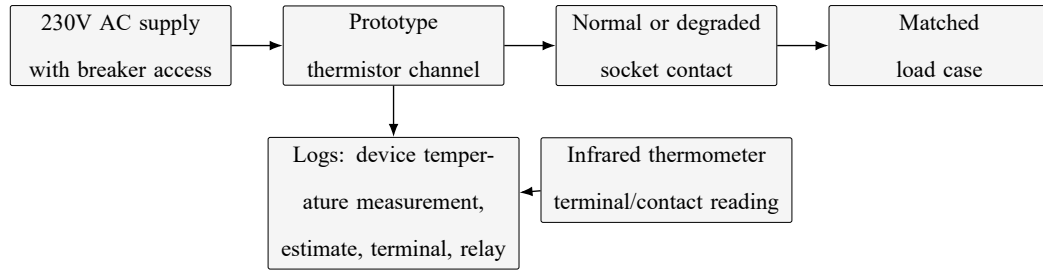


Figure 3.27. Testing setup block diagram for socket-heating evaluation.

Figure 3.27 shows that both the prototype device temperature measurement and a reference socket or terminal temperature measurement were observed during testing. This made it possible to distinguish device temperature measurement, estimated socket temperature, and reference terminal temperature in the results.

The degraded socket contacts are shown in *Figure 3.28*.



Figure 3.28. Corroded socket contacts used for degraded-contact testing.

Figure 3.28 documents the degraded-contact condition used to accelerate contact heating. The degraded socket was compared against a normal socket under matched load combinations.

The actual socket-heating test setup is shown in *Figure 3.29*.

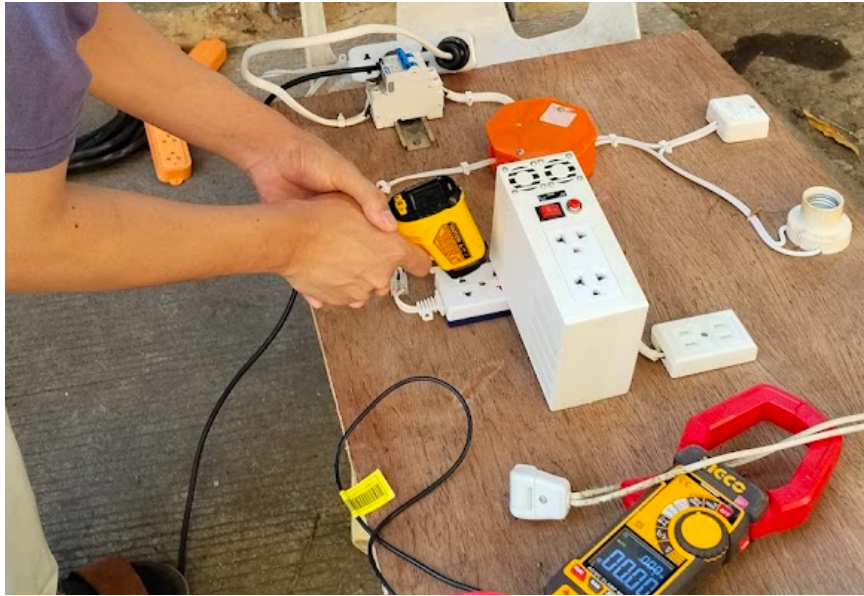


Figure 3.29. Actual test setup for socket-heating evaluation.

Figure 3.29 shows the bench configuration used to record time, current, device temperature measurement, estimated socket temperature, reference terminal temperature, and relay behavior. The characterization data in *Appendix C* were used to support the thermal model, while the protection validation data in *Appendix D* were used to evaluate relay response.

3.9.4 Series Arc-Fault Test Procedure

Series arc-fault testing used a controlled arc platform and appliance loads. The goal was to create repeatable waveform disturbances while recording current features, model outputs, relay behavior, and event labels. The assistance letter and safety proposal for this testing activity are included in *Appendix G*. The arc platform is shown in Figure 3.30.

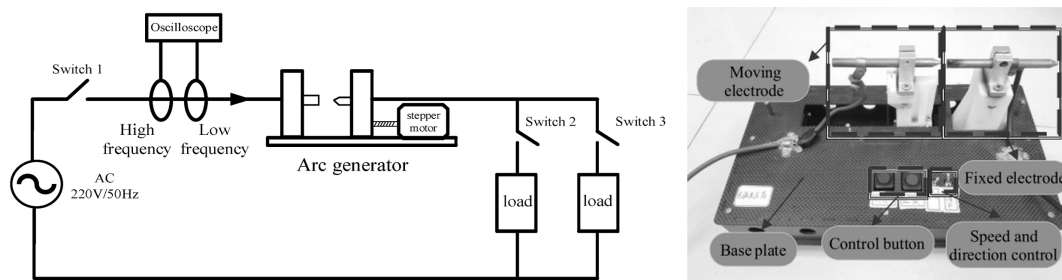


Figure 3.30. Series arc-fault test platform [48].

Figure 3.30 shows the principle of the series arc test: the current path is interrupted by an adjustable contact gap, allowing a controlled arc disturbance to appear in series with the load. The adapted source describes a 220V/50Hz platform using a fixed electrode and a moving electrode controlled by a stepper motor [48]. The prototype testing used the same series-interruption principle as a guide, while the actual bench setup used in this study is documented in *Figure 3.32*.

The block diagram of the series arc-fault test is shown in *Figure 3.31*.

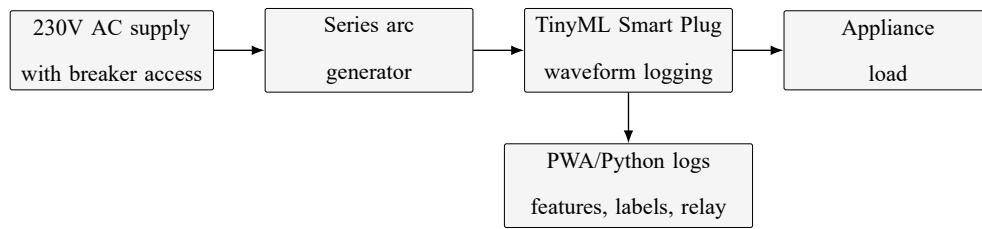


Figure 3.31. Testing setup block diagram for series arc-fault evaluation.

Figure 3.31 shows the smart plug measuring the load current while the arc generator is placed in series with the appliance.

The actual series arc-fault test setup is shown in *Figure 3.32*.

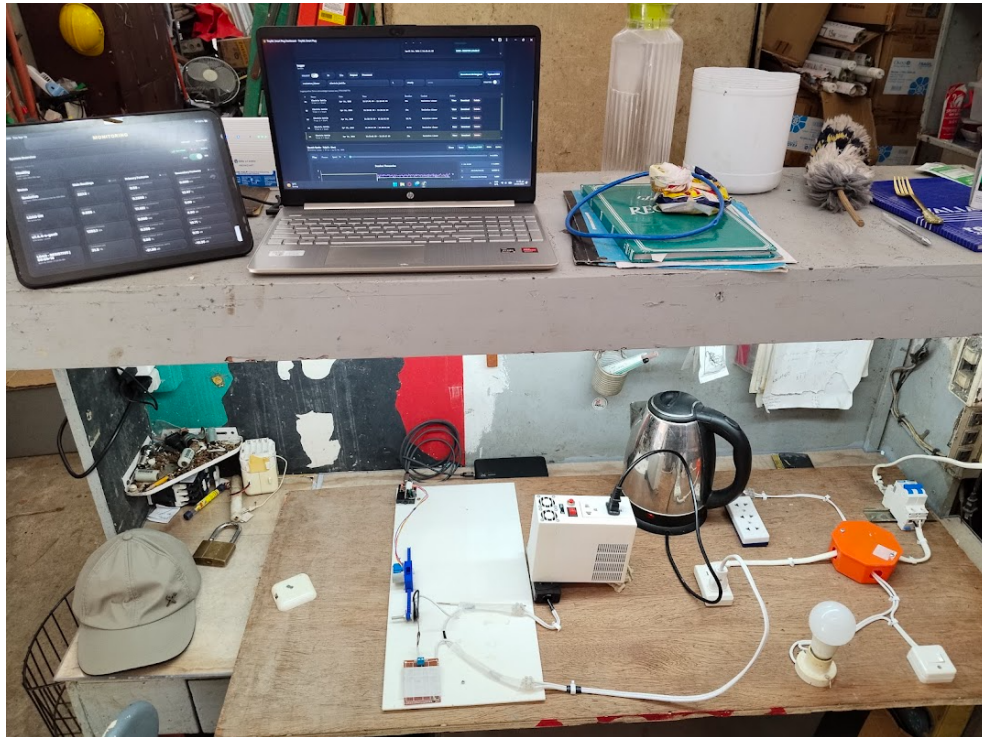


Figure 3.32. Actual test setup for series arc-fault evaluation.

Figure 3.32 documents the laboratory arrangement used to collect arc and normal trials. Complete arc-fault test records are referenced in *Appendices E and F*.

3.9.5 Confusion Matrix and Classification Metrics

For arc-fault evaluation, the positive class was the arc-fault condition and the negative class was normal operation. The model output was assessed according to whether each evaluated frame was correctly or incorrectly assigned to these two classes. The four possible outcomes are represented through a confusion matrix, as shown in *Table 3.10*.

Table 3.10. Confusion matrix for arc-fault classification

Actual class	Predicted arc fault	Predicted normal
Arc fault	True positive (TP)	False negative (FN)
Normal operation	False positive (FP)	True negative (TN)

As shown in *Table 3.10*, the confusion matrix defines the four possible classification outcomes used in the evaluation. A true positive (TP) represents an arc-fault frame correctly detected as an arc fault, while a true negative (TN) represents a normal frame correctly identified as normal. A false positive (FP) represents a normal frame incorrectly classified as an arc fault, which corresponds to a false-trip tendency. A false negative (FN) represents an arc-fault frame incorrectly classified as normal, which corresponds to a missed-fault tendency.

The raw data collection produced an imbalanced distribution of normal and arc-fault trials, as shown in *Appendix F*. Because of this, the classification evaluation in *Appendix E* used balanced normal and arc-fault trials so that accuracy would represent performance across both classes more reliably. This step was necessary because, under an imbalanced dataset, a model may obtain high accuracy by favoring the majority class even when its detection of arc-fault cases is weak.

Accuracy was computed as

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \quad (3.45)$$

Accuracy represents the proportion of all evaluated samples that were classified correctly. In this study, it indicates the general correctness of the classifier after the normal and arc-fault trials were balanced for evaluation.

Precision was computed as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \quad (3.46)$$

Precision measures how many arc-positive decisions were actually arc faults. A higher precision value means that fewer normal samples were incorrectly treated as arc faults, which is important for reducing unnecessary trip decisions during healthy operation.

Recall was computed as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (3.47)$$

Recall measures how many actual arc-fault samples were detected by the model. A higher recall value means that fewer arc-fault samples were missed, which is important because an undetected arc fault may allow the hazardous condition to continue.

The F1-score was computed as

$$F_1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (3.48)$$

The F1-score combines precision and recall into one measure. It was included because arc-fault detection must consider both false trips and missed faults, rather than relying on accuracy alone. These metrics are commonly used for supervised classification and confusion-matrix evaluation [74].

Chapter 4

RESULTS AND DISCUSSION

This chapter presents the results following the sequence established in the methodology. The discussion begins with sensor validation, then proceeds to overload protection, socket-heating characterization and validation, series arc-fault detection, TinyML model performance, and prototype feasibility.

4.1 Sensor Validation

The sensor-validation results determine whether the calibrated voltage, current, and device-temperature measurements were reliable enough for the implemented protection decisions. The results were evaluated using MAE, MSE, and RMSE to show both average measurement error and larger protection-relevant deviations [72].

4.1.1 Voltage Sensor Validation

The voltage sensor was compared with a Fluke 115 True RMS multimeter at ten test points.

Table 4.1. Validation results for the voltage sensor

Test	Device (V)	Fluke (V)	Abs. error (V)	Squared error (V ²)
1	181.0	184.7	3.7	13.69
2	196.1	192.8	3.3	10.89
3	198.4	201.9	3.5	12.25
4	210.3	213.6	3.3	10.89
5	223.8	219.7	4.1	16.81
6	221.0	225.4	4.4	19.36
7	234.6	230.9	3.7	13.69
8	232.7	236.5	3.8	14.44
9	247.5	243.1	4.4	19.36
10	244.3	249.2	4.9	24.01

Table 4.1 shows the validation results for the voltage sensor. The corrected prototype readings produced an MAE of 3.910V, MSE of 15.539V², and RMSE of 3.94V. These errors are small relative to the firmware voltage bands summarized in *Table 3.8*. Therefore,

the calibrated voltage channel is adequate for distinguishing normal operation from the undervoltage and overvoltage regions used in this prototype. This result supports voltage-state discrimination for the implemented supervision logic, but it does not certify the prototype as a utility-grade voltage meter.

4.1.2 Current Sensor Validation

The current sensor was compared with an Ingco clamp meter.

Table 4.2. Validation results for the current sensor

Test	Device (A)	Clamp meter (A)	Abs. error (A)	Squared error (A ²)
1	0.66	0.68	0.02	0.0004
2	1.98	1.94	0.04	0.0016
3	5.00	5.11	0.11	0.0121
4	7.80	7.63	0.17	0.0289
5	8.97	9.14	0.17	0.0289
6	10.76	10.58	0.18	0.0324
7	11.84	12.06	0.22	0.0484
8	14.00	13.74	0.26	0.0676
9	14.34	14.67	0.33	0.1089
10	16.14	15.86	0.28	0.0784

Table 4.2 shows that the current-channel error increased slightly at higher currents but remained much smaller than the distance between the warning and sustained-trip thresholds. The current channel produced an MAE of 0.178A, MSE of 0.04076A², and RMSE of 0.202A. This level of error is acceptable for the implemented overload logic because the warning threshold is 10.0A and the sustained-overload threshold is 14.5A. The same current channel is also used by the socket thermal model and the TinyML feature extractor, so low RMS-current error improves both threshold-based protection and feature reliability. This supports the overload-protection interpretation discussed in *Section 4.2*.

4.1.3 Device Temperature Measurement Validation

The TTC05103JSY 10k Ω thermistor was compared with a reference infrared thermometer at the sensor measurement point.

Table 4.3. Validation results for the device temperature measurement thermistor channel

Test	Device (°C)	IR thermometer (°C)	Abs. error (°C)	Squared error (°C ²)
1	27.9	27.4	0.5	0.25
2	31.1	31.8	0.7	0.49
3	37.7	36.6	1.1	1.21
4	41.1	41.9	0.8	0.64
5	48.2	47.3	0.9	0.81
6	51.2	52.5	1.3	1.69
7	59.0	58.1	0.9	0.81
8	61.6	63.2	1.6	2.56
9	67.8	66.8	1.0	1.00
10	68.3	69.7	1.4	1.96

Table 4.3 supports the use of the thermistor channel as a stable observable input for the thermal estimator. The device temperature measurement channel produced an MAE of 1.02°C, MSE of 1.142°C², and RMSE of 1.07°C. This validates the thermistor measurement at the sensor location, but it must not be interpreted as a direct validation of the internal contact temperature. This distinction is consistent with the reviewed literature, which explains that external thermistor placement, thermal lag, and hidden contact hotspots can cause the measured sensor temperature to under-read the hottest plug-socket interface region [27]-[31]. It also supports the methodological need for the compensated socket-temperature model because degraded contacts may generate localized heating while the external sensor observes only a delayed and spatially separated temperature response [46], [47]. The socket-heating tests later show that the socket terminal can be hotter than either the external thermistor reading or the compensated estimate.

4.2 Overload Protection

Five overload-related trials were conducted using high-power load combinations. All trials exceeded the overload-warning threshold, but only the most severe case remained above the sustained-overload criterion long enough to trip the relay. This follows the firmware method in *Section 3.8.2*, where warning and sustained relay trip are intentionally separated to distinguish short current excursions from persistent thermal stress. This behavior is consistent with the reviewed literature, which treats overload protection as a

time-dependent overcurrent condition rather than a purely instantaneous cutoff [22], [44], [45].

Table 4.4. Overload evaluation results

Trial	Load setup	Mean V	Mean A	Peak A	Warning	Trip latency
1	1300W electric kettle + water heater	231.8	12.87	14.85	Yes	–
2	1300W electric kettle + 1500W electric kettle	229.5	12.33	13.63	Yes	–
3	1500W electric kettle + water heater	229.6	13.36	14.86	Yes	–
4	1300W electric kettle + water heater + 2 × 500W halogen lamp	224.0	17.30	19.73	Yes	61.3s
5	1300W electric kettle + water heater + 500W halogen lamp	229.1	14.42	16.23	Yes	–

Table 4.4 shows that the prototype issued overload warnings in all heavy-load trials, while relay disconnection occurred only in Trial 4. Trial 4 had a mean current of 17.30A, which was clearly above the 14.5A sustained-overload threshold. Its trip latency of 61.3s also agrees with the intended approximately 60s persistence behavior of the overload score. Trials 1, 3, and 5 reached peak currents near or above the sustained-overload threshold, but their current levels did not remain above the criterion long enough for the persistence score to reach the relay-trip value.

These results support the overload-protection objective because the prototype operated as a warning-and-persistence device rather than as an instantaneous current cutoff. The warning response gave early indication of heavy loading, while the delayed relay trip was reserved for a sustained overcurrent condition. Within the tested laboratory load combinations, this behavior reduced the chance of tripping on short-duration peaks while still disconnecting the load when the overload condition persisted.

4.3 Socket-Heating Characterization

The socket-heating characterization compared a normal socket with a degraded socket under similar current levels. This section presents only the flat NTC temperature measurements obtained after resistance-to-temperature conversion using the Steinhart-

Hart/Beta equation described in *Section 3.5.3*; the thermal constants, expected normal socket temperature, and estimated socket temperature were not yet applied to the plotted data.

The characterization was performed to verify that socket heating is affected not only by current, but also by contact condition. This follows the reviewed literature showing that poor plug-socket contact can produce localized heating while the current path remains functional [17], [24], [46], [47], and that an external temperature measurement may lag or under-read the internal hotspot because the hottest region is usually inside the contact interface [27]-[31]. The graphs presented in this section are derived from the characterization records in *Appendix C*.

The 38.2°C comparison point was used only as a common reference marker for comparing how quickly each socket condition approached the same measured thermistor temperature level.

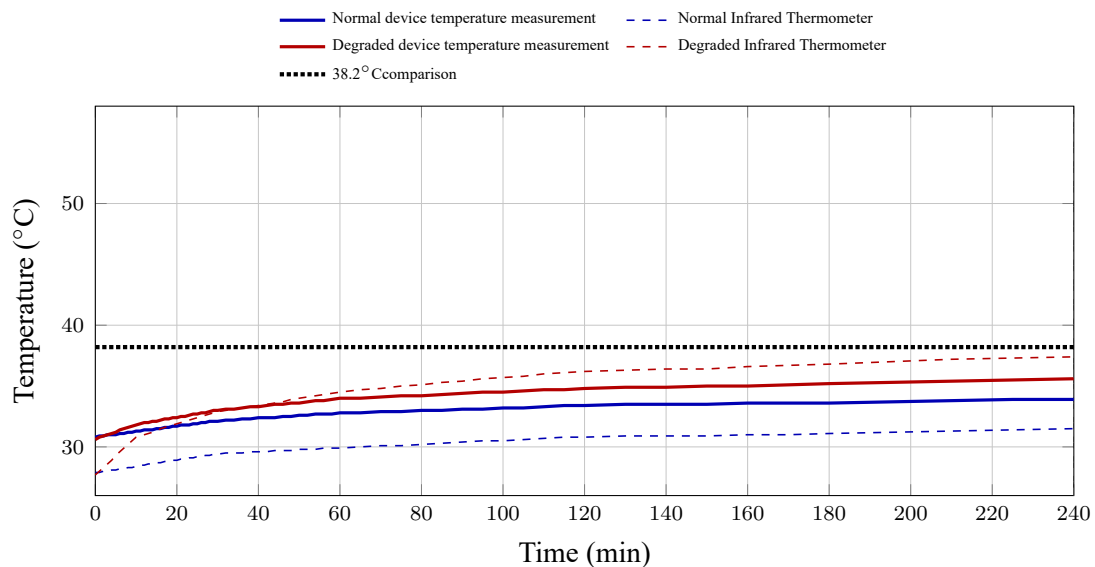


Figure 4.1. Water heater temperature trend during socket-heating characterization.

Figure 4.1 shows that at about 5.2A, both socket conditions stayed below the 38.2°C comparison point within the 240-minute window. However, the corroded socket still ended hotter than the normal socket in both the device temperature measurement and the infrared terminal reading. This indicates that the contact-condition effect was already present at the lowest tested load.

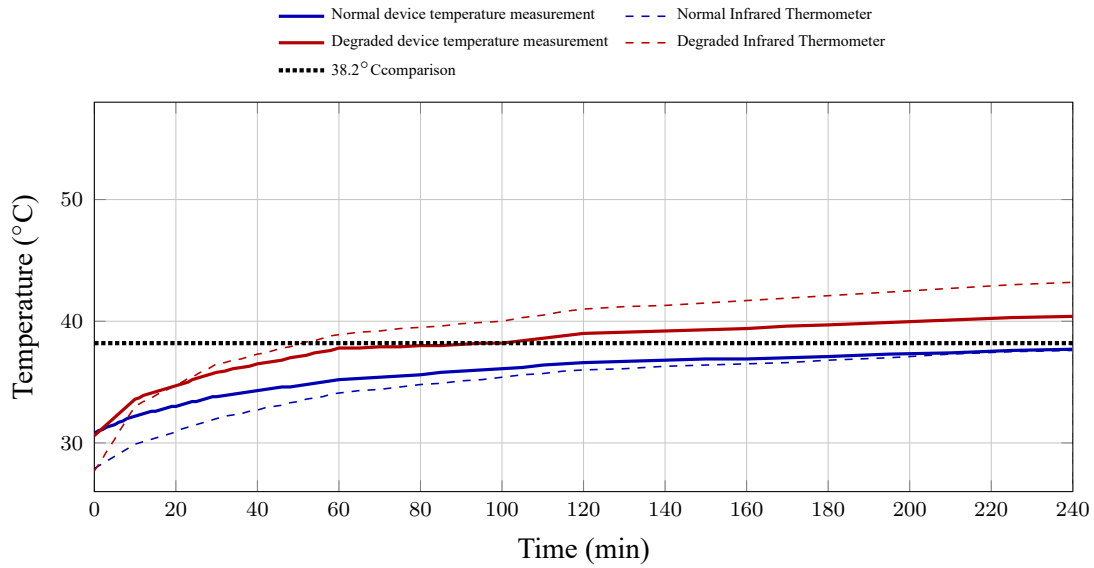


Figure 4.2. Water heater plus one halogen lamp temperature trend during characterization.

Figure 4.2 shows clearer separation between normal and corroded behavior. The corroded socket reached the 38.2°C comparison point, while the normal socket did not reach that point within the same window.

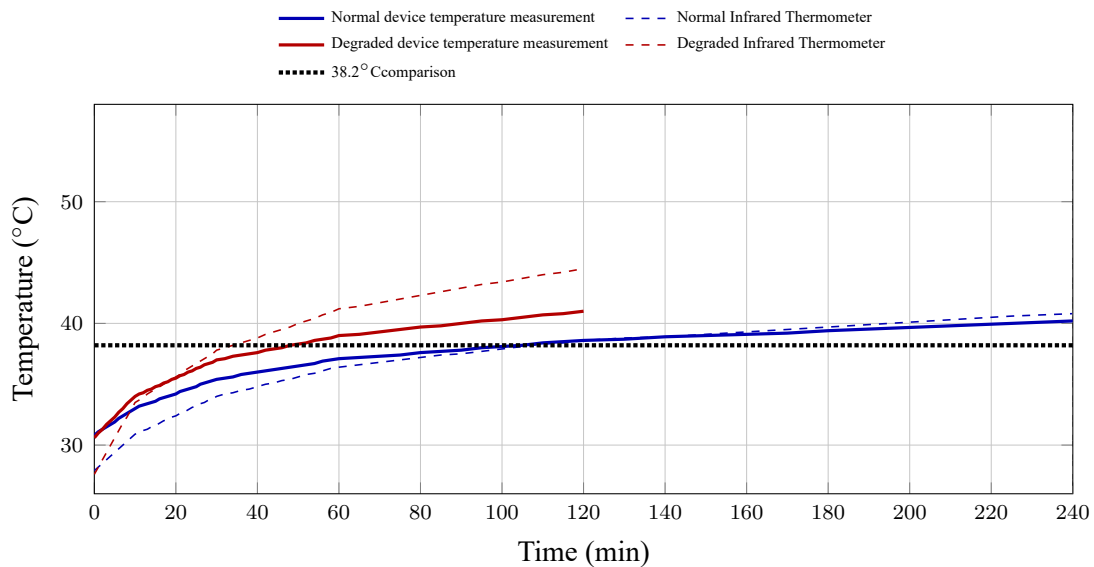


Figure 4.3. Water heater plus two halogen lamps temperature trend during characterization.

Figure 4.3 shows that the degraded socket reached the comparison temperature much earlier than the normal socket. The infrared terminal temperature also rose above the device measurement, especially in the corroded case. This supports treating the external thermistor as an indirect reading rather than the actual hotspot temperature [27]-[31].

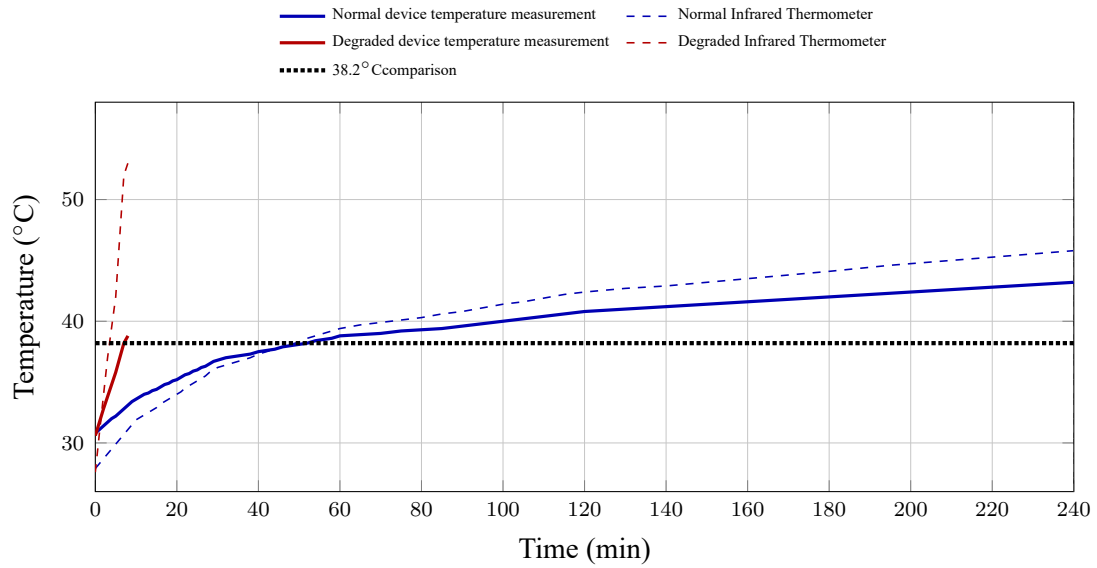


Figure 4.4. Water heater plus three halogen lamps temperature trend during characterization.

Figure 4.4 shows the strongest time-to-threshold contrast in the characterization set. The corroded socket reached the comparison point after 7 minutes, while the normal socket reached it after 52 minutes. Since the current levels were similar, this difference is attributed mainly to degraded contact condition, supporting the reviewed finding that poor contact can accelerate localized heating [17], [24], [46], [47].

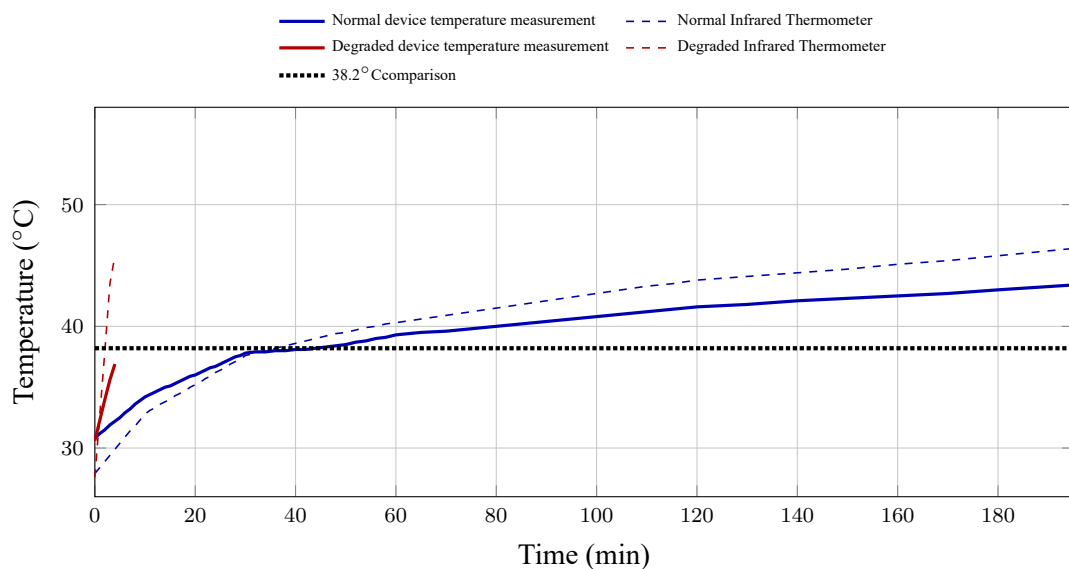


Figure 4.5. Water heater plus four halogen lamps temperature trend during characterization.

Figure 4.5 shows that the high-current degraded trial developed rapidly and was

stopped after a short observation window. The terminal temperature reached 45.8°C within 4 minutes even though the final device temperature measurement was only 36.9°C. This difference shows that the NTC thermistor, as placed in the prototype, could under-read the contact-region temperature during severe degraded-contact heating.

Table 4.5 summarizes the characterization data used to interpret these trends.

Table 4.5. Summary of socket-heating characterization data

Load case	Socket	Time to 38.2°C	Mean A	Final device temp.	Final IR temp.	Window (min)
Water heater	Normal	–	5.18	33.9	31.5	240
Water heater	Corroded	–	5.16	35.6	37.4	240
Water heater + 1 lamp	Normal	–	7.12	37.7	37.6	240
Water heater + 1 lamp	Corroded	95	7.10	40.4	43.2	240
Water heater + 2 lamps	Normal	105	9.07	40.2	40.8	240
Water heater + 2 lamps	Corroded	48	9.05	41.0	44.5	120
Water heater + 3 lamps	Normal	52	11.06	43.2	45.8	240
Water heater + 3 lamps	Corroded	7	11.15	38.8	53.0	8
Water heater + 4 lamps	Normal	44	13.00	43.4	46.4	195
Water heater + 4 lamps	Corroded	–	13.10	36.9	45.8	4

Table 4.5 confirms that degraded contact changed both heating speed and hotspot severity. At about 9A, the corroded socket reached 38.2°C in 48 minutes compared with 105 minutes for the normal socket. At about 11A, the corroded socket reached the same point in 7 minutes compared with 52 minutes for the normal socket. These differences show that similar current levels did not produce identical thermal behavior when the socket condition changed. This supports the reviewed finding that poor contact condition increases localized heating risk through contact resistance and thermal concentration [17], [24], [46], [47].

The table also shows why the compensated thermal model was needed after characterization. In the higher-temperature and degraded-contact cases, the infrared terminal reading was consistently higher than the device temperature measurement. For example, in the water heater plus three-lamp corroded case, the device temperature measurement was 38.8°C, while the infrared terminal reading reached 53.0°C. In the water heater plus four-lamp corroded case, the device temperature measurement was 36.9°C, while the infrared terminal reading reached 45.8°C. These observations demonstrate that the flat NTC thermistor measurement alone did not fully represent the hotter contact region. This is consistent

with the reviewed literature explaining that external temperature sensing can under-read an internal hotspot because of thermal separation, sensor placement, and response delay [27]-[31].

These characterization findings verify why socket-level thermal monitoring was included in the prototype. The flat NTC measurements were able to show the heating trend of each socket, but they did not fully represent the hotter contact-region temperature. This supports the use of a compensated protection-facing temperature estimate, since localized socket heating may develop in a region that conventional current-based protection and external temperature measurement do not directly observe.

4.4 Socket-Heating Protection Validation

Socket-heating protection validation evaluated the relay-facing estimated socket temperature under the same general load progression used during characterization. This validation was necessary because the socket-heating function does not trip from the raw thermistor value alone; instead, the firmware uses the compensated estimated socket temperature described in *Section 3.6* and evaluated by the protection logic in *Section 3.8*. The methodology compared the device temperature measurement, infrared thermometer reading, estimated socket temperature, and expected normal socket temperature: the thermistor channel provided the device temperature, the infrared thermometer served as the contact-region reference, the estimated socket temperature served as the protection-facing value, and the expected normal socket temperature served as the estimator's internal comparator.

This distinction is important because poor plug-socket contact can create localized heating even when total current does not necessarily force conventional protection to operate [17], [24], [46], [47]. It is also consistent with the reviewed literature showing that an external temperature sensor can lag or under-read the true internal hotspot because the hottest point is usually inside the contact interface [27]-[31]. For this reason, the trip threshold was applied to the estimated socket temperature, while the infrared thermometer reading was used to interpret the contact-region temperature during the same trial. The plotted trends in this section were generated from the full protection validation records in *Appendix D*.

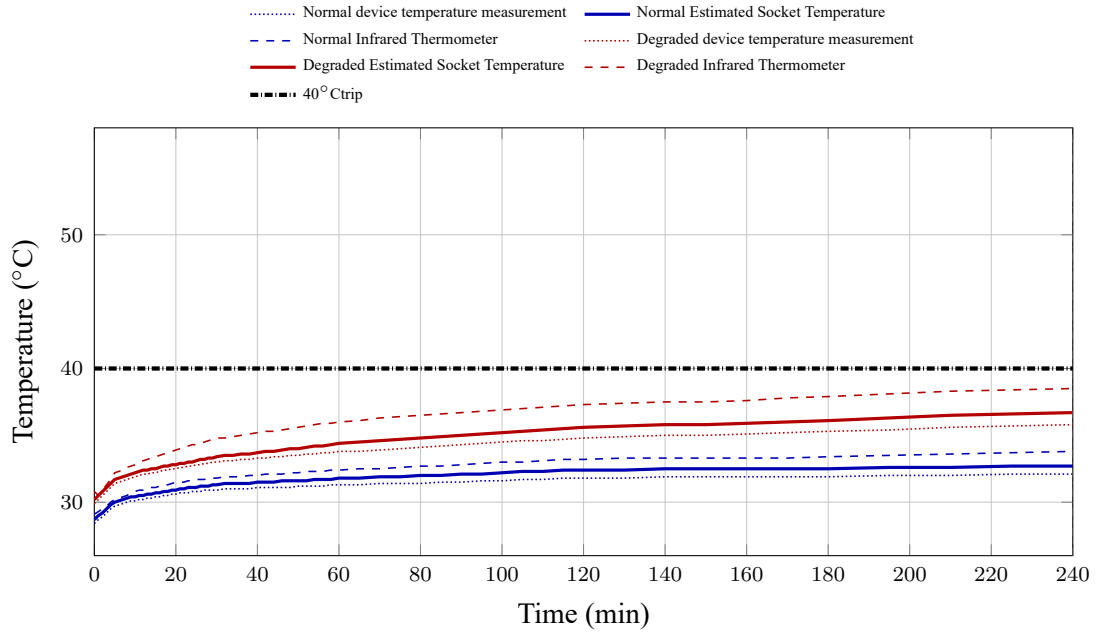


Figure 4.6. Water heater temperature trend during socket-heating protection validation.

Figure 4.6 shows that the low-current case did not trip for either socket condition within the observation window. Although the corroded terminal ended hotter than the normal terminal, the estimated socket temperature remained below the 40°C trip threshold.

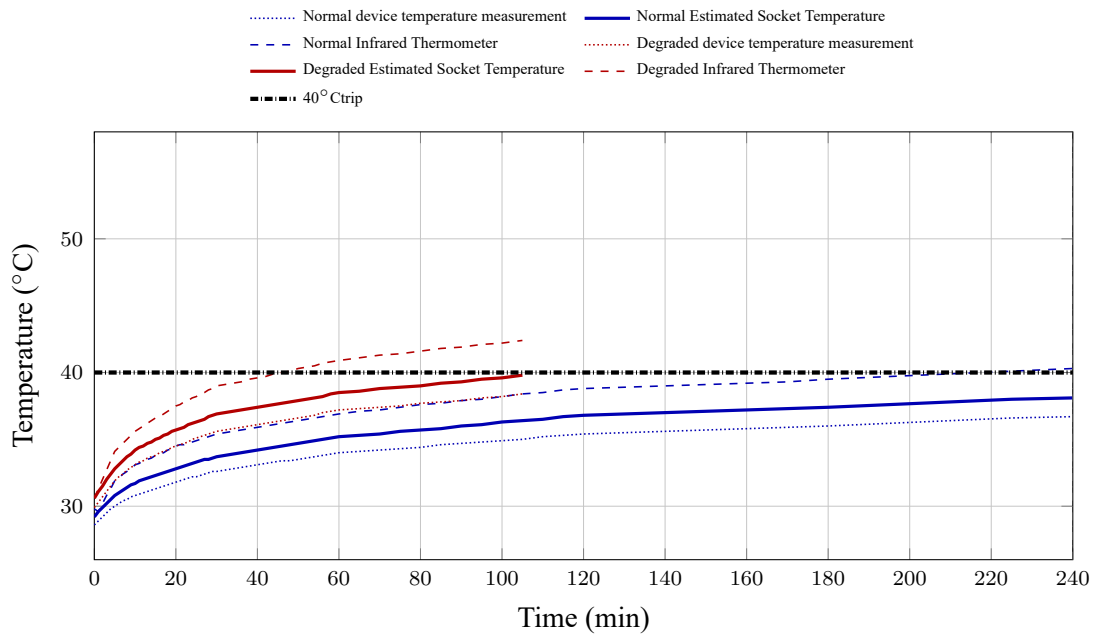


Figure 4.7. Water heater plus one halogen lamp trend during protection validation.

Figure 4.7 shows that the normal socket did not trip, while the corroded socket tripped after reaching the estimated 40°C threshold. This indicates that the relay response

followed the compensated thermal estimate rather than current magnitude alone.

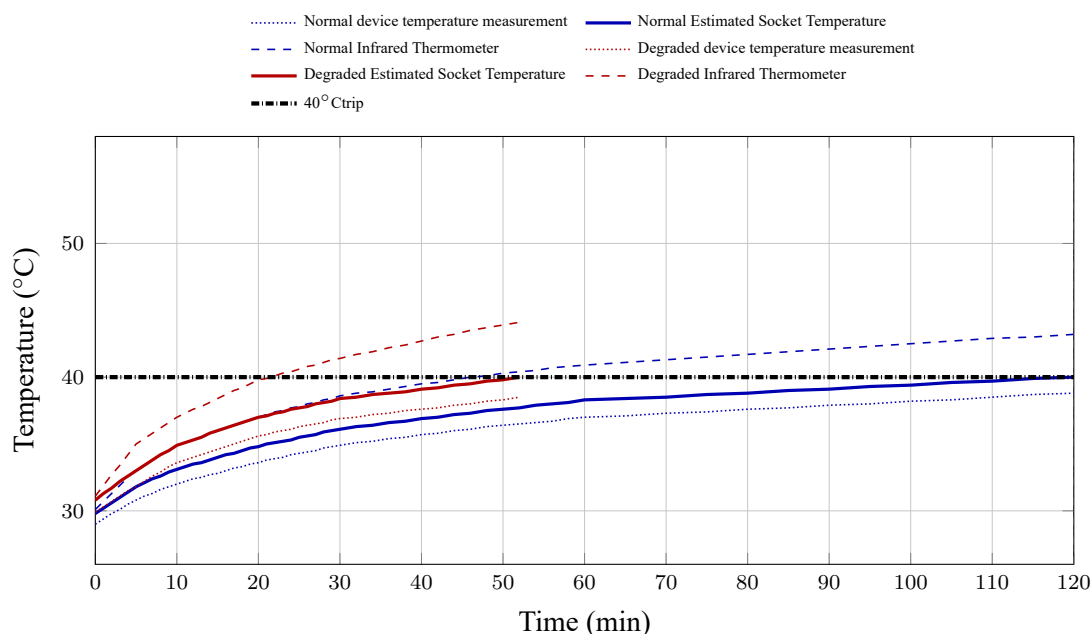


Figure 4.8. Water heater plus two halogen lamps trend during protection validation.

Figure 4.8 shows that higher current increased socket heating for both socket conditions, but the corroded socket tripped earlier because the degraded contact produced additional localized heating beyond the current effect. This supports the interpretation that contact condition influenced the trip time, not current magnitude alone.

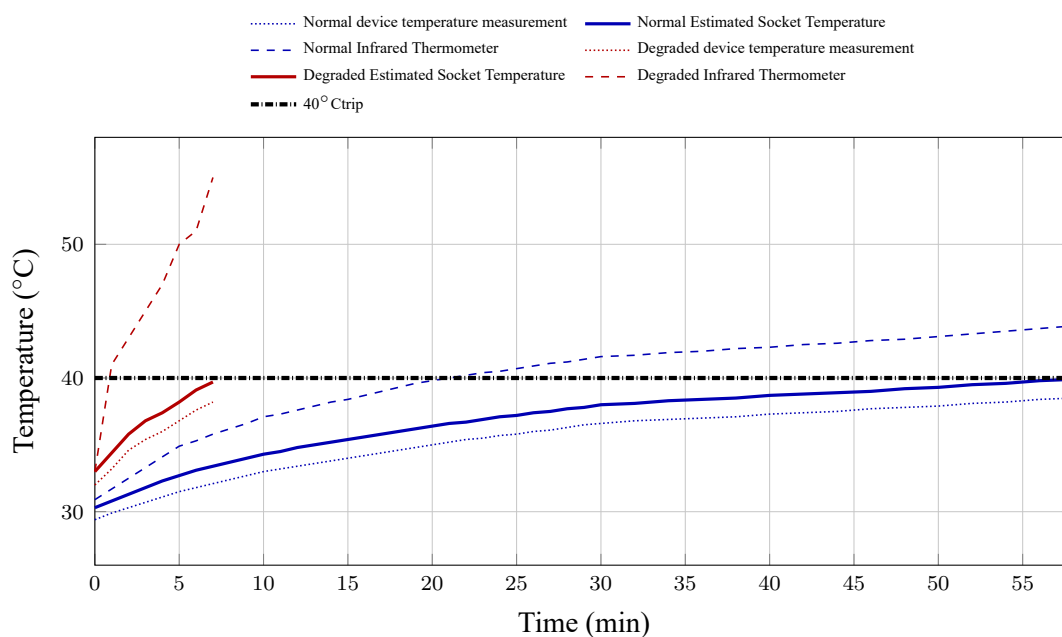


Figure 4.9. Water heater plus three halogen lamps trend during protection validation.

Figure 4.9 shows fast thermal development in the corroded case. At this load level, the degraded socket reached a hazardous contact-region temperature within a short observation window, with the reference terminal temperature reaching 55.0°C. This result shows that the poor contact condition accelerated the thermal rise beyond what would be expected from current level alone. It also supports the use of the estimated socket temperature as a conservative relay-facing value, since the external measurement path may still represent a lower and delayed view of the actual contact-region heating [27]-[31].

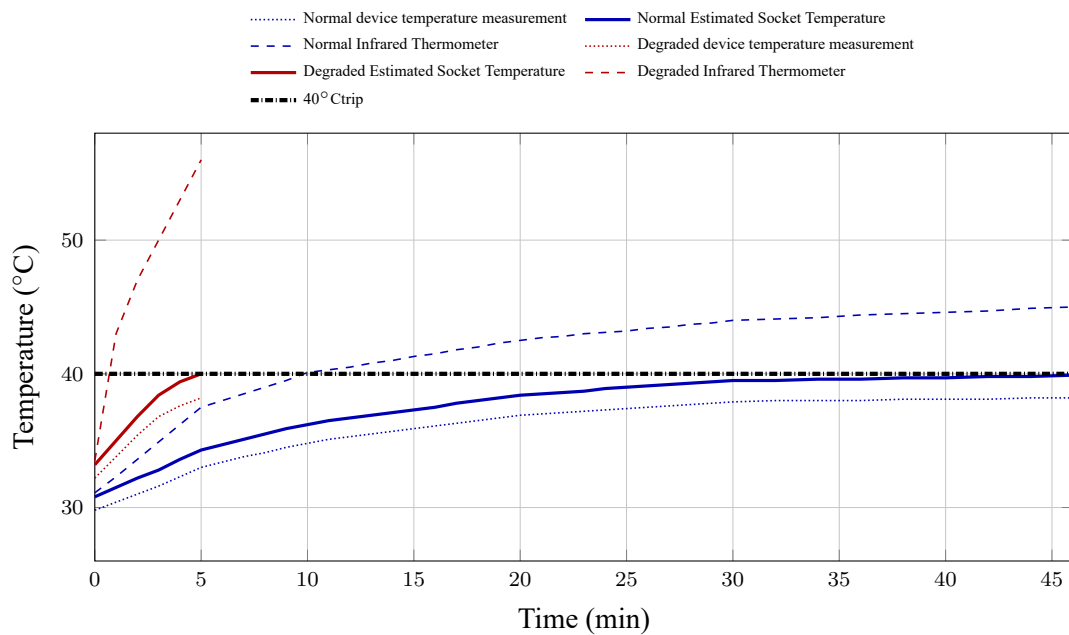


Figure 4.10. Water heater plus four halogen lamps trend during protection validation.

Figure 4.10 shows the most severe heating case in the protection validation set. The corroded socket tripped after 5min when the estimated socket temperature reached 40.0°C, while the reference terminal temperature had already reached 56.0°C. Compared with the lower-current trials, this case shows that high current increased the thermal stress, but the degraded contact made the heating rise much faster. This is consistent with the reviewed literature showing that poor plug-socket contact can create localized heating that is not fully represented by external temperature measurement or current magnitude alone [17], [24], [27]-[31], [46], [47].

Table 4.6 summarizes the protection validation results.

Table 4.6. Socket-heating protection validation results

Load case	Socket	Mean A	Device 38.2	Est. 40	Est. temp	IR temp	Relay
Water heater	Normal	5.17	–	–	32.7	33.8	No
Water heater	Corroded	4.85	–	–	36.7	38.5	No
Water heater + 1 lamp	Normal	7.11	–	–	38.1	40.3	No
Water heater + 1 lamp	Corroded	7.10	100	–	39.8	42.4	Yes
Water heater + 2 lamps	Normal	9.06	100	120	40.0	43.2	Yes
Water heater + 2 lamps	Corroded	9.04	48	52	40.0	44.1	Yes
Water heater + 3 lamps	Normal	11.07	54	–	39.9	43.9	Yes
Water heater + 3 lamps	Corroded	11.14	7	–	39.7	55.0	Yes
Water heater + 4 lamps	Normal	13.02	44	–	39.9	45.0	Yes
Water heater + 4 lamps	Corroded	13.08	5	5	40.0	56.0	Yes

Table 4.6 summarizes how the relay response changed with socket condition and load level. At low current, the corroded socket did not trip because the estimated socket temperature remained below the 40°C threshold. As the load current increased, the corroded socket tripped earlier than the normal socket because degraded contact added localized heating at the plug-socket interface. This is consistent with the reviewed literature showing that poor contact pressure, oxidation, and degraded contact condition can increase local thermal stress without requiring a classical overcurrent event [17], [24], [46], [47].

The table also shows why the relay-facing estimate must be interpreted as an indirect protection value. In the severe degraded-contact cases, the reference terminal temperature was higher than both the device temperature measurement and the estimated socket temperature. This follows the reviewed literature showing that external temperature sensing can under-read a hidden contact hotspot due to sensor placement, thermal separation, and response delay [27]-[31]. Thus, the 40°C estimated-temperature threshold served as a conservative socket-heating trip point for the prototype, not as the actual internal contact temperature.

These results support the study’s objective of detecting contact heating at the plug-socket interface through complementary socket-level protection. The relay action was not based on RMS current alone, and the results show that similar current levels can lead to different thermal behavior when the socket condition changes. This confirms the need for local thermal supervision, since localized contact heating may develop before conventional current-based protection can directly observe the fault.

4.5 Series Arc-Fault Detection

The series arc-fault tests evaluated whether waveform disturbance and conduction intermittence could be converted into a relay decision. During arcing, the current waveform was expected to show low-current gaps, restrike spikes, unstable cycle shape, and changes in mid/high-frequency content. This follows the reviewed literature showing that series arcs may remain below conventional breaker trip levels while still producing waveform abnormalities that require multi-feature interpretation rather than RMS-current magnitude alone [16], [33]-[39], [49]-[54]. These behaviors correspond to the features defined in *Section 3.7*, especially `max_low_current_run_ms`, `low_current_ratio`, `zero_dwell_ratio`, `midband_residual_ratio`, and `spectral_flux_midhf`.

The trial-level results are summarized as confusion matrices in *Table 4.7*.

Table 4.7. Trial-level confusion matrices for series arc-fault testing

Device group	Prototype response	Arcing event	Normal event
Recognized loads	Relay trips	49	0
Recognized loads	Relay remains on	1	50
Unrecognized loads	Relay trips	47	1
Unrecognized loads	Relay remains on	3	49
All tested loads	Relay trips	96	1
All tested loads	Relay remains on	4	99

Table 4.7 shows that the prototype detected 96 of 100 arc trials and produced one relay trip during 100 normal trials. The recognized-load group had no normal-event false trip and one missed arc event, while the unrecognized-load group had three missed arc

events and one false trip. This difference is consistent with the reviewed literature showing that healthy nonlinear loads can produce waveform distortion, conduction gaps, or spectral behavior that may resemble fault behavior when load context is not fully represented [8], [33], [40]-[42]. Thus, the unrecognized-load results show the practical importance of context-aware inference in reducing nuisance operation while still preserving sensitivity to arc-related waveform changes.

The result supports the arc-detection objective under the tested laboratory conditions. The classifier response was strongest when the arc created persistent low-current gaps and spectral disturbance, allowing the leaky arc counter to reach the relay-trip threshold. The missed trials indicate that not all arcs produced equally strong or persistent feature responses, especially when the load type differed from the fitted set. This interpretation is consistent with the reviewed finding that series-arc detection is stochastic and load-dependent, and that reliable detection depends on combining several time-domain and frequency-domain indicators rather than treating one feature as conclusive evidence [33]-[39], [49]-[54]. Therefore, the prototype is best interpreted as a controlled feasibility demonstration of socket-level complementary arc-fault protection, not as a certified AFCI-equivalent device.

4.6 Arc-Fault Dataset Summary

The cleaned arc-training dataset contained 49,296 rows, of which 283 were arc markers, equivalent to about 0.577% of the retained data. This confirms that the arc fault dataset was highly imbalanced: most frames represented normal operation even in sessions that contained an arc event. The imbalance explains why frame-level accuracy alone can be misleading, since a classifier can appear accurate by predicting mostly normal frames. Precision, recall, F1-score, confusion matrices, and event-level metrics were therefore used to interpret the safety behavior of the prototype. This follows the methodology in *Section 3.4*, where the arc model was evaluated not only by row-level classification but also by its ability to support relay decisions during grouped arc episodes.

The detailed trial records are presented in *Appendices E and F*. These records ex-

tend the summarized results by showing the arc markers, feature-frame behavior, and relay-state transitions used to verify the trial outcomes.

4.7 TinyML Model Performance

The TinyML results were interpreted at two levels. First, the embedded prototype was evaluated through trial-level relay behavior in the laboratory. Second, the exported Random Forest report was evaluated through held-out frame-level and event-level metrics. This two-level interpretation is necessary because the relay decision represents the complete protection sequence, while the frame-level metrics show the behavior of the model before persistence counting and relay actuation are applied. This follows the methodology in *Section 3.7.1*, where the deployed arc model was defined as a feature-based TinyML classifier whose output is further processed by firmware protection logic.

This distinction is important because series arc-fault detection in the prototype was not based on a single instantaneous model output. The Random Forest classifier identified arc-like waveform frames from the selected features, but the relay response still depended on the protection manager confirming repeated or persistent evidence before actuation. This arrangement connects the TinyML model to the study objective of socket-level complementary protection: the model provides compact embedded inference for waveform disturbance, while the firmware logic reduces the chance that isolated transient behavior will immediately become a relay trip. This is consistent with the reviewed literature showing that series arcs can be intermittent, load-dependent, and difficult to separate from normal nonlinear-load distortion using RMS current alone [33]-[39], [49]-[54].

Table 4.8 gives the trial-level metrics for recognized loads, unrecognized loads, and all loads combined.

Table 4.8. Trial-level arc-fault performance metrics

Load group	Accuracy	Precision	Recall	F1-score
Recognized loads	99.00%	100.00%	98.00%	98.99%
Unrecognized loads	96.00%	97.92%	94.00%	95.92%
All tested loads	97.50%	98.97%	96.00%	97.46%

Figure 4.11 graphs the same four metrics by load setup. The figure shows how performance changed from recognized loads to unrecognized loads while keeping the same metric definitions for accuracy, precision, recall, and F1-score.

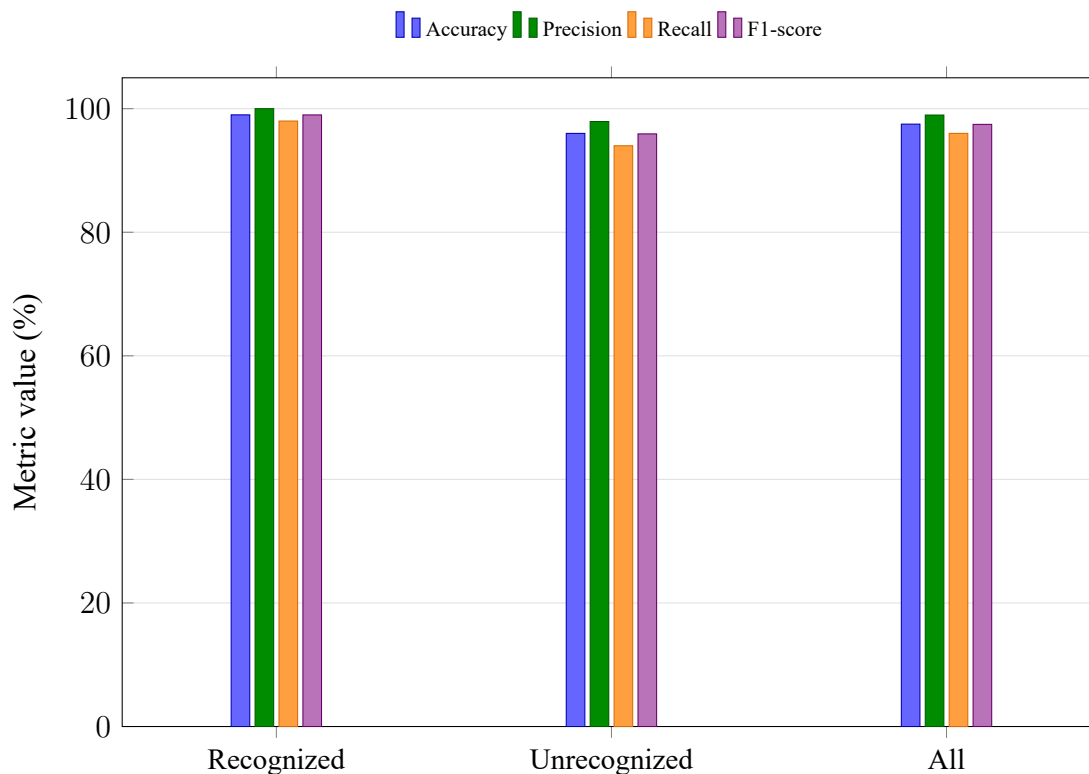


Figure 4.11. Trial-level performance metrics by recognized and unrecognized load setup.

Table 4.8 and Figure 4.11 show high overall performance across 200 trial outcomes. Precision is related to nuisance-trip reduction: the 98.97% overall precision means that only one normal event was incorrectly tripped in the trial set. Recall is related to safety sensitivity: the 96.00% overall recall means that four arc trials were missed. The recognized-load group had slightly higher recall and no normal-event false trip, while the unrecognized-load group accounted for three of the four missed arc trials and the single false trip. This is consistent with the reviewed literature showing that arc detection is load-dependent and that normal nonlinear loads can produce waveform behavior that complicates fault interpretation [8], [33], [40]-[42].

The trial-level classifier and persistence logic therefore produced balanced performance for the tested conditions. Precision remained high for nuisance-trip reduction, while recall remained high enough to detect most induced arc trials. These results apply to the de-

ployed feature contract described in *Section 3.7.1*: six selected waveform features were used by the arc model, with context fields appended to the model vector. This supports the design choice of treating series arc-fault detection as a multi-feature and context-aware inference problem rather than as a direct RMS-current threshold decision [33]-[39], [49]-[54].

The per-load trial metrics are shown in *Table 4.9*.

Table 4.9. Per-load trial-level performance metrics

Device	Device type	Accuracy	Precision	Recall	F1-score
Water heater	Resistive	100%	100%	100%	100%
Halogen lamp	Resistive	100%	100%	100%	100%
Electric kettle	Resistive	100%	100%	100%	100%
40W stand fan	Inductive	100%	100%	100%	100%
Handheld vacuum	Brushed	95%	100%	90%	94.74%
Electrolux vacuum	Brushed	95%	100%	90%	94.74%
Angle grinder	Brushed	95%	100%	90%	94.74%
Bench cutter	Brushed	95%	100%	90%	94.74%
Impact drill	Brushed	95%	90.91%	100%	95.24%
Heat gun	Resistive	100%	100%	100%	100%

Table 4.9 identifies the load cases that contributed to the remaining trial-level errors. The impact drill is the only load listed with precision below 100%, meaning it contributed the normal-event false trip. The handheld vacuum, Electrolux vacuum, angle grinder, and bench cutter had lower recall, meaning those load cases contributed missed arc trials. These results show that the residual errors were concentrated in brushed or universal-motor loads rather than distributed equally across all tested devices.

The trial records were also aggregated by load setup and load family to show where the errors occurred at a higher level. For load setup, recognized loads contributed 49 detected arc trials, one missed arc trial, 50 correct normal trials, and no false trip; unrecognized loads contributed 47 detected arc trials, three missed arc trials, 49 correct normal trials, and one false trip. For family grouping, water heater, halogen lamp, electric kettle, and heat gun were treated as resistive loads; the 40W stand fan was treated as the inductive load; and the handheld vacuum, Electrolux vacuum, angle grinder, bench cutter, and impact drill were treated as brushed or universal-motor loads. These aggregations summarize the recorded trial tables in *Appendix E* and clarify the distribution of the trial-level errors.

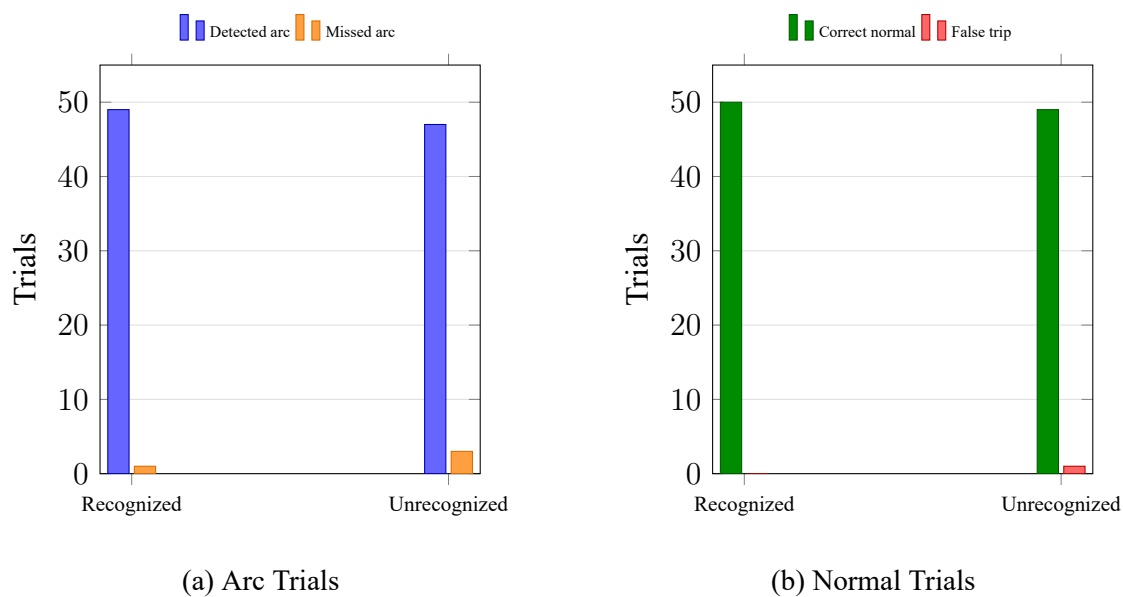


Figure 4.12. Grouped trial outcomes for recognized and unrecognized load setups.

Figure 4.12(a) shows that the unrecognized group had more missed arc trials than the recognized group. Figure 4.12(b) shows that the only normal-event false trip also came from the unrecognized group. These results are consistent with the per-load results in Table 4.9. The protection sequence still handled most generalization cases, but the remaining errors were associated with loads whose normal waveform behavior was more difficult to separate from arc-like disturbance.

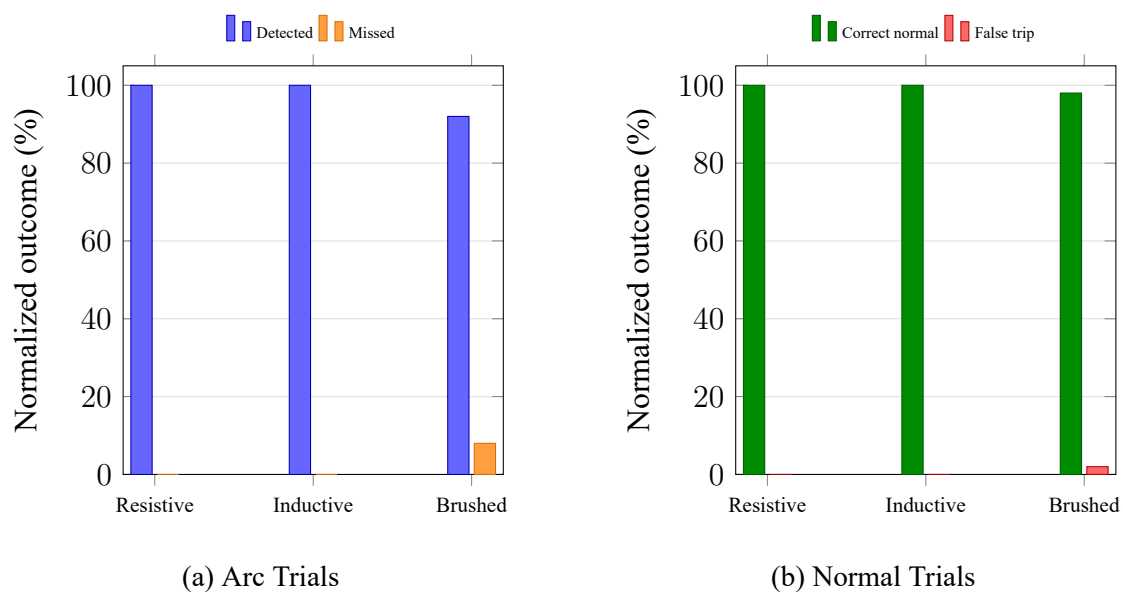


Figure 4.13. Normalized grouped trial outcomes by load family.

Figure 4.13 is normalized because the family groups did not contain the same num-

ber of devices or trials. In *Figure 4.13(a)*, resistive and inductive load-family trials had 100% arc detection, while the brushed-load family had 92% arc detection and 8% missed arc trials. In *Figure 4.13(b)*, the brushed-load family also contained the only false trip, giving 98% correct normal classification and 2% false-trip outcome for that family. The normalized view therefore shows that the brushed or universal-motor group was the main source of residual model error rather than simply the largest group in raw counts. This supports the reviewed finding that motor-type and nonlinear loads can produce commutation effects, waveform irregularity, and harmonic behavior that require context-aware interpretation [8], [40]-[42].

The exported Random Forest report provides a second view of performance. The held-out test split contained 10,268 feature rows, including 90 positive rows. The frame-level confusion matrix contained 10,173 true negatives, 5 false positives, 25 false negatives, and 65 true positives. The corresponding frame-level accuracy was 99.7078%, precision was 92.8571%, recall was 72.2222%, and F1-score was 81.25%. At event level, the same report showed 25 true arc events, 21 detected events, 4 missed events, 31 predicted events, 1 false-alarm event, 84.0% event recall, 96.7742% event precision, and 89.9358% event F1-score. The data-preparation, training-report, and exported-model details supporting these metrics are summarized in *Appendix F.2*.

The model-report values are lower than the final trial-level relay metrics because they evaluate individual frames and grouped arc events before the full firmware protection sequence is interpreted at trial level. This difference clarifies the role of the persistence counter: isolated model outputs are not treated as final relay decisions until they satisfy the firmware’s protection logic. The deployed arc-detection result should therefore be interpreted as a model-plus-firmware protection behavior, where the Random Forest identifies arc-like frames and the persistence logic converts repeated evidence into a relay action.

4.8 Model Limitations

The main limitation is dataset coverage. The tests included recognized and unrecognized loads, but they did not cover all appliance families, wiring conditions, electrode

materials, contact gaps, ambient conditions, or utility waveform variations. This limitation is consistent with the reviewed arc-fault literature, which identifies load dependence, limited data coverage, false alarms, and missed detections as recurring challenges in AI-based arc-fault detection [33]. The results therefore support controlled prototype feasibility within the tested scope, but they do not establish AFCI standard compliance or replacement equivalence with certified protective devices.

The limitation is also important for interpreting the unrecognized-load and brushed-load results. The prototype generalized to most tested cases, but the remaining missed arcs and the single false trip show that wider load-family coverage would be required before the model could be treated as broadly representative of residential appliance behavior. This follows the reviewed literature showing that healthy nonlinear loads can produce waveform disturbance that overlaps with arc-related indicators, making load context an important part of false-positive control [8], [40]-[42].

4.9 Prototype Feasibility Assessment

Taken together, the overload, socket-heating, and series arc-fault evaluations show that the TinyML Smart Plug achieved the intended hybrid protection behavior under controlled laboratory conditions. The calibrated voltage, current, and temperature measurements were sufficient to support the protection thresholds used in the prototype. The overload tests showed warning behavior followed by relay trip only when sustained overload persisted, which is consistent with the reviewed literature describing overload as a time-dependent current and thermal-stress condition [12]-[14], [22]. The socket-heating tests showed that degraded contact can produce faster and more severe heating than current magnitude alone suggests, supporting the reviewed finding that contact heating at the plug-socket interface can remain localized and hidden from conventional upstream protection [17], [18], [24], [30], [31]. The arc-fault tests showed that waveform features and Random Forest inference can support local relay action for series-arc events in the tested load set, following the reviewed literature showing that series arcs may remain below ordinary breaker trip levels while still producing waveform disturbance and conduction intermittence [16],

[33]-[39].

The feasibility of the prototype comes from matching each target fault condition to the form of evidence that best represents it. Overload was treated as a current-stress condition, so the protection response used threshold logic with persistence. Contact heating was treated as a localized thermal condition at the plug-socket interface, so the response used local temperature supervision and compensated socket-temperature estimation. Series arcing was treated as a load-dependent waveform condition, so the response used multi-feature interpretation with load-family context rather than current magnitude alone. This division follows the reviewed literature showing that overload, contact heating, and series arcing are related electrical-fire precursors, but they are not equivalent fault classes [34]-[39], [44]-[47]. The prototype therefore addressed the study objective by combining calibrated sensing, threshold-based protection, load-family recognition, TinyML-supported series-arc interpretation, and relay latching in one complementary socket-level protection device.

The prototype remains a complementary protection device rather than a replacement for circuit breakers, AFCIs, GFCIs, proper wiring practice, contact inspection, or certified protection devices. This interpretation is necessary because the evaluation was performed on a laboratory bench, the thermal estimator was intentionally conservative, and the TinyML model was trained and tested using a finite set of loads and induced arc conditions. Within that boundary, the results show that the proposed smart plug can provide earlier local warning, relay disconnection, and monitoring for selected progressive fire-inducing electrical faults before they develop into more severe electrical-fire events.

These findings support the main research objective by showing that the developed TinyML Smart Plug can integrate calibrated voltage, current, and temperature sensing with threshold-based protection, compensated socket-temperature estimation, TinyML-supported series-arc interpretation, relay actuation, local notification, and PWA monitoring. Through these combined functions, the system responded to overload, socket heating, and series arc-fault events in a manner consistent with its intended role as a complementary socket-level protection device for progressive fire-inducing electrical faults.

Chapter 5

CONCLUSION AND RECOMMENDATION

This chapter presents the conclusions and recommendations drawn from the development and evaluation of the TinyML Smart Plug. The conclusions summarize how the prototype addressed the study objectives through calibrated sensing, threshold-based overload and socket-heating protection, TinyML-supported series arc-fault detection, relay actuation, local notification, and PWA monitoring. The recommendations identify the remaining validation, dataset, thermal, hardware-safety, logging, and embedded-platform improvements needed before the prototype can be considered for residential application.

5.1 Conclusions

The study demonstrated that the TinyML Smart Plug can operate as a complementary socket-level protection prototype under controlled laboratory conditions. The calibrated voltage, current, and temperature channels produced errors low enough to support the implemented protection decisions, with the sensor-validation results reporting mean absolute errors of 3.910V, 0.178A, and 1.02°C, respectively. These results support the first objective because the prototype was able to acquire the required electrical and thermal quantities and use them in embedded decision logic for monitoring, warning, and relay control.

The overload tests showed that the prototype could distinguish warning-level current from sustained overload. All overload trials generated warnings, while only the trial with a 17.30A mean current produced a relay trip after 61.3s. This agrees with the implemented persistence method and supports the objective of adding time-aware overload response rather than using an instantaneous cutoff for every current excursion. This is consistent with the reviewed literature describing overload as a time-dependent current and thermal-stress condition, where the protection response must distinguish short transients from sustained heating risk [12]-[14], [22], [44], [45].

The socket-heating tests confirmed that degraded contact can create faster and more severe heating than a normal socket at comparable current levels. The estimated socket tem-

perature reached the 40°C trip threshold earlier in corroded-socket trials, and the reference terminal temperature was already higher than the device temperature reading at trip. This supports the need for a thermal protection layer that is separate from current-only protection. This follows the reviewed literature showing that poor plug-socket contact can produce localized and hidden heating even when the total current does not necessarily force conventional upstream protection to operate [17], [24], [46], [47]. It also supports the reviewed finding that external temperature measurement can under-read the hottest contact region because of sensor placement, thermal lag, and thermal separation from the internal hotspot [27]-[31].

The series arc-fault tests showed that embedded TinyML inference was practical on the implemented platform. Across the tested devices, the prototype achieved 97.50% accuracy, 98.97% precision, 96.00% recall, and a 97.46% F1-score. These results support the objective of detecting arc-related waveform disturbance using a feature-based embedded model. This is consistent with the reviewed literature showing that series arcs may remain below ordinary breaker trip levels while still producing conduction intermittence, waveform disturbance, and spectral changes that require multi-feature interpretation rather than RMS current magnitude alone [16], [33]-[39], [49]-[54]. The use of load context also supports the reviewed finding that healthy nonlinear loads can produce distortion that overlaps with arc-like behavior, making context-aware inference important for reducing nuisance operation [8], [40]-[43].

The prototype is therefore best interpreted as a complementary protection device for selected progressive fire-inducing fault conditions. It was able to combine calibrated voltage, current, and temperature sensing with deterministic overload and socket-heating protection, TinyML-supported series arc-fault interpretation, relay actuation, notification, event logging, and PWA monitoring in one working prototype. However, the results must be interpreted within the tested load set, induced-fault procedures, and controlled laboratory conditions only. The prototype is not a replacement for circuit breakers, AFCIs, GFCIs, proper wiring practice, certified receptacles, inspection, or other mandatory protection devices.

5.2 Recommendations

Longer validation should be performed in residentially realistic or semi-field conditions. Additional tests should include ordinary daily load transitions, shared branch circuits, extension cords, aged receptacles, different plug-fit conditions, and different wiring lengths.

The arc-fault dataset should be expanded to include more load families, additional appliance models, more arc severities, and non-fault disturbances such as motor startup, switching noise, loose plugs without confirmed arcing, and intermittent contact. This would give the Random Forest model stronger coverage of nuisance events and would help improve recall for less obvious arc-fault cases.

The socket-temperature subsystem should be improved through better thermal sensor placement and additional model validation. Subsequent prototypes should evaluate thermistors placed closer to the plug-contact interface, multiple temperature sensors, and longer degraded-contact trials so that the relationship among device temperature measurement, estimated socket temperature, and actual terminal temperature can be fitted with stronger evidence.

The enclosure, creepage and clearance distances, insulation barriers, relay rating, wire gauge, strain relief, and thermal isolation should be redesigned for certification-oriented testing. Additional tests should be aligned with applicable electrical safety and appliance standards before any field or consumer deployment is considered.

The monitoring system should add stronger event logging and data export. Exportable CSV logs, synchronized timestamps across waveform and telemetry events, firmware-version fields, and clearer fault-state records would make later validation easier and would support model retraining from collected test data.

Finally, later implementations may use a microcontroller platform with more memory and data-handling margin. Additional resources would allow longer waveform buffers, larger validation logs, broader feature comparison, and more robust embedded inference while keeping the protection logic responsive.

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Appendix A

SENSOR CALIBRATION

This appendix presents the raw calibration datasets used to derive the correction equations for the voltage, current, and temperature sensing channels. The data support the calibration procedures discussed in *Chapter 3* and the validation results presented in *Chapter 4*. The tables are retained in the appendix to show the measured sensor values against the corresponding reference readings used during calibration.

Table A.1. Voltage Sensor Calibration Data

ZMPT101B (V)	Fluke Multimeter (V)	ZMPT101B (V)	Fluke Multimeter (V)
34.1	35.2	179.1	180.5
43.9	44.79	189.0	190.0
58.6	60.1	199.2	200.4
73.0	74.5	204.7	205.3
83.3	85.0	210.7	210.0
99.3	100.2	214.2	214.8
113.6	115.1	220.0	220.2
129.0	130.0	225.3	225.1
139.6	140.3	228.4	230.8
149.0	150.0	235.3	235.1
159.1	160.0	238.8	240.7
169.2	169.9	242.9	244.6

The voltage calibration data show the uncorrected ZMPT101B readings and the corresponding reference readings from the Fluke multimeter. These values were used to generate the voltage correction equation applied in the firmware.

Table A.2. Current Sensor Calibration Data

ACS37035 (A)	Ingco Clamp Meter (A)	ACS37035 (A)	Ingco Clamp Meter (A)
0.08	0.08	3.28	4.09
0.15	0.15	3.31	4.15
0.17	0.18	3.86	4.70
0.19	0.20	3.97	4.90
0.21	0.22	5.89	7.27
0.53	0.64	4.62	5.72
0.56	0.66	4.84	6.11
0.58	0.69	4.95	6.22
0.63	0.77	5.04	6.33
0.73	0.86	5.11	6.46
0.76	0.88	5.33	6.60
0.70	0.89	5.36	6.76
0.76	1.02	5.41	6.88
1.55	1.90	6.51	8.12
1.58	1.91	6.79	8.35
1.57	1.94	6.85	8.48
1.76	2.18	7.07	8.80
1.77	2.18	8.24	10.32
1.96	2.32	8.35	10.46
2.02	2.47	8.41	10.47
2.20	2.77	9.40	12.20
3.12	3.82	10.30	13.30
3.11	3.87	12.40	16.00

The current calibration data compare the ACS37035-derived current values with the reference clamp-meter readings. These points were used to fit the current correction equation used by the overload logic, socket-heating estimator, and waveform-feature pipeline.

Table A.3. Reference Temperature and Thermistor Resistance Calibration Data

Reference Reading (°C)	Absolute Temperature (K)	Measured Thermistor Resistance (Ω)	ln(R)
38	311.15 K	5.8k	8.66072
50	323.15 K	3.6k	8.18869
100	373.15 K	0.7k	6.55108

The temperature calibration data were used to support the thermistor conversion discussed in *Chapter 3*. Since the thermistor measures the temperature at its own placement point, these calibration values validate the sensor channel and do not directly represent the hidden contact-spot temperature inside the socket interface.

Appendix B BOARD LAYOUT

This appendix presents the board-layout views used to document the internal hardware implementation of the prototype.

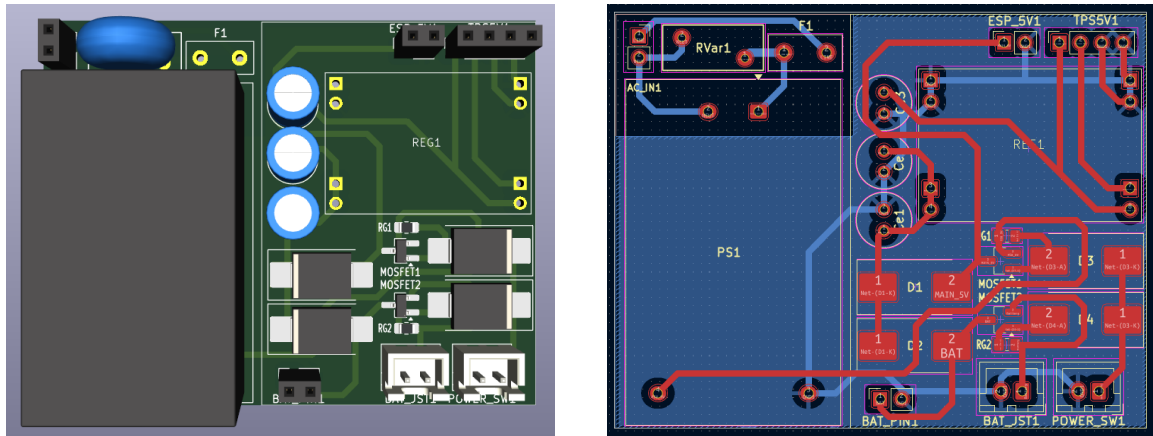


Figure B.1. PCB layout for the power-supply section.

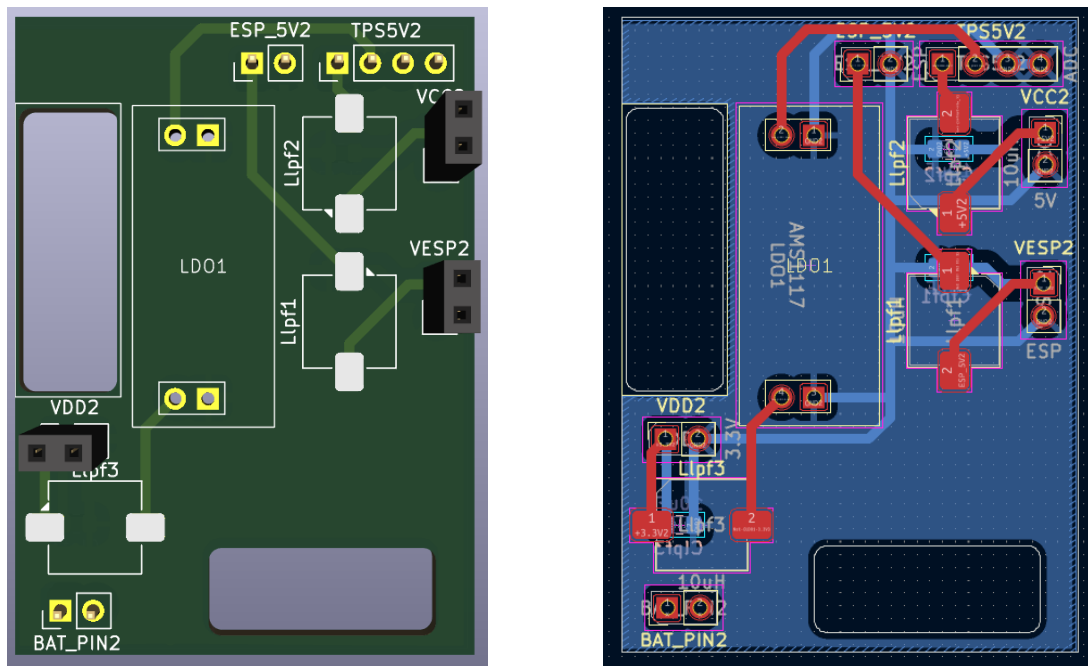


Figure B.2. PCB layout for the voltage-regulator section.

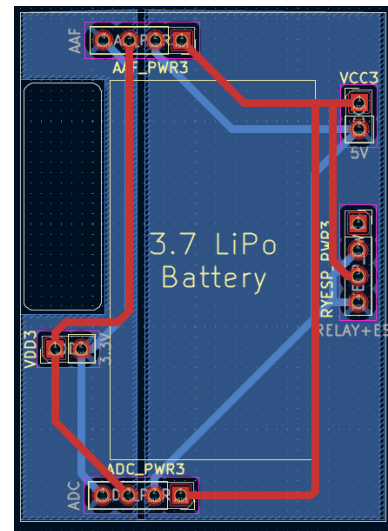
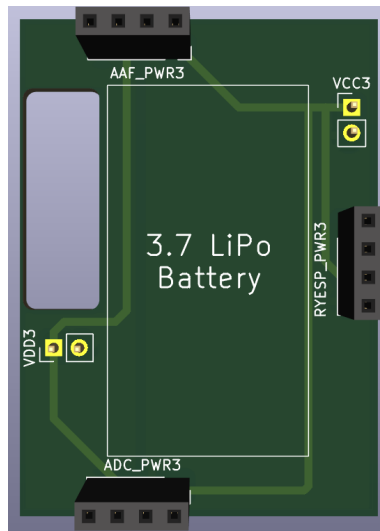


Figure B.3. PCB layout for the battery and power-distribution section.

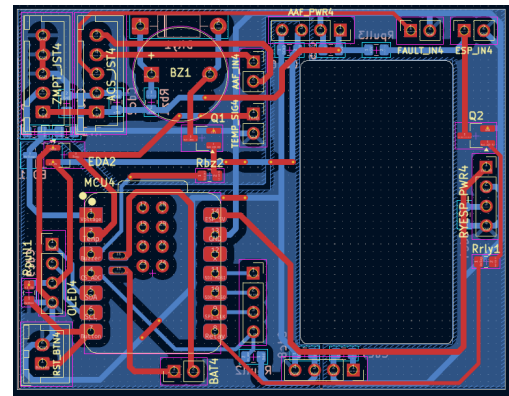
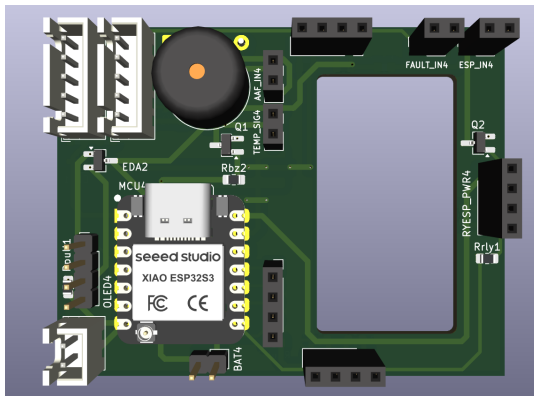


Figure B.4. PCB layout for the processor section.

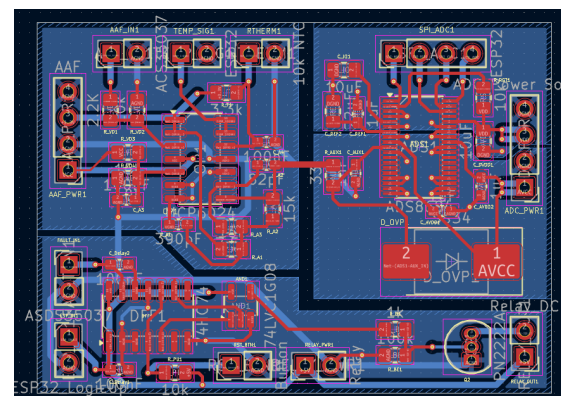
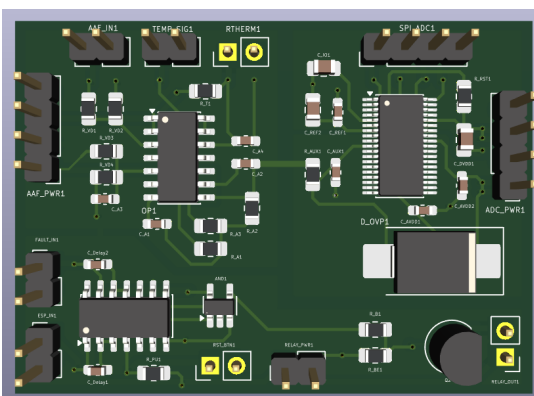


Figure B.5. PCB layout for the signal conditioner and relay latch.

Appendix C

SOCKET HEATING CHARACTERIZATION

This appendix contains the full socket-heating characterization tables for the normal and degraded socket conditions discussed in Chapter 4.

C.1 Water Heater

Table C.1. Characterization Data of Water Heater Load in Normal Socket

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
0 (Ambient)	30.8	27.9	5.37	5.25
1min	30.9	27.9	5.37	5.25
2min	30.9	28	5.37	5.25
3min	31	28	5.37	5.25
4min	31	28.1	5.37	5.26
5min	31	28.1	5.37	5.26
6min	31.1	28.2	5.37	5.26
7min	31.1	28.2	5.38	5.26
8min	31.2	28.3	5.38	5.26
9min	31.2	28.3	5.38	5.26
10min	31.3	28.4	5.38	5.26
11min	31.3	28.5	5.38	5.26
12min	31.4	28.5	5.38	5.26
13min	31.4	28.6	5.38	5.26
14min	31.4	28.6	5.38	5.26
15min	31.5	28.7	5.37	5.26
16min	31.5	28.7	5.37	5.26
17min	31.6	28.8	5.37	5.25
18min	31.6	28.8	5.37	5.25
19min	31.7	28.9	5.37	5.25
20min	31.7	28.9	5.36	5.24
21min	31.8	29	5.36	5.24
22min	31.8	29	5.36	5.23
23min	31.8	29.1	5.35	5.23
24min	31.9	29.1	5.35	5.22
25min	31.9	29.2	5.34	5.22
26min	32	29.2	5.34	5.21
27min	32	29.3	5.33	5.21
28min	32.1	29.3	5.33	5.20
29min	32.1	29.4	5.32	5.19
30min	32.1	29.4	5.32	5.19

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
32min	32.2	29.5	5.30	5.17
34min	32.2	29.5	5.29	5.16
36min	32.3	29.5	5.28	5.14
38min	32.3	29.6	5.27	5.13
40min	32.4	29.6	5.26	5.12
42min	32.4	29.6	5.25	5.11
44min	32.4	29.7	5.24	5.10
46min	32.5	29.7	5.24	5.09
48min	32.5	29.7	5.23	5.08
50min	32.6	29.8	5.23	5.08
52min	32.6	29.8	5.23	5.08
54min	32.7	29.8	5.23	5.07
56min	32.7	29.9	5.23	5.08
58min	32.7	29.9	5.23	5.08
60min	32.8	29.9	5.24	5.08
65min	32.8	30	5.26	5.10
70min	32.9	30.1	5.27	5.13
75min	32.9	30.1	5.29	5.15
80min	33	30.2	5.30	5.16
85min	33	30.3	5.30	5.17
90min	33.1	30.4	5.29	5.17
95min	33.1	30.5	5.28	5.16
100min	33.2	30.5	5.26	5.14
105min	33.2	30.6	5.24	5.12
110min	33.3	30.7	5.22	5.10
115min	33.4	30.8	5.21	5.09
120min	33.4	30.8	5.21	5.09
130min	33.5	30.9	5.25	5.11
140min	33.5	30.9	5.30	5.15
150min	33.5	30.9	5.34	5.18
160min	33.6	31	5.34	5.19
170min	33.6	31	5.30	5.16
180min	33.6	31.1	5.26	5.14
195min	33.7	31.2	5.26	5.14
210min	33.8	31.3	5.31	5.18
225min	33.9	31.4	5.32	5.18
240min	33.9	31.5	5.25	5.10

Table C.2. Characterization Data of Water Heater Load in Corroded Socket

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
0 (Ambient)	30.6	27.7	5.40	5.28
1min	30.8	28	5.40	5.28
2min	30.9	28.3	5.40	5.28
3min	31	28.6	5.39	5.28
4min	31.1	28.9	5.39	5.27
5min	31.2	29.3	5.39	5.27
6min	31.4	29.6	5.38	5.26
7min	31.5	29.9	5.38	5.26
8min	31.6	30.2	5.38	5.26
9min	31.7	30.5	5.37	5.25
10min	31.8	30.8	5.37	5.25
11min	31.9	30.9	5.37	5.24
12min	32	31	5.36	5.24
13min	32	31.1	5.36	5.23
14min	32.1	31.2	5.35	5.22
15min	32.1	31.3	5.35	5.22
16min	32.2	31.5	5.34	5.21
17min	32.3	31.6	5.34	5.20
18min	32.3	31.7	5.33	5.20
19min	32.4	31.8	5.32	5.19
20min	32.4	31.9	5.32	5.18
21min	32.5	32	5.31	5.18
22min	32.5	32.1	5.31	5.17
23min	32.6	32.2	5.30	5.16
24min	32.7	32.3	5.29	5.15
25min	32.7	32.4	5.29	5.15
26min	32.8	32.5	5.28	5.14
27min	32.8	32.6	5.28	5.13
28min	32.9	32.7	5.27	5.13
29min	33	32.8	5.27	5.12
30min	33	32.9	5.26	5.11
32min	33.1	33	5.25	5.10
34min	33.1	33.1	5.24	5.09
36min	33.2	33.2	5.24	5.08
38min	33.3	33.3	5.23	5.08
40min	33.3	33.4	5.23	5.07
42min	33.4	33.5	5.23	5.07
44min	33.5	33.6	5.23	5.07
46min	33.5	33.7	5.23	5.08
48min	33.6	33.9	5.23	5.08
50min	33.6	34	5.24	5.09
52min	33.7	34.1	5.24	5.09
54min	33.8	34.2	5.25	5.10
56min	33.8	34.3	5.26	5.11

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
58min	33.9	34.4	5.27	5.12
60min	34	34.5	5.27	5.13
65min	34	34.7	5.29	5.15
70min	34.1	34.8	5.30	5.17
75min	34.2	35	5.30	5.18
80min	34.2	35.1	5.30	5.18
85min	34.3	35.3	5.28	5.16
90min	34.4	35.4	5.26	5.15
95min	34.5	35.6	5.24	5.13
100min	34.5	35.7	5.23	5.11
105min	34.6	35.8	5.22	5.09
110min	34.7	36	5.22	5.09
115min	34.7	36.1	5.23	5.09
120min	34.8	36.2	5.25	5.11
130min	34.9	36.3	5.30	5.15
140min	34.9	36.4	5.34	5.19
150min	35	36.4	5.34	5.19
160min	35	36.6	5.31	5.17
170min	35.1	36.7	5.26	5.14
180min	35.2	36.8	5.25	5.13
195min	35.3	37	5.29	5.17
210min	35.4	37.2	5.32	5.18
225min	35.5	37.3	5.28	5.12
240min	35.6	37.4	5.20	5.05

C.2 Water Heater + 1 Halogen Lamp

Table C.3. Characterization Data of Water Heater + 1 Halogen Load in Normal Socket

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
0 (Ambient)	30.8	27.9	7.33	7.21
1min	31	28.1	7.33	7.21
2min	31.1	28.3	7.33	7.21
3min	31.3	28.5	7.33	7.21
4min	31.4	28.7	7.33	7.21
5min	31.5	28.9	7.33	7.21
6min	31.7	29.1	7.33	7.21
7min	31.8	29.3	7.33	7.21
8min	32	29.5	7.33	7.21
9min	32.1	29.7	7.33	7.21

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
10min	32.2	29.9	7.33	7.21
11min	32.3	30	7.33	7.21
12min	32.4	30.1	7.33	7.21
13min	32.5	30.2	7.33	7.21
14min	32.6	30.3	7.32	7.20
15min	32.6	30.4	7.32	7.20
16min	32.7	30.5	7.32	7.20
17min	32.8	30.6	7.32	7.19
18min	32.9	30.7	7.31	7.19
19min	33	30.8	7.31	7.19
20min	33	30.9	7.31	7.18
21min	33.1	31.1	7.30	7.18
22min	33.2	31.2	7.30	7.17
23min	33.3	31.3	7.29	7.16
24min	33.4	31.4	7.29	7.16
25min	33.4	31.5	7.28	7.15
26min	33.5	31.6	7.28	7.14
27min	33.6	31.7	7.27	7.14
28min	33.7	31.8	7.26	7.13
29min	33.8	31.9	7.26	7.12
30min	33.8	32	7.25	7.12
32min	33.9	32.2	7.24	7.10
34min	34	32.3	7.23	7.09
36min	34.1	32.4	7.22	7.07
38min	34.2	32.6	7.21	7.06
40min	34.3	32.7	7.20	7.05
42min	34.4	32.9	7.19	7.04
44min	34.5	33	7.19	7.03
46min	34.6	33.1	7.18	7.03
48min	34.6	33.3	7.18	7.02
50min	34.7	33.4	7.18	7.02
52min	34.8	33.6	7.18	7.02
54min	34.9	33.7	7.18	7.02
56min	35	33.8	7.18	7.03
58min	35.1	34	7.19	7.03
60min	35.2	34.1	7.19	7.04
65min	35.3	34.3	7.21	7.06
70min	35.4	34.4	7.23	7.08
75min	35.5	34.6	7.24	7.10
80min	35.6	34.8	7.25	7.12
85min	35.8	34.9	7.25	7.12
90min	35.9	35.1	7.24	7.11
95min	36	35.2	7.22	7.10
100min	36.1	35.4	7.20	7.08

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
105min	36.2	35.6	7.18	7.06
110min	36.4	35.7	7.17	7.04
115min	36.5	35.9	7.16	7.04
120min	36.6	36	7.17	7.04
130min	36.7	36.1	7.21	7.06
140min	36.8	36.3	7.26	7.10
150min	36.9	36.4	7.29	7.13
160min	36.9	36.5	7.28	7.13
170min	37	36.6	7.24	7.10
180min	37.1	36.8	7.21	7.08
195min	37.3	37	7.21	7.09
210min	37.4	37.3	7.26	7.13
225min	37.6	37.5	7.26	7.11
240min	37.7	37.6	7.19	7.03

Table C.4. Characterization Data of Water Heater + 1 Halogen Load in Corroded Socket

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
0 (Ambient)	30.6	27.7	7.35	7.23
1min	30.9	28.2	7.35	7.23
2min	31.2	28.7	7.35	7.22
3min	31.5	29.3	7.34	7.22
4min	31.8	29.8	7.34	7.22
5min	32.1	30.3	7.33	7.21
6min	32.4	30.9	7.33	7.21
7min	32.7	31.4	7.33	7.20
8min	33	32	7.32	7.20
9min	33.3	32.5	7.32	7.19
10min	33.6	33	7.31	7.18
11min	33.7	33.2	7.31	7.18
12min	33.9	33.4	7.30	7.17
13min	34	33.5	7.30	7.16
14min	34.1	33.7	7.29	7.16
15min	34.2	33.9	7.28	7.15
16min	34.3	34.1	7.28	7.14
17min	34.4	34.2	7.27	7.14
18min	34.5	34.4	7.27	7.13
19min	34.6	34.6	7.26	7.12
20min	34.7	34.7	7.25	7.11
21min	34.8	34.9	7.25	7.11

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
22min	34.9	35.1	7.24	7.10
23min	35	35.3	7.24	7.09
24min	35.2	35.4	7.23	7.08
25min	35.3	35.6	7.23	7.08
26min	35.4	35.8	7.22	7.07
27min	35.5	35.9	7.21	7.06
28min	35.6	36.1	7.21	7.06
29min	35.7	36.3	7.20	7.05
30min	35.8	36.5	7.20	7.05
32min	35.9	36.6	7.19	7.04
34min	36.1	36.8	7.19	7.03
36min	36.2	36.9	7.18	7.03
38min	36.3	37.1	7.18	7.02
40min	36.5	37.3	7.18	7.02
42min	36.6	37.4	7.18	7.02
44min	36.7	37.6	7.18	7.02
46min	36.8	37.8	7.18	7.03
48min	37	37.9	7.19	7.03
50min	37.1	38.1	7.19	7.04
52min	37.2	38.3	7.20	7.05
54min	37.4	38.4	7.21	7.06
56min	37.5	38.6	7.22	7.07
58min	37.6	38.8	7.22	7.08
60min	37.8	38.9	7.23	7.09
65min	37.8	39.1	7.24	7.11
70min	37.9	39.2	7.25	7.12
75min	37.9	39.4	7.25	7.13
80min	38	39.5	7.24	7.12
85min	38	39.6	7.22	7.10
90min	38.1	39.8	7.20	7.08
95min	38.2	39.9	7.18	7.06
100min	38.2	40	7.17	7.05
105min	38.4	40.3	7.17	7.04
110min	38.6	40.5	7.17	7.04
115min	38.8	40.8	7.19	7.04
120min	39	41	7.21	7.06
130min	39.1	41.2	7.26	7.11
140min	39.2	41.3	7.29	7.14
150min	39.3	41.5	7.28	7.14
160min	39.4	41.7	7.24	7.11
170min	39.6	41.9	7.21	7.08
180min	39.7	42.1	7.20	7.08
195min	39.9	42.4	7.24	7.12
210min	40.1	42.7	7.27	7.12

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
225min	40.3	43	7.21	7.06
240min	40.4	43.2	7.14	6.99

C.3 Water Heater + 2 Halogen Lamps

Table C.5. Characterization Data of Water Heater + 2 Halogen Load in Normal Socket

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
0 (Ambient)	30.8	27.9	9.29	9.17
1min	31.1	28.2	9.29	9.17
2min	31.3	28.5	9.29	9.17
3min	31.5	28.8	9.29	9.17
4min	31.7	29.1	9.29	9.17
5min	31.9	29.4	9.28	9.17
6min	32.2	29.7	9.28	9.16
7min	32.4	30	9.28	9.16
8min	32.6	30.3	9.28	9.16
9min	32.8	30.6	9.28	9.16
10min	33	30.9	9.28	9.16
11min	33.2	31	9.28	9.16
12min	33.3	31.2	9.28	9.15
13min	33.4	31.3	9.27	9.15
14min	33.5	31.5	9.27	9.15
15min	33.6	31.6	9.27	9.14
16min	33.8	31.8	9.26	9.14
17min	33.9	32	9.26	9.13
18min	34	32.1	9.25	9.13
19min	34.1	32.3	9.25	9.12
20min	34.2	32.4	9.25	9.12
21min	34.4	32.6	9.24	9.11
22min	34.5	32.7	9.24	9.10
23min	34.6	32.9	9.23	9.10
24min	34.7	33.1	9.22	9.09
25min	34.8	33.2	9.22	9.08
26min	35	33.4	9.21	9.08
27min	35.1	33.5	9.21	9.07
28min	35.2	33.7	9.20	9.06
29min	35.3	33.8	9.20	9.05
30min	35.4	34	9.19	9.05
32min	35.5	34.2	9.18	9.03

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
34min	35.6	34.3	9.17	9.02
36min	35.8	34.5	9.16	9.01
38min	35.9	34.6	9.15	9.00
40min	36	34.8	9.14	8.99
42min	36.1	35	9.13	8.98
44min	36.2	35.1	9.13	8.97
46min	36.3	35.3	9.13	8.97
48min	36.4	35.4	9.13	8.97
50min	36.5	35.6	9.13	8.97
52min	36.6	35.8	9.13	8.97
54min	36.7	35.9	9.13	8.97
56min	36.9	36.1	9.14	8.98
58min	37	36.3	9.14	8.99
60min	37.1	36.4	9.15	9.00
65min	37.2	36.6	9.17	9.02
70min	37.3	36.8	9.19	9.04
75min	37.4	37	9.20	9.06
80min	37.6	37.2	9.20	9.07
85min	37.7	37.4	9.19	9.07
90min	37.8	37.5	9.18	9.06
95min	38	37.7	9.16	9.04
100min	38.1	37.9	9.14	9.02
105min	38.2	38.1	9.12	9.00
110min	38.4	38.3	9.11	8.99
115min	38.5	38.4	9.11	8.98
120min	38.6	38.6	9.12	8.99
130min	38.7	38.8	9.17	9.02
140min	38.9	38.9	9.22	9.06
150min	39	39.1	9.24	9.08
160min	39.1	39.3	9.22	9.07
170min	39.2	39.5	9.18	9.04
180min	39.4	39.7	9.15	9.02
195min	39.6	40	9.16	9.04
210min	39.8	40.3	9.21	9.08
225min	40	40.6	9.20	9.05
240min	40.2	40.8	9.12	8.96

Table C.6. Characterization Data of Water Heater +2 Halogen Load in Corroded Socket

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
0 (Ambient)	30.6	27.6	9.30	9.18
1min	31	28.2	9.30	9.17
2min	31.3	28.8	9.29	9.17
3min	31.7	29.4	9.29	9.16
4min	32	30	9.28	9.16
5min	32.3	30.6	9.28	9.15
6min	32.7	31.2	9.27	9.15
7min	33	31.8	9.27	9.14
8min	33.4	32.3	9.26	9.13
9min	33.7	32.9	9.26	9.13
10min	34	33.5	9.25	9.12
11min	34.2	33.7	9.25	9.11
12min	34.3	33.9	9.24	9.10
13min	34.5	34.2	9.23	9.10
14min	34.6	34.4	9.23	9.09
15min	34.8	34.6	9.22	9.08
16min	34.9	34.8	9.22	9.07
17min	35.1	35	9.21	9.07
18min	35.2	35.2	9.20	9.06
19min	35.4	35.4	9.20	9.05
20min	35.5	35.6	9.19	9.04
21min	35.7	35.8	9.19	9.04
22min	35.8	36.1	9.18	9.03
23min	36	36.3	9.17	9.02
24min	36.1	36.5	9.17	9.02
25min	36.3	36.7	9.16	9.01
26min	36.4	36.9	9.16	9.00
27min	36.6	37.1	9.15	9.00
28min	36.7	37.3	9.15	8.99
29min	36.8	37.5	9.14	8.99
30min	37	37.8	9.14	8.98
32min	37.1	38	9.13	8.98
34min	37.3	38.2	9.13	8.97
36min	37.4	38.4	9.13	8.97
38min	37.5	38.6	9.13	8.97
40min	37.6	38.8	9.13	8.97
42min	37.8	39.1	9.13	8.97
44min	37.9	39.3	9.13	8.98
46min	38	39.5	9.14	8.98
48min	38.2	39.7	9.14	8.99
50min	38.3	40	9.15	9.00
52min	38.4	40.2	9.16	9.01
54min	38.6	40.4	9.17	9.02

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
56min	38.7	40.7	9.17	9.03
58min	38.8	40.9	9.18	9.04
60min	39	41.2	9.19	9.05
65min	39.1	41.4	9.20	9.06
70min	39.3	41.7	9.20	9.07
75min	39.5	42	9.20	9.07
80min	39.7	42.3	9.18	9.06
85min	39.8	42.6	9.16	9.04
90min	40	42.9	9.14	9.02
95min	40.2	43.2	9.13	9
100min	40.3	43.4	9.12	8.99
105min	40.5	43.7	9.12	8.98
110min	40.7	44	9.13	8.98
115min	40.8	44.2	9.15	9.00
120min	41	44.5	9.17	9.02
Equipment damage is observed				

C.4 Water Heater + 3 Halogen Lamps

Table C.7. Characterization Data of Water Heater + 3 Halogen Load in Normal Socket

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
0 (Ambient)	30.8	27.9	11.29	11.17
1min	31.1	28.3	11.29	11.17
2min	31.4	28.7	11.29	11.17
3min	31.7	29.1	11.29	11.17
4min	32	29.5	11.29	11.17
5min	32.2	29.9	11.29	11.17
6min	32.5	30.3	11.29	11.16
7min	32.8	30.7	11.28	11.16
8min	33.1	31.1	11.28	11.16
9min	33.4	31.5	11.28	11.16
10min	33.6	31.9	11.28	11.15
11min	33.8	32.1	11.27	11.15
12min	34	32.3	11.27	11.14
13min	34.1	32.5	11.27	11.14
14min	34.3	32.7	11.26	11.14
15min	34.4	32.9	11.26	11.13
16min	34.6	33.2	11.25	11.12

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
17min	34.8	33.4	11.25	11.12
18min	34.9	33.6	11.25	11.11
19min	35.1	33.8	11.24	11.11
20min	35.2	34	11.23	11.10
21min	35.4	34.2	11.23	11.09
22min	35.6	34.5	11.22	11.09
23min	35.7	34.7	11.22	11.08
24min	35.9	34.9	11.21	11.07
25min	36	35.1	11.21	11.06
26min	36.2	35.3	11.20	11.06
27min	36.3	35.5	11.19	11.05
28min	36.5	35.8	11.19	11.04
29min	36.7	36	11.18	11.04
30min	36.8	36.2	11.18	11.03
32min	37	36.4	11.17	11.01
34min	37.1	36.6	11.16	11.00
36min	37.2	36.8	11.15	10.99
38min	37.3	37	11.14	10.98
40min	37.5	37.3	11.13	10.98
42min	37.6	37.5	11.13	10.97
44min	37.7	37.7	11.13	10.97
46min	37.9	37.9	11.13	10.96
48min	38	38.1	11.13	10.97
50min	38.1	38.3	11.13	10.97
52min	38.2	38.6	11.13	10.97
54min	38.4	38.8	11.14	10.98
56min	38.5	39	11.14	10.99
58min	38.6	39.2	11.15	10.99
60min	38.8	39.4	11.16	11.00
65min	38.9	39.7	11.18	11.03
70min	39	39.9	11.19	11.05
75min	39.2	40.1	11.20	11.06
80min	39.3	40.3	11.20	11.07
85min	39.4	40.6	11.19	11.06
90min	39.6	40.8	11.17	11.05
95min	39.8	41.1	11.15	11.03
100min	40	41.4	11.13	11.01
105min	40.2	41.6	11.12	10.99
110min	40.4	41.9	11.11	10.98
115min	40.6	42.2	11.11	10.98
120min	40.8	42.4	11.13	10.99
130min	41	42.7	11.18	11.02
140min	41.2	42.9	11.22	11.06
150min	41.4	43.2	11.24	11.08

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
160min	41.6	43.5	11.21	11.06
170min	41.8	43.8	11.17	11.03
180min	42	44.1	11.14	11.02
195min	42.3	44.6	11.17	11.05
210min	42.6	45	11.21	11.07
225min	42.9	45.4	11.19	11.03
240min	43.2	45.8	11.11	10.95

Table C.8. Characterization Data of Water Heater +3 Halogen Load in Corroded Socket

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
0 (Ambient)	30.6	27.6	11.30	11.17
1min	31.7	30.8	11.29	11.17
2min	32.8	33.9	11.29	11.16
3min	33.8	37	11.28	11.15
4min	34.8	39.5	11.28	11.15
5min	35.8	42	11.27	11.14
6min	37	47	11.27	11.13
7min	38.2	52	11.26	11.12
8min	38.8	53	11.25	11.12
Equipment damage is observed				

C.5 Water Heater + 4 Halogen Lamps

Table C.9. Characterization Data of Water Heater + 4 Halogen Load in Normal Socket

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
0 (Ambient)	30.8	27.9	13.25	13.13
1min	31.2	28.4	13.25	13.12
2min	31.5	28.9	13.24	13.12
3min	31.9	29.4	13.24	13.12
4min	32.2	29.9	13.24	13.12
5min	32.5	30.4	13.24	13.11
6min	32.9	30.9	13.23	13.11
7min	33.2	31.4	13.23	13.11
8min	33.6	31.8	13.23	13.10

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
9min	33.9	32.3	13.23	13.10
10min	34.2	32.8	13.22	13.09
11min	34.4	33.1	13.22	13.09
12min	34.6	33.3	13.21	13.08
13min	34.8	33.6	13.21	13.08
14min	35	33.8	13.20	13.07
15min	35.1	34	13.20	13.07
16min	35.3	34.3	13.19	13.06
17min	35.5	34.5	13.19	13.05
18min	35.7	34.7	13.18	13.05
19min	35.9	35	13.18	13.04
20min	36	35.2	13.17	13.03
21min	36.2	35.4	13.17	13.02
22min	36.4	35.7	13.16	13.02
23min	36.6	35.9	13.15	13.01
24min	36.7	36.2	13.15	13.00
25min	36.9	36.4	13.14	12.99
26min	37.1	36.6	13.14	12.99
27min	37.3	36.9	13.13	12.98
28min	37.5	37.1	13.12	12.97
29min	37.6	37.3	13.12	12.97
30min	37.8	37.6	13.11	12.96
32min	37.9	37.8	13.10	12.95
34min	37.9	38	13.10	12.94
36min	38	38.2	13.09	12.93
38min	38	38.4	13.08	12.92
40min	38.1	38.6	13.08	12.92
42min	38.1	38.8	13.08	12.91
44min	38.2	39	13.07	12.91
46min	38.3	39.2	13.08	12.91
48min	38.4	39.4	13.08	12.92
50min	38.5	39.5	13.08	12.92
52min	38.7	39.7	13.09	12.93
54min	38.8	39.9	13.09	12.93
56min	39	40	13.10	12.94
58min	39.1	40.2	13.11	12.95
60min	39.3	40.3	13.11	12.96
65min	39.5	40.6	13.13	12.98
70min	39.6	40.9	13.14	13.00
75min	39.8	41.2	13.15	13.01
80min	40	41.5	13.14	13.01
85min	40.2	41.8	13.13	13.00
90min	40.4	42.1	13.11	12.99
95min	40.6	42.4	13.09	12.97

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
100min	40.8	42.7	13.07	12.95
105min	41	43	13.06	12.93
110min	41.2	43.3	13.06	12.92
115min	41.4	43.5	13.07	12.93
120min	41.6	43.8	13.09	12.94
130min	41.8	44.1	13.14	12.98
140min	42.1	44.4	13.18	13.02
150min	42.3	44.7	13.18	13.03
160min	42.5	45.1	13.15	13.00
170min	42.7	45.4	13.11	12.98
180min	43	45.8	13.09	12.97
195min	43.4	46.4	13.12	13.00
Equipment damage is observed				

Table C.10. Characterization Data of Water Heater +4 Halogen Load in Corroded Socket

Time	Temperature (°C)		Current (A)	
	10k NTC Thermistor	Socket Metal Terminal	Device	Ingco Clamp Meter
0 (Ambient)	30.6	27.6	13.24	13.11
1min	32.3	32.8	13.24	13.10
2min	34	38	13.23	13.10
3min	35.6	43.5	13.23	13.09
4min	36.9	45.8	13.22	13.08
Equipment damage is observed				

Appendix D

SOCKET HEATING TESTING

This appendix contains the full socket-heating protection testing tables used to support the socket-heating evaluation in Chapter 4.

D.1 Water Heater

Table D.1. Socket Heating Testing Data of Water Heater in Normal Socket

Time	NTC	Temp. (°C) Est. Socket Temp.	Socket Terminal	Current (A) Device	Clamp Meter
0 (Ambient)	28.4	28.7	29.1	5.37	5.24
1min	28.7	29	29.3	5.37	5.24
2min	28.9	29.2	29.5	5.37	5.24
3min	29.2	29.5	29.7	5.37	5.24
4min	29.5	29.8	30	5.37	5.25
5min	29.7	30	30.2	5.37	5.25
6min	29.8	30.1	30.3	5.37	5.25
7min	29.9	30.2	30.4	5.38	5.25
8min	30	30.3	30.5	5.38	5.25
9min	30.1	30.4	30.7	5.38	5.25
10min	30.1	30.4	30.8	5.38	5.25
11min	30.2	30.5	30.9	5.38	5.25
12min	30.2	30.5	30.9	5.38	5.25
13min	30.3	30.6	31	5.38	5.25
14min	30.3	30.6	31.1	5.38	5.25
15min	30.4	30.7	31.1	5.37	5.25
16min	30.4	30.7	31.2	5.37	5.25
17min	30.5	30.8	31.3	5.37	5.24
18min	30.5	30.8	31.3	5.37	5.24
19min	30.6	30.9	31.4	5.37	5.24
20min	30.6	30.9	31.5	5.36	5.23
21min	30.7	31	31.5	5.36	5.23
22min	30.7	31	31.5	5.36	5.22
23min	30.7	31.1	31.6	5.35	5.22
24min	30.8	31.1	31.6	5.35	5.21
25min	30.8	31.1	31.7	5.34	5.21
26min	30.8	31.2	31.7	5.34	5.20
27min	30.9	31.2	31.7	5.33	5.20
28min	30.9	31.2	31.8	5.33	5.19
29min	30.9	31.3	31.8	5.32	5.18

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
30min	30.9	31.3	31.8	5.32	5.18
32min	31	31.4	31.9	5.30	5.16
34min	31	31.4	31.9	5.29	5.15
36min	31	31.4	31.9	5.28	5.13
38min	31	31.4	32	5.27	5.12
40min	31.1	31.5	32	5.26	5.11
42min	31.1	31.5	32.1	5.25	5.10
44min	31.1	31.5	32.1	5.24	5.09
46min	31.1	31.6	32.1	5.24	5.08
48min	31.1	31.6	32.2	5.23	5.07
50min	31.2	31.6	32.2	5.23	5.07
52min	31.2	31.6	32.3	5.23	5.07
54min	31.2	31.7	32.3	5.23	5.06
56min	31.2	31.7	32.3	5.23	5.07
58min	31.3	31.7	32.4	5.23	5.07
60min	31.3	31.8	32.4	5.24	5.07
65min	31.3	31.8	32.5	5.26	5.09
70min	31.4	31.9	32.5	5.27	5.12
75min	31.4	31.9	32.6	5.29	5.14
80min	31.4	32	32.7	5.30	5.15
85min	31.5	32	32.7	5.30	5.16
90min	31.5	32.1	32.8	5.29	5.16
95min	31.6	32.1	32.9	5.28	5.15
100min	31.6	32.2	33	5.26	5.13
105min	31.7	32.3	33	5.24	5.11
110min	31.7	32.3	33.1	5.22	5.09
115min	31.8	32.4	33.2	5.21	5.08
120min	31.8	32.4	33.2	5.21	5.08
130min	31.8	32.4	33.3	5.25	5.10
140min	31.9	32.5	33.3	5.30	5.14
150min	31.9	32.5	33.3	5.34	5.17
160min	31.9	32.5	33.3	5.34	5.18
170min	31.9	32.5	33.3	5.30	5.15
180min	31.9	32.5	33.4	5.26	5.13
195min	32	32.6	33.5	5.26	5.13
210min	32	32.6	33.6	5.31	5.17
225min	32.1	32.7	33.7	5.32	5.17
240min	32.1	32.7	33.8	5.25	5.09

Table D.2. Socket Heating Testing Data of Water Heater in Corroded Socket

Time	Temp. (°C)			Current (A)	
	NTC	Socket Est. Temp.	Socket Terminal	Device	Clamp Meter
0 (Ambient)	29.9	30.2	30.8	5.10	4.97
1min	30.2	30.5	30.5	5.10	4.97
2min	30.5	30.8	30.9	5.10	4.97
3min	30.8	31.1	31.3	5.09	4.97
4min	31.1	31.4	31.7	5.09	4.96
5min	31.4	31.7	32.2	5.09	4.96
6min	31.5	31.8	32.3	5.08	4.96
7min	31.6	31.9	32.4	5.08	4.95
8min	31.7	32	32.6	5.08	4.95
9min	31.8	32.1	32.7	5.07	4.94
10min	31.9	32.2	32.8	5.07	4.94
11min	32	32.3	33	5.07	4.93
12min	32.1	32.4	33.1	5.06	4.93
13min	32.1	32.4	33.2	5.06	4.92
14min	32.2	32.5	33.3	5.05	4.91
15min	32.2	32.5	33.4	5.05	4.91
16min	32.3	32.6	33.5	5.04	4.90
17min	32.4	32.7	33.6	5.04	4.89
18min	32.4	32.7	33.7	5.03	4.89
19min	32.5	32.8	33.8	5.02	4.88
20min	32.5	32.8	33.9	5.02	4.87
21min	32.6	32.9	34	5.01	4.87
22min	32.6	32.9	34.1	5.01	4.86
23min	32.7	33	34.2	5.00	4.85
24min	32.7	33	34.3	4.99	4.84
25min	32.8	33.1	34.3	4.99	4.84
26min	32.8	33.2	34.4	4.98	4.83
27min	32.9	33.2	34.5	4.98	4.82
28min	32.9	33.3	34.6	4.97	4.82
29min	33	33.3	34.7	4.97	4.81
30min	33	33.4	34.8	4.96	4.80
32min	33.1	33.5	34.8	4.95	4.79
34min	33.1	33.5	34.9	4.94	4.78
36min	33.2	33.6	35	4.94	4.78
38min	33.2	33.6	35.1	4.93	4.77
40min	33.3	33.7	35.2	4.93	4.77
42min	33.3	33.8	35.3	4.93	4.76
44min	33.4	33.8	35.3	4.93	4.76
46min	33.4	33.9	35.4	4.93	4.77
48min	33.5	34	35.5	4.93	4.77

Time	Temp. (°C)		Current (A)		
	NTC	Est. Socket Temp.	Socket Terminal	Device	Clamp Meter
50min	33.5	34	35.6	4.94	4.78
52min	33.6	34.1	35.7	4.94	4.78
54min	33.6	34.2	35.8	4.95	4.79
56min	33.7	34.2	35.9	4.96	4.80
58min	33.7	34.3	35.9	4.97	4.81
60min	33.8	34.4	36	4.97	4.82
65min	33.8	34.5	36.1	4.99	4.84
70min	33.9	34.6	36.3	5.00	4.86
75min	34	34.7	36.4	5.00	4.87
80min	34.1	34.8	36.5	5.00	4.87
85min	34.2	34.9	36.6	4.98	4.86
90min	34.3	35	36.7	4.96	4.84
95min	34.4	35.1	36.8	4.94	4.82
100min	34.5	35.2	36.9	4.93	4.80
105min	34.6	35.3	37	4.92	4.78
110min	34.6	35.4	37.1	4.92	4.78
115min	34.7	35.5	37.2	4.93	4.78
120min	34.8	35.6	37.3	4.95	4.80
130min	34.9	35.7	37.4	5.00	4.84
140min	35	35.8	37.5	5.04	4.88
150min	35	35.8	37.5	5.04	4.88
160min	35.1	35.9	37.6	5.01	4.86
170min	35.2	36	37.8	4.96	4.83
180min	35.3	36.1	37.9	4.95	4.82
195min	35.4	36.3	38.1	4.99	4.86
210min	35.6	36.5	38.3	5.02	4.87
225min	35.7	36.6	38.4	4.98	4.81
240min	35.8	36.7	38.5	4.90	4.74

D.2 Water Heater + 1 Halogen Lamp

Table D.3. Socket Heating Testing Data of Water Heater + 1 Halogen Lamp in Normal Socket

Time	Temp. (°C)		Current (A)		
	NTC	Est. Socket Temp.	Socket Terminal	Device	Clamp Meter
0					
(Ambient)	28.6	29.2	29.8	7.33	7.20
1min	28.9	29.6	29.6	7.33	7.20
2min	29.2	29.9	30.2	7.33	7.20

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
3min	29.5	30.2	30.8	7.33	7.20
4min	29.8	30.5	31.3	7.33	7.20
5min	30	30.8	31.9	7.33	7.20
6min	30.2	31	32.1	7.33	7.20
7min	30.4	31.2	32.4	7.33	7.20
8min	30.5	31.4	32.6	7.33	7.20
9min	30.7	31.6	32.8	7.33	7.20
10min	30.8	31.7	33.1	7.33	7.20
11min	30.9	31.9	33.2	7.33	7.20
12min	31	32	33.4	7.33	7.20
13min	31.1	32.1	33.5	7.33	7.20
14min	31.2	32.2	33.6	7.32	7.19
15min	31.3	32.3	33.8	7.32	7.19
16min	31.4	32.4	33.9	7.32	7.19
17min	31.5	32.5	34	7.32	7.18
18min	31.6	32.6	34.2	7.31	7.18
19min	31.7	32.7	34.3	7.31	7.18
20min	31.8	32.8	34.5	7.31	7.17
21min	31.9	32.9	34.6	7.30	7.17
22min	32	33	34.6	7.30	7.16
23min	32.1	33.1	34.7	7.29	7.15
24min	32.2	33.2	34.8	7.29	7.15
25min	32.2	33.3	34.9	7.28	7.14
26min	32.3	33.4	35	7.28	7.13
27min	32.4	33.5	35.1	7.27	7.13
28min	32.5	33.5	35.2	7.26	7.12
29min	32.6	33.6	35.3	7.26	7.11
30min	32.6	33.7	35.4	7.25	7.11
32min	32.7	33.8	35.5	7.24	7.09
34min	32.8	33.9	35.6	7.23	7.08
36min	32.9	34	35.7	7.22	7.06
38min	33	34.1	35.8	7.21	7.05
40min	33.1	34.2	35.9	7.20	7.04
42min	33.2	34.3	36	7.19	7.03
44min	33.3	34.4	36.1	7.19	7.02
46min	33.4	34.5	36.2	7.18	7.02
48min	33.4	34.6	36.3	7.18	7.01
50min	33.5	34.7	36.4	7.18	7.01
52min	33.6	34.8	36.5	7.18	7.01
54min	33.7	34.9	36.6	7.18	7.01
56min	33.8	35	36.7	7.18	7.02
58min	33.9	35.1	36.8	7.19	7.02
60min	34	35.2	36.9	7.19	7.03
65min	34.1	35.3	37.1	7.21	7.05

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
70min	34.2	35.4	37.2	7.23	7.07
75min	34.3	35.6	37.4	7.24	7.09
80min	34.4	35.7	37.6	7.25	7.11
85min	34.6	35.8	37.7	7.25	7.11
90min	34.7	36	37.9	7.24	7.10
95min	34.8	36.1	38	7.22	7.09
100min	34.9	36.3	38.2	7.20	7.07
105min	35	36.4	38.4	7.18	7.05
110min	35.2	36.5	38.5	7.17	7.03
115min	35.3	36.7	38.7	7.16	7.03
120min	35.4	36.8	38.8	7.17	7.03
130min	35.5	36.9	38.9	7.21	7.05
140min	35.6	37	39	7.26	7.09
150min	35.7	37.1	39.1	7.29	7.12
160min	35.8	37.2	39.2	7.28	7.12
170min	35.9	37.3	39.3	7.24	7.09
180min	36	37.4	39.5	7.21	7.07
195min	36.2	37.6	39.7	7.21	7.08
210min	36.4	37.8	39.9	7.26	7.12
225min	36.6	38	40.1	7.26	7.10
240min	36.7	38.1	40.3	7.19	7.02

Table D.4. Socket Heating Testing Data of Water Heater + 1 Halogen Lamp in Corroded Socket

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
0 (Ambient)	29.9	30.6	30.6	7.35	7.22
1min	30.3	31.1	31.3	7.35	7.22
2min	30.7	31.5	32	7.35	7.21
3min	31.1	32	32.7	7.34	7.21
4min	31.5	32.4	33.4	7.34	7.21
5min	31.9	32.8	34.1	7.33	7.20
6min	32.2	33.1	34.4	7.33	7.20
7min	32.4	33.4	34.7	7.33	7.19
8min	32.7	33.7	35	7.32	7.19
9min	32.9	33.9	35.3	7.32	7.18
10min	33.1	34.2	35.6	7.31	7.17
11min	33.3	34.4	35.8	7.31	7.17
12min	33.4	34.5	36	7.30	7.16
13min	33.6	34.7	36.2	7.30	7.15

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
14min	33.7	34.8	36.4	7.29	7.15
15min	33.8	35	36.6	7.28	7.14
16min	34	35.1	36.7	7.28	7.13
17min	34.1	35.3	36.9	7.27	7.13
18min	34.2	35.4	37.1	7.27	7.12
19min	34.4	35.6	37.3	7.26	7.11
20min	34.5	35.7	37.5	7.25	7.10
21min	34.6	35.8	37.6	7.25	7.10
22min	34.7	35.9	37.8	7.24	7.09
23min	34.8	36.1	37.9	7.24	7.08
24min	35	36.2	38.1	7.23	7.07
25min	35.1	36.3	38.2	7.23	7.07
26min	35.2	36.4	38.4	7.22	7.06
27min	35.3	36.5	38.5	7.21	7.05
28min	35.4	36.7	38.7	7.21	7.05
29min	35.5	36.8	38.8	7.20	7.04
30min	35.6	36.9	39	7.20	7.04
32min	35.7	37	39.1	7.19	7.03
34min	35.8	37.1	39.2	7.19	7.02
36min	35.9	37.2	39.3	7.18	7.02
38min	36	37.3	39.5	7.18	7.01
40min	36.1	37.4	39.6	7.18	7.01
42min	36.2	37.5	39.7	7.18	7.01
44min	36.3	37.6	39.9	7.18	7.01
46min	36.4	37.7	40	7.18	7.02
48min	36.5	37.8	40.1	7.19	7.02
50min	36.6	37.9	40.3	7.19	7.03
52min	36.7	38	40.4	7.20	7.04
54min	36.8	38.1	40.5	7.21	7.05
56min	37	38.2	40.7	7.22	7.06
58min	37.1	38.4	40.8	7.22	7.07
60min	37.2	38.5	40.9	7.23	7.08
65min	37.3	38.6	41.1	7.24	7.10
70min	37.4	38.8	41.3	7.25	7.11
75min	37.5	38.9	41.4	7.25	7.12
80min	37.7	39	41.6	7.24	7.11
85min	37.8	39.2	41.8	7.22	7.09
90min	37.9	39.3	41.9	7.20	7.07
95min	38.1	39.5	42.1	7.18	7.05
100min	38.2	39.6	42.2	7.17	7.04
105min	38.4	39.8	42.4	7.17	7.03
Relay has tripped					

D.3 Water Heater + 2 Halogen Lamps

Table D.5. Socket Heating Testing Data of Water Heater + 2 Halogen Lamp in Normal Socket

Time	NTC	Temp. (°C) Est. Socket Temp.	Socket Terminal	Current (A) Device	Clamp Meter
0 (Ambient)	29	29.8	30.1	9.29	9.16
1min	29.4	30.2	30.7	9.29	9.16
2min	29.8	30.6	31.3	9.29	9.16
3min	30.1	31	31.9	9.29	9.16
4min	30.5	31.4	32.5	9.29	9.16
5min	30.8	31.8	33.1	9.28	9.16
6min	31.1	32.1	33.4	9.28	9.15
7min	31.3	32.4	33.8	9.28	9.15
8min	31.6	32.6	34.2	9.28	9.15
9min	31.8	32.9	34.5	9.28	9.15
10min	32	33.1	34.9	9.28	9.15
11min	32.2	33.3	35.1	9.28	9.15
12min	32.4	33.5	35.3	9.28	9.14
13min	32.5	33.6	35.5	9.27	9.14
14min	32.7	33.8	35.7	9.27	9.14
15min	32.8	34	36	9.27	9.13
16min	33	34.2	36.2	9.26	9.13
17min	33.2	34.3	36.4	9.26	9.12
18min	33.3	34.5	36.6	9.25	9.12
19min	33.5	34.7	36.8	9.25	9.11
20min	33.6	34.8	37	9.25	9.11
21min	33.8	35	37.2	9.24	9.10
22min	33.9	35.1	37.4	9.24	9.09
23min	34	35.2	37.5	9.23	9.09
24min	34.2	35.3	37.7	9.22	9.08
25min	34.3	35.5	37.8	9.22	9.07
26min	34.4	35.6	38	9.21	9.07
27min	34.5	35.7	38.1	9.21	9.06
28min	34.7	35.9	38.3	9.20	9.05
29min	34.8	36	38.4	9.20	9.04
30min	34.9	36.1	38.6	9.19	9.04
32min	35.1	36.3	38.8	9.18	9.02
34min	35.2	36.4	38.9	9.17	9.01
36min	35.4	36.6	39.1	9.16	9.00
38min	35.5	36.7	39.3	9.15	8.99
40min	35.7	36.9	39.5	9.14	8.98
42min	35.8	37	39.6	9.13	8.97

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
44min	36	37.2	39.8	9.13	8.96
46min	36.1	37.3	40	9.13	8.96
48min	36.3	37.5	40.1	9.13	8.96
50min	36.4	37.6	40.3	9.13	8.96
52min	36.5	37.7	40.4	9.13	8.96
54min	36.6	37.9	40.5	9.13	8.96
56min	36.7	38	40.7	9.14	8.97
58min	36.9	38.1	40.8	9.14	8.98
60min	37	38.3	40.9	9.15	8.99
65min	37.1	38.4	41.1	9.17	9.01
70min	37.3	38.5	41.3	9.19	9.03
75min	37.4	38.7	41.5	9.20	9.05
80min	37.6	38.8	41.7	9.20	9.06
85min	37.7	39	41.9	9.19	9.06
90min	37.9	39.1	42.1	9.18	9.05
95min	38	39.3	42.3	9.16	9.03
100min	38.2	39.4	42.5	9.14	9.01
105min	38.3	39.6	42.7	9.12	8.99
110min	38.5	39.7	42.9	9.11	8.98
115min	38.7	39.9	43	9.11	8.97
120min	38.8	40	43.2	9.12	8.98
Relay has tripped					

Table D.6. Socket Heating Testing Data of Water Heater + 2 Halogen Lamp in Corroded Socket

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
0 (Ambient)	29.9	30.8	31.1	9.30	9.17
1min	30.3	31.3	31.9	9.30	9.16
2min	30.7	31.7	32.7	9.29	9.16
3min	31.1	32.2	33.4	9.29	9.15
4min	31.5	32.6	34.2	9.28	9.15
5min	31.8	33	35	9.28	9.14
6min	32.2	33.4	35.4	9.27	9.14
7min	32.6	33.8	35.8	9.27	9.13
8min	32.9	34.2	36.2	9.26	9.12
9min	33.3	34.5	36.6	9.26	9.12
10min	33.6	34.9	37	9.25	9.11
11min	33.8	35.1	37.3	9.25	9.10
12min	34	35.3	37.6	9.24	9.09

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
13min	34.2	35.5	37.8	9.23	9.09
14min	34.4	35.8	38.1	9.23	9.08
15min	34.6	36	38.4	9.22	9.07
16min	34.8	36.2	38.7	9.22	9.06
17min	35	36.4	39	9.21	9.06
18min	35.2	36.6	39.2	9.20	9.05
19min	35.4	36.8	39.5	9.20	9.04
20min	35.6	37	39.8	9.19	9.03
21min	35.7	37.1	39.9	9.19	9.03
22min	35.9	37.3	40.1	9.18	9.02
23min	36	37.4	40.3	9.17	9.01
24min	36.1	37.6	40.4	9.17	9.01
25min	36.3	37.7	40.6	9.16	9.00
26min	36.4	37.8	40.8	9.16	8.99
27min	36.5	38	40.9	9.15	8.99
28min	36.6	38.1	41.1	9.15	8.98
29min	36.8	38.2	41.3	9.14	8.98
30min	36.9	38.4	41.4	9.14	8.97
32min	37	38.5	41.7	9.13	8.97
34min	37.2	38.7	41.9	9.13	8.96
36min	37.3	38.8	42.2	9.13	8.96
38min	37.5	38.9	42.4	9.13	8.96
40min	37.6	39.1	42.7	9.13	8.96
42min	37.7	39.2	43	9.13	8.96
44min	37.9	39.4	43.2	9.13	8.97
46min	38	39.5	43.5	9.14	8.97
48min	38.2	39.7	43.7	9.14	8.98
50min	38.3	39.8	43.9	9.15	8.99
52min	38.5	40	44.1	9.16	9.00
Relay has tripped					

D.4 Water Heater + 3 Halogen Lamps

Table D.7. Socket Heating Testing Data of Water Heater + 3 Halogen Lamp in Normal Socket

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
0					
(Ambient)	29.4	30.3	30.9	11.29	11.16
1min	29.9	30.8	31.7	11.29	11.16

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
2min	30.3	31.3	32.5	11.29	11.16
3min	30.7	31.8	33.3	11.29	11.16
4min	31.1	32.3	34.1	11.29	11.16
5min	31.5	32.7	34.9	11.29	11.16
6min	31.8	33.1	35.3	11.29	11.15
7min	32.1	33.4	35.8	11.28	11.15
8min	32.4	33.7	36.2	11.28	11.15
9min	32.7	34	36.6	11.28	11.15
10min	33	34.3	37.1	11.28	11.14
11min	33.2	34.5	37.3	11.27	11.14
12min	33.4	34.8	37.6	11.27	11.13
13min	33.6	35	37.9	11.27	11.13
14min	33.8	35.2	38.2	11.26	11.13
15min	34	35.4	38.4	11.26	11.12
16min	34.2	35.6	38.7	11.25	11.11
17min	34.4	35.8	39	11.25	11.11
18min	34.6	36	39.3	11.25	11.10
19min	34.8	36.2	39.6	11.24	11.10
20min	35	36.4	39.8	11.23	11.09
21min	35.2	36.6	40	11.23	11.08
22min	35.4	36.7	40.2	11.22	11.08
23min	35.5	36.9	40.4	11.22	11.07
24min	35.7	37.1	40.5	11.21	11.06
25min	35.8	37.2	40.7	11.21	11.05
26min	36	37.4	40.9	11.20	11.05
27min	36.1	37.5	41.1	11.19	11.04
28min	36.3	37.7	41.2	11.19	11.03
29min	36.5	37.8	41.4	11.18	11.03
30min	36.6	38	41.6	11.18	11.02
32min	36.8	38.1	41.7	11.17	11.00
34min	36.9	38.3	41.9	11.16	10.99
36min	37	38.4	42	11.15	10.98
38min	37.1	38.5	42.2	11.14	10.97
40min	37.3	38.7	42.3	11.13	10.97
42min	37.4	38.8	42.5	11.13	10.96
44min	37.5	38.9	42.6	11.13	10.96
46min	37.7	39	42.8	11.13	10.95
48min	37.8	39.2	42.9	11.13	10.96
50min	37.9	39.3	43.1	11.13	10.96
52min	38.1	39.5	43.3	11.13	10.96
54min	38.2	39.6	43.5	11.14	10.97
56min	38.4	39.8	43.7	11.14	10.98
58min	38.5	39.9	43.9	11.15	10.98
Relay has tripped					

Table D.8. Socket Heating Testing Data of Water Heater + 3 Halogen Lamp in Corroded Socket

Time	NTC	Temp. (°C) Est. Socket Temp.	Socket Terminal	Current (A) Device	Clamp Meter
0 (Ambient)	32	33	33	11.30	11.16
1min	33.2	34.4	41	11.29	11.16
2min	34.6	35.8	43	11.29	11.15
3min	35.4	36.8	45	11.28	11.14
4min	36	37.4	47	11.28	11.14
5min	36.8	38.2	50	11.27	11.13
6min	37.6	39.1	51	11.27	11.12
7min	38.2	39.7	55	11.26	11.11
Relay has tripped					

D.5 Water Heater + 4 Halogen Lamps

Table D.9. Socket Heating Testing Data of Water Heater + 4 Halogen Lamp in Normal Socket

Time	NTC	Temp. (°C) Est. Socket Temp.	Socket Terminal	Current (A) Device	Clamp Meter
0 (Ambient)	29.8	30.8	31.1	13.25	13.12
1min	30.4	31.5	32.3	13.25	13.11
2min	31	32.2	33.6	13.24	13.11
3min	31.6	32.8	34.9	13.24	13.11
4min	32.3	33.6	36.2	13.24	13.11
5min	33	34.3	37.5	13.24	13.10
6min	33.4	34.7	38	13.23	13.10
7min	33.8	35.1	38.5	13.23	13.10
8min	34.1	35.5	39	13.23	13.09
9min	34.5	35.9	39.5	13.23	13.09
10min	34.8	36.2	40.1	13.22	13.08
11min	35.1	36.5	40.3	13.22	13.08
12min	35.3	36.7	40.5	13.21	13.07
13min	35.5	36.9	40.8	13.21	13.07
14min	35.7	37.1	41	13.20	13.06
15min	35.9	37.3	41.3	13.20	13.06
16min	36.1	37.5	41.5	13.19	13.05
17min	36.3	37.8	41.8	13.19	13.04

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
18min	36.5	38	42	13.18	13.04
19min	36.7	38.2	42.3	13.18	13.03
20min	36.9	38.4	42.5	13.17	13.02
21min	37	38.5	42.7	13.17	13.01
22min	37.1	38.6	42.8	13.16	13.01
23min	37.2	38.7	43	13.15	13.00
24min	37.3	38.9	43.1	13.15	12.99
25min	37.4	39	43.2	13.14	12.98
26min	37.5	39.1	43.4	13.14	12.98
27min	37.6	39.2	43.5	13.13	12.97
28min	37.7	39.3	43.7	13.12	12.96
29min	37.8	39.4	43.8	13.12	12.96
30min	37.9	39.5	44	13.11	12.95
32min	38	39.5	44.1	13.10	12.94
34min	38	39.6	44.2	13.10	12.93
36min	38	39.6	44.4	13.09	12.92
38min	38.1	39.7	44.5	13.08	12.91
40min	38.1	39.7	44.6	13.08	12.91
42min	38.1	39.8	44.7	13.08	12.90
44min	38.2	39.8	44.9	13.07	12.90
46min	38.2	39.9	45	13.08	12.90
Relay has tripped					

Table D.10. Socket Heating Testing Data of Water Heater + 4 Halogen Lamp in Corroded Socket

Time	Temp. (°C)		Socket Terminal	Current (A)	
	NTC	Est. Socket Temp.		Device	Clamp Meter
0 (Ambient)	32.2	33.2	33.5	13.24	13.10
1min	33.8	35	43	13.24	13.09
2min	35.4	36.8	47	13.23	13.09
3min	36.8	38.4	50	13.23	13.08
4min	37.6	39.4	53	13.22	13.07
5min	38.2	40	56	13.21	13.06
Relay has tripped					

Appendix E

ARC FAULT TESTING

This appendix contains the detailed arc-fault testing results, confusion matrices, and performance metrics for the recorded and unrecorded devices evaluated in Chapter 4.

Recorded Appliances

E.1 Water Heater

Table E.1. Arc Fault Evaluation of Water Heater Test Result

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
1	FALSE	TRUE	1	TRUE	FALSE
2	FALSE	TRUE	2	TRUE	FALSE
3	FALSE	TRUE	3	TRUE	FALSE
4	FALSE	TRUE	4	TRUE	FALSE
5	FALSE	TRUE	5	TRUE	FALSE
6	FALSE	TRUE	6	TRUE	FALSE
7	FALSE	TRUE	7	TRUE	FALSE
8	FALSE	TRUE	8	TRUE	FALSE
9	FALSE	TRUE	9	TRUE	FALSE
10	FALSE	TRUE	10	TRUE	FALSE
Total	0	10	Total	10	0

Table E.2. Arc Fault Evaluation of Water Heater Test Confusion Matrix

	Arcing Event	Normal Event
Relay Trips	10	0
Relay remains on	0	10

Table E.3. Arc Fault Evaluation of Water Heater Test Performance Matrix

Accuracy	Precision	Recall	F1-Score
100%	100%	100%	100%

E.2 Halogen Lamp

Table E.4. Arc Fault Evaluation of Halogen Lamp Test Result

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
1	FALSE	TRUE	1	TRUE	FALSE
2	FALSE	TRUE	2	TRUE	FALSE
3	FALSE	TRUE	3	TRUE	FALSE
4	FALSE	TRUE	4	TRUE	FALSE
5	FALSE	TRUE	5	TRUE	FALSE
6	FALSE	TRUE	6	TRUE	FALSE
7	FALSE	TRUE	7	TRUE	FALSE
8	FALSE	TRUE	8	TRUE	FALSE
9	FALSE	TRUE	9	TRUE	FALSE
10	FALSE	TRUE	10	TRUE	FALSE
Total	0	10	Total	10	0

Table E.5. Arc Fault Evaluation of Halogen Lamp Test Confusion Matrix

	Arcing Event	Normal Event
Relay Trips	10	0
Relay remains on	0	10

Table E.6. Arc Fault Evaluation of Halogen Lamp Test Performance Matrix

Accuracy	Precision	Recall	F1-Score
100%	100%	100%	100%

E.3 Electric Kettle

Table E.7. Arc Fault Evaluation of Electric Kettle Test Result

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
1	FALSE	TRUE	1	TRUE	FALSE
2	FALSE	TRUE	2	TRUE	FALSE
3	FALSE	TRUE	3	TRUE	FALSE

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
4	FALSE	TRUE	4	TRUE	FALSE
5	FALSE	TRUE	5	TRUE	FALSE
6	FALSE	TRUE	6	TRUE	FALSE
7	FALSE	TRUE	7	TRUE	FALSE
8	FALSE	TRUE	8	TRUE	FALSE
9	FALSE	TRUE	9	TRUE	FALSE
10	FALSE	TRUE	10	TRUE	FALSE
Total	0	10	Total	10	0

Table E.8. Arc Fault Evaluation of Electric Kettle Test Confusion Matrix

	Arcing Event	Normal Event
Relay Trips	10	0
Relay remains on	0	10

Table E.9. Arc Fault Evaluation of Electric Kettle Test Performance Matrix

Accuracy	Precision	Recall	F1-Score
100%	100%	100%	100%

E.4 40W Electric Fan

Table E.10. Arc Fault Evaluation of 40W Electric Fan Test Result

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
1	FALSE	TRUE	1	TRUE	FALSE
2	FALSE	TRUE	2	TRUE	FALSE
3	FALSE	TRUE	3	TRUE	FALSE
4	FALSE	TRUE	4	TRUE	FALSE
5	FALSE	TRUE	5	TRUE	FALSE
6	FALSE	TRUE	6	TRUE	FALSE
7	FALSE	TRUE	7	TRUE	FALSE
8	FALSE	TRUE	8	TRUE	FALSE
9	FALSE	TRUE	9	TRUE	FALSE
10	FALSE	TRUE	10	TRUE	FALSE
Total	0	10	Total	10	0

Table E.11. Arc Fault Evaluation of 40W Electric Fan Test Confusion Matrix

	Arcing Event	Normal Event
Relay Trips	10	0
Relay remains on	0	10

Table E.12. Arc Fault Evaluation of 40W Electric Fan Test Performance Matrix

Accuracy	Precision	Recall	F1-Score
100%	100%	100%	100%

E.5 Handheld Vacuum

Table E.13. Arc Fault Evaluation of Handheld Vacuum Test Result

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
1	FALSE	TRUE	1	TRUE	FALSE
2	FALSE	TRUE	2	TRUE	FALSE
3	FALSE	TRUE	3	TRUE	FALSE
4	FALSE	TRUE	4	FALSE	TRUE
5	FALSE	TRUE	5	TRUE	FALSE
6	FALSE	TRUE	6	TRUE	FALSE
7	FALSE	TRUE	7	TRUE	FALSE
8	FALSE	TRUE	8	TRUE	FALSE
9	FALSE	TRUE	9	TRUE	FALSE
10	FALSE	TRUE	10	TRUE	FALSE
Total	0	10	Total	9	1

Table E.14. Arc Fault Evaluation of Handheld Vacuum Test Confusion Matrix

	Arcing Event	Normal Event
Relay Trips	9	0
Relay remains on	1	10

Table E.15. Arc Fault Evaluation of Handheld Vacuum Test Performance Matrix

Accuracy	Precision	Recall	F1-Score
95%	100%	90%	94.74%

E.6 Vacuum Cleaner (Electrolux)

Table E.16. Arc Fault Evaluation of Vacuum Cleaner (Electrolux) Test Result

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
1	FALSE	TRUE	1	TRUE	FALSE
2	FALSE	TRUE	2	TRUE	FALSE
3	FALSE	TRUE	3	TRUE	FALSE
4	FALSE	TRUE	4	FALSE	TRUE
5	FALSE	TRUE	5	TRUE	FALSE
6	FALSE	TRUE	6	TRUE	FALSE
7	FALSE	TRUE	7	TRUE	FALSE
8	FALSE	TRUE	8	TRUE	FALSE
9	FALSE	TRUE	9	TRUE	FALSE
10	FALSE	TRUE	10	TRUE	FALSE
Total	0	10	Total	9	1

Table E.17. Arc Fault Evaluation of Vacuum Cleaner (Electrolux) Test Confusion Matrix

	Arcing Event	Normal Event
Relay Trips	9	0
Relay remains on	1	10

Table E.18. Arc Fault Evaluation of Vacuum Cleaner (Electrolux) Test Performance Matrix

Accuracy	Precision	Recall	F1-Score
95%	100%	90%	94.74%

E.7 Angle Grinder

Table E.19. Arc Fault Evaluation of Angle Grinder Test Result

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
1	FALSE	TRUE	1	TRUE	FALSE
2	FALSE	TRUE	2	TRUE	FALSE
3	FALSE	TRUE	3	TRUE	FALSE
4	FALSE	TRUE	4	TRUE	FALSE
5	FALSE	TRUE	5	TRUE	FALSE
6	FALSE	TRUE	6	TRUE	FALSE
7	FALSE	TRUE	7	TRUE	FALSE
8	FALSE	TRUE	8	TRUE	FALSE
9	FALSE	TRUE	9	FALSE	TRUE
10	FALSE	TRUE	10	TRUE	FALSE
Total	0	10	Total	9	1

Table E.20. Arc Fault Evaluation of Angle Grinder Test Confusion Matrix

	Arcing Event	Normal Event
Relay Trips	9	0
Relay remains on	1	10

Table E.21. Arc Fault Evaluation of Angle Grinder Test Performance Matrix

Accuracy	Precision	Recall	F1-Score
95%	100%	90%	94.74%

E.8 Bench Cutter

Table E.22. Arc Fault Evaluation of Bench Cutter Test Result

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
1	FALSE	TRUE	1	TRUE	FALSE
2	FALSE	TRUE	2	TRUE	FALSE
3	FALSE	TRUE	3	TRUE	FALSE
4	FALSE	TRUE	4	TRUE	FALSE
5	FALSE	TRUE	5	TRUE	FALSE
6	FALSE	TRUE	6	FALSE	TRUE
7	FALSE	TRUE	7	TRUE	FALSE
8	FALSE	TRUE	8	TRUE	FALSE

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
9	FALSE	TRUE	9	TRUE	FALSE
10	FALSE	TRUE	10	TRUE	FALSE
Total	0	10	Total	9	1

Table E.23. Arc Fault Evaluation of Bench Cutter Test Confusion Matrix

	Arcing Event	Normal Event
Relay Trips	9	0
Relay remains on	1	10

Table E.24. Arc Fault Evaluation of Bench Cutter Test Performance Matrix

Accuracy	Precision	Recall	F1-Score
95%	100%	90%	94.74%

E.9 Impact Drill

Table E.25. Arc Fault Evaluation of Impact Drill Test Result

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
1	FALSE	TRUE	1	TRUE	FALSE
2	FALSE	TRUE	2	TRUE	FALSE
3	FALSE	TRUE	3	TRUE	FALSE
4	FALSE	TRUE	4	TRUE	FALSE
5	FALSE	TRUE	5	TRUE	FALSE
6	FALSE	TRUE	6	TRUE	FALSE
7	FALSE	TRUE	7	TRUE	FALSE
8	FALSE	TRUE	8	TRUE	FALSE
9	TRUE	FALSE	9	TRUE	FALSE
10	FALSE	TRUE	10	TRUE	FALSE
Total	1	9	Total	10	0

Table E.26. Arc Fault Evaluation of Impact Drill Test Confusion Matrix

	Arcing Event	Normal Event
Relay Trips	10	1
Relay remains on	0	9

Table E.27. Arc Fault Evaluation of Impact Drill Test Performance Matrix

Accuracy	Precision	Recall	F1-Score
95%	90.91%	100%	95.24%

E.10 Heat Gun**Table E.28. Arc Fault Evaluation of Heat Gun Test Result**

Normal Trials			Arc Trials		
	Relay Tripped? (FP)	Relay Still On? (TN)		Relay Tripped? (TP)	Relay Still On? (FN)
Trial			Trial		
1	FALSE	TRUE	1	TRUE	FALSE
2	FALSE	TRUE	2	TRUE	FALSE
3	FALSE	TRUE	3	TRUE	FALSE
4	FALSE	TRUE	4	TRUE	FALSE
5	FALSE	TRUE	5	TRUE	FALSE
6	FALSE	TRUE	6	TRUE	FALSE
7	FALSE	TRUE	7	TRUE	FALSE
8	FALSE	TRUE	8	TRUE	FALSE
9	FALSE	TRUE	9	TRUE	FALSE
10	FALSE	TRUE	10	TRUE	FALSE
Total	0	10	Total	10	0

Table E.29. Arc Fault Evaluation of Heat Gun Test Confusion Matrix

	Arcing Event	Normal Event
Relay Trips	10	0
Relay remains on	0	10

Table E.30. Arc Fault Evaluation of Heat Gun Test Performance Matrix

Accuracy	Precision	Recall	F1-Score
100%	100%	100%	100%

Appendix F

ARC FAULT DATA COLLECTION LOGS

This appendix summarizes the arc-fault data-preparation logs, training report, and exported embedded-model settings used by the TinyML Smart Plug. The raw CSV and JSON artifacts remain in the TinyML project directory; this appendix summarizes the values needed to connect the source artifacts to this paper without reproducing long console dumps or oversized tables.

F.1 Dataset Preparation Summary

Table F.1 summarizes the cleaned dataset reported by the data-preparation workflow. The values were taken from the generated preparation report for the selected training generation.

Table F.1. Arc-fault dataset preparation summary

Item	Count
Raw rows before cleaning	55,085
Rows after cleaning	49,296
Rows removed	5,789
Trainable rows	45,010
Trusted normal rows	17,762
Conflict-marked rows	1
Context-only placeholder rows preserved	16,353
Arc-truth available rows	32,943

The cleaning stage removed invalid or unusable rows while preserving rows that were still needed for context modeling. The resulting set was strongly imbalanced, which is why this paper reports precision, recall, F1-score, and event-level metrics instead of relying only on accuracy.

The retained rows were grouped by division and label as shown in Table F.2. These counts connect the laboratory collection procedure in Appendix E to the training and evaluation workflow described in Chapter 3.

Table F.2. Prepared data division and label counts

Category	Count
Start-division rows	18,475
Steady-division rows	17,762
Arc-division rows	13,059
Normal-label rows	49,013
Arc-label rows	283
Arc-training rows	30,821
Load-context rows	18,475
Conflict policy: none	49,295
Conflict policy: prefer more trusted normal	1

The division counts show that the context model was trained from start segments, while the arc model used the steady and arc divisions. The small number of arc-labeled rows confirms that series-arc detection was a sparse event-classification task.

F.2 TinyML Training Report and Export Summary

Table F.3 summarizes the selected Random Forest settings from the training report. These settings describe the offline model selected before export to the embedded firmware.

Table F.3. Selected Random Forest training settings

Parameter	Selected value
Model family	Random Forest
Number of trees	160
Maximum tree depth	16
Minimum samples per leaf	4
Minimum samples per split	12
Maximum feature fraction	0.35
Bootstrap sampling	Enabled
Split criterion	Entropy
Cost-complexity pruning alpha	0.0001
Minimum impurity decrease	0.000001
Cross-validation splits	5

The final training view contained 35,512 training rows, including 344 positive rows and 35,168 negative rows. The held-out validation split contained 9,434 rows with 90 positive rows, and the held-out test split contained 10,268 rows with 90 positive rows.

Table F.4 gives the held-out test metrics reported for the offline Random Forest model. These frame-level metrics are the source of the TinyML model-performance discussion in Chapter 4.

Table F.4. Held-out frame-level Random Forest test metrics

Metric	Value
Offline report threshold	0.25
Accuracy	0.997078
Precision	0.928571
Recall	0.722222
F1-score	0.812500
True negatives	10,173
False positives	5
False negatives	25
True positives	65

Table F.4 shows that the frame-level classifier had high precision but lower recall. This means that false positive frames were rare, while some positive arc frames were not detected at the frame level. For this reason, Chapter 4 also interprets event-level and trial-level behavior.

The event-level report grouped adjacent positive frames into arc episodes. Table F.5 summarizes this view.

Table F.5. Held-out event-level Random Forest test metrics

Metric	Value
True arc events	25
Detected arc events	21
Missed arc events	4
Predicted events	31
Overlapping predicted events	30
False-alarm events	1
False-alarm sessions	1
False alarms per session	0.1
Event recall	0.840000
Event precision	0.967742
Event F1-score	0.899358

The event-level metrics better match the protection problem because a series arc

appears as a short disturbance episode rather than a single independent frame. The one false-alarm event and four missed events explain why the prototype should be interpreted as complementary protection and not as a certified replacement for standard protective devices.

F.3 Embedded Model Export Summary

The exported embedded arc model used a context-aware input vector. The source header defines feature-version 6, an input dimension of 13, six base waveform features, and seven context fields. The embedded input order is shown in Table F.6.

Table F.6. Embedded Random Forest arc-model input order

Index	Feature
0	Maximum low-current run duration
1	Zero-dwell ratio
2	Low-current ratio
3	Absolute RMS-current z-score versus baseline
4	Midband residual ratio
5	Mid/high-frequency spectral flux
6	Context family: resistive linear
7	Context family: inductive motor
8	Context family: rectifier SMPS
9	Context family: phase-angle controlled
10	Context family: brush universal motor
11	Context family: other mixed
12	Context-family confidence

The deployed header used an embedded arc threshold of 0.4800 for supported known-context decisions, a context-confidence minimum of 0.4500, an unknown-context threshold of 0.7800, and a minimum of three corroborating arc-like feature votes under unknown context.

The exported context model is a prototype classifier with six inputs: ΔI_{rms} , half-cycle asymmetry, midband residual ratio, current THD, mid/high-frequency spectral flux, and high-frequency energy delta. It standardizes each input, computes distance to six family centroids, and accepts a family only when the best context confidence is at least 0.4500. Otherwise, the arc model is evaluated under the stricter unknown-context threshold.

Appendix G

LETTERS AND PROPOSALS

G.1 Assistance Letter Request for the PPA

January 22, 2026

ENGR. KRIS LLOYD T. VILLAMONTE

Director, Physical Plant Administration (PPA)

Ateneo de Naga University

Naga City 4400 Philippines

Dear Engr. Villamonte,

Greetings of Peace!

We are writing to you upon the specific recommendation of our thesis panel to request the technical assistance of the Physical Plant Administration (PPA) for our thesis study, *TinyML Smart Plug for Complementary Protection Against Progressive Fire-Inducing Electrical Faults in Residential Circuits*.

Our research aims to develop a device that detects electrical faults that commonly cause fire, specifically series arc faults, socket heating, and overload that standard breakers often miss. To acquire data for machine learning model training and to validate the performance of the resulting device, we must simulate hazardous conditions, including generating arc faults and inducing overload and heat in the electrical system.

Given the risks involved, we need the support of personnel who are skilled in electrical safety to execute, control, and supervise the simulation of electrical faults.

Attached is our Proposed Safety Guidelines and Testing Protocols, which details the specific dates, times, and locations for the testing and our general plan in conducting the testing.

We would be deeply grateful for your support in safely finishing our thesis study.
Respectfully yours,

FRANCIS M. PEÑA

4 - BS ECE Student

NOTED BY:

DR. REFERENDO D. SORIANO

Thesis Adviser

Date Signed: _____

DR. CELESTE M. OJEDA

Chairperson, ECE/CpE Department

Date Signed: _____

G.2 Safety Guideline Proposal to the PPA

PROPOSED SAFETY GUIDELINES AND TESTING PROTOCOLS

For the Conduct of Electrical Fault Simulation during Data Collection and Testing

I. Title and Proponents

The research project, titled *TinyML Smart Plug for Complementary Protection Against Progressive Fire-Inducing Electrical Faults in Residential Circuits*, is presented by **Alex A. Alferez, Jett M. Lumabi, and Francis M. Peña**, fourth-year BS Electronics Engineering students from the Department of Electronics and Computer Engineering at Ateneo de Naga University. Conducted under the advisement of Dr. Referendo D. Soriano, the study is scheduled for its primary development and validation phase from **January 2026 to May 2026**.

II. Overview

The study aims to develop an intelligent, ESP32-based smart plug capable of detecting fire-inducing electrical faults that standard circuit breakers often miss. These faults are series arc faults, silent sustained overloads, and contact heating.

Since these faults are inherently dangerous, this proposal seeks the assistance of

trained PPA personnel who are skilled in safely handling electrical circuits to ensure the safe and controlled execution of laboratory simulations required for data collection and system validation.

III. Rationale

Validating a solution for these electrical faults requires generating them in a controlled environment. However, creating actual arcs, inducing overloads, and artificially degrading electrical sockets involve significant risks including fire, shock, and equipment damage. To mitigate potential risks and protect students, laboratory equipment, and facilities, all high-voltage procedures and fault simulations will be conducted under the direct supervision and execution of authorized PPA personnel.

IV. Objectives

The objective of this proposal is to safely collect actual electrical waveform data of hazardous faults to train and test the TinyML fault-detection system, specifically:

1. To simulate sustained overloads from 10A to 15A to test overload behavior, current-sensor accuracy, and protective response;
2. To induce contact heating above 70°C on test terminals or a degraded socket to validate thermal sensing and estimation;
3. To simulate series arc faults using a controlled arc generator setup under PPA supervision; and
4. To establish a safe protocol where PPA personnel manage live wires and fault simulations while students handle data logging.

V. Scope and Limitations

1. The request covers the period from January 2026 to May 2026.
2. PPA assistance is requested specifically for live fault generation, emergency switching, and setup inspection.

3. Testing is limited to single-phase 230V AC in a controlled laboratory environment.
4. PPA staff are not expected to debug ESP32 code or electronics; their role is to manage the AC mains supply and fault-generation rigs for safety.

VI. Methodology and Safety Protocols

The following protocols will be observed throughout the testing period:

1. No high-voltage testing or fault simulation will commence without an authorized PPA staff member present.
2. The testing setup will use an accessible circuit breaker, and the D315 workshop breaker must remain accessible to laboratory custodians and PPA personnel.
3. All personnel inside the testing perimeter must wear rubber-soled safety shoes, insulated gloves, and safety glasses.
4. Students will handle data logging and software monitoring. PPA staff will handle live connections and fault triggers.

Table G.1. Hazard identification and mitigation strategy

Fault type	Identified hazard	Mitigation strategy
Overload	Wire overheating and potential fire	Use a controlled load bank or source, maintain real-time thermal monitoring, and keep breaker access available.
Arc fault	Plasma burns, UV radiation, and fire ignition	Require safety goggles and insulated gloves, keep a fire extinguisher on standby, and use an arc containment cover or box.
Contact heating	Burns and fumes from heated or melting materials	Provide ventilation, use non-contact temperature checks, use face masks as needed, and follow hot-work precautions.

VII. Proposed Schedule

The following timeline details the activities requiring PPA presence. Dates are subject to personnel availability, institutional announcements, weather interruptions, and

holidays.

Table G.2. Proposed schedule of activities requiring PPA assistance

Activity	Dates	Location	PPA involvement
Checking the testing setup	February 26–27, 2026	D315 Workshop	Confirm the safety of the researchers' testing setup.
Data collection for overload	March 2–6, 2026	D315 Workshop	Assist in simulating overload using appliances or a high-power load while breaker access is maintained.
Data collection for contact heating	March 9–13, 2026	D315 Workshop	Assist in simulating contact heating through a degraded socket or controlled heating of contacts.
Data collection for arc fault	March 16–April 1, 2026	D315 Workshop	Assist in simulating a series arc fault by operating the covered arc-fault generator.
Model and device testing	April 6–20, 2026	D315 Workshop and Electrical Warehouse	Assist during induced-fault tests used to evaluate the device algorithm.

Table G.3. Proposed daily testing hours

Monday	Tuesday	Wednesday	Thursday	Friday
9AM–3PM	9AM–5PM	9AM–3PM	9AM–5PM	9AM–3PM

NOTED BY:

DR. REFERENDO D. SORIANO

Thesis Adviser

Date Signed: _____

DR. CELESTE M. OJEDA

Chairperson, ECE/CpE Department

Date Signed: _____

APPROVED BY:

ENGR. KRIS LLOYD T. VILLAMONTE

Director, Physical Plant Administration (PPA)

Date Signed: _____

Appendix H GANTT CHART

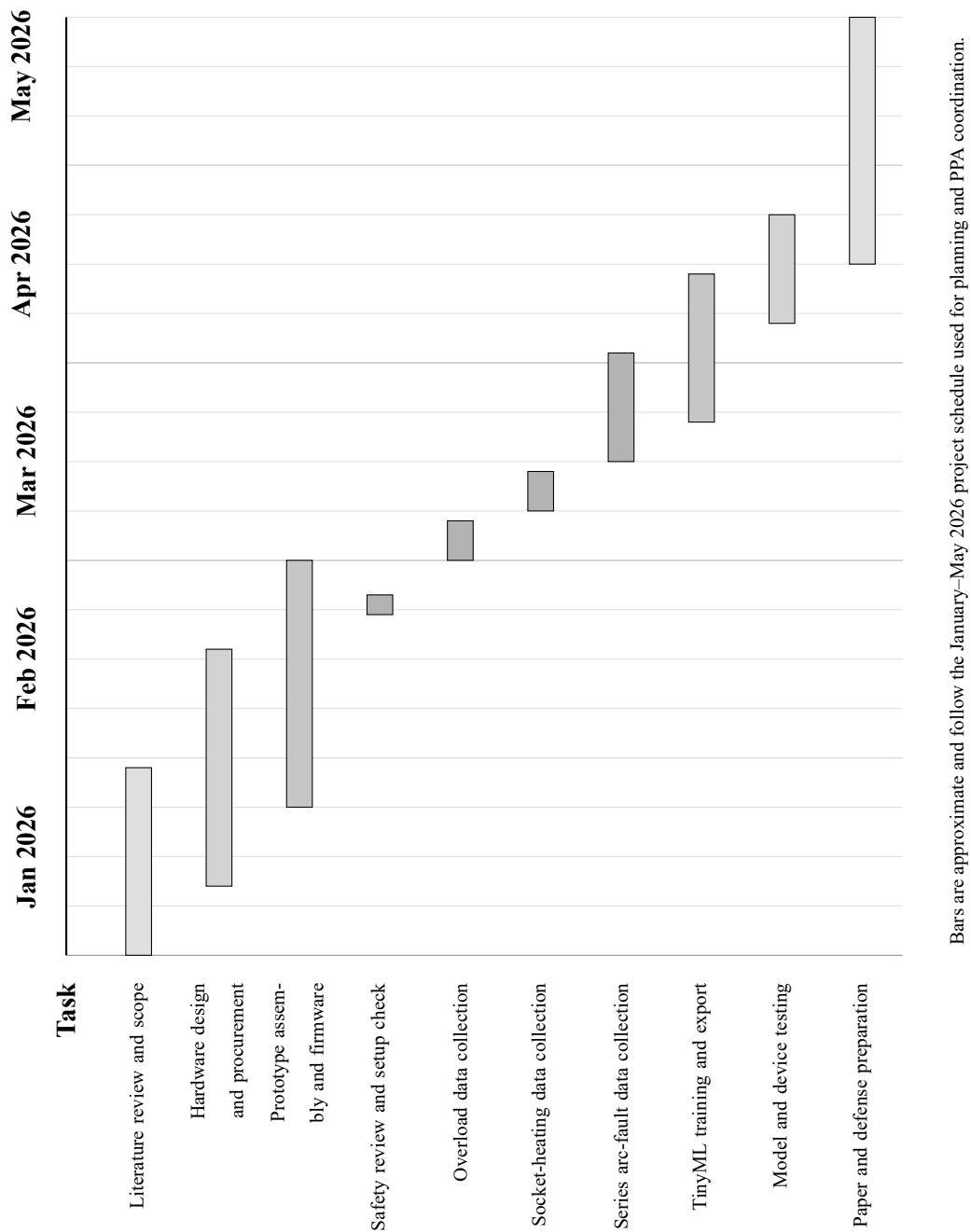


Figure H.1. Gantt chart of project activities.

Appendix I

COST BREAKDOWN

Table I.1. Prototype hardware cost breakdown

Item	Component material	or Function in the prototype	Qty.	Source	Cost (PHP)
1	Seeed Studio XIAO ESP32S3	Main ESP32-S3 controller for sensing, computation, TinyML inference, Wi-Fi, Firebase, and PWA communication	1	DigiKey	250.00
2	ACS37035 KLACTR-020B5	Current measurement for RMS current, overload detection, socket-heating estimation, and waveform-feature extraction	1	DigiKey	173.00
3	MCP3204-CI/P	External ADC for digitizing the current-sensor signal	1	DigiKey	250.00
4	MCP6024 op-amp	Anti-aliasing filter stage before ADC sampling	1	DigiKey	145.06
5	ZMPT101B	AC 220V AC line-voltage monitoring and protection input module	1	Shopee	156.00
6	10k Ω NTC thermistor	Socket-interface temperature sensing for heating protection	1	Shopee	6.00
7	SSR-40DA	40A Load-disconnection ac-tuator in the protection module	1	Shopee	128.00
8	74HC74 IC	D-latch Hardware latch for stable trip-control behavior	1	DigiKey	18.01
9	Hi-Link PM01	HLK- Isolated 5V supply for control electronics	1	Shopee	179.00
10	TPS63020	buck- Low-voltage power-path regulation for battery-backed operation	1	Shopee	219.00
11	AMS1117	3.3V 3.3V rail for the ESP32-S3 and low-voltage logic	1	Shopee	40.00
12	3.7V 600mAh lithium-polymer battery	Backup supply for uninterrupted powered section	1	Shopee	210.00
13	0.96in OLED display	SSD1306 Local display for readings, relay status, and fault messages	1	Shopee	149.00

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Table I.1. Prototype hardware cost breakdown (continued)

Item	Component or Function in the prototype material	Qty.	Source	Cost (PHP)	
14	5V passive buzzer module	Local audible alert for detected fault conditions	1	Shopee	49.00
15	220V female socket/outlet	Protected load-side outlet for appliance connection	1	Shopee	86.00
16	220V male plug	Line-side input plug for the prototype	1	Shopee	20.00
17	Universal PCB/prototyping board	Mounting support for sensing, filtering, latch, and support circuitry	1	Shopee	120.00
18	Terminal crimp blocks, headers, and wiring allowance	Internal AC/DC wiring and detachable interconnections	1 lot	Shopee	194.00
19	40mm cooling fan	Internal cooling support for enclosed operation	1	Shopee	79.00
20	SMD ceramic capacitors, SMD resistors, and small passives	Anti-aliasing, pull-up, divider, decoupling, and support networks	1 lot	Shopee	180.00
21	3D printed enclosure	Protective housing for the assembled smart-plug electronics	1	Electronics shop	Work- 1,009.00
22	PCB fabrication service for device modules	Fabrication of main device module boards	1	Shopee	950.00
23	PCB fabrication service for ACS37035 module	Fabrication of current-sense for sensor carrier board	1	Shopee	300.00
24	PCB fabrication service for ADC module	Fabrication of ADC module board	1	Shopee	350.00
Total estimated prototype hardware cost					5,260.07